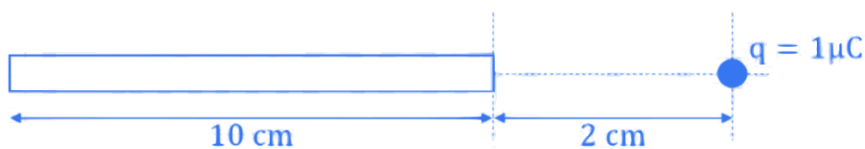


Electric Charges and Fields

Question1

A point charge $q = 1\mu\text{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu\text{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is ____ N.

(Use : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)



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Answer: 90

Solution:

Because the charge on the wire is distributed, we consider a small element dx on the wire at a distance x from the point charge q .



Linear charge density of the line charge is $\lambda = \frac{Q}{L}$

Charge on element $dQ = \lambda dx$

Small force acting between the charges is,

$$dF = \frac{1}{4\pi\epsilon_0} \frac{q \cdot dQ}{x^2} = \frac{kq\lambda dx}{x^2}$$

To find the total force F , we integrate from the nearest end ($x = 2 \text{ cm} = 0.02 \text{ m}$) to the farthest end ($x = 2 \text{ cm} + 10 \text{ m} = 12 \text{ cm} = 0.12 \text{ m}$):

$$F = \int_{0.02}^{0.12} \frac{kq\lambda}{x^2} dx = kq\lambda \left[-\frac{1}{x} \right]_{0.02}^{0.12}$$

$$\Rightarrow F = kq\lambda \left(\frac{1}{0.02} - \frac{1}{0.12} \right) = kq\lambda \left(\frac{6-1}{0.12} \right) = \frac{5kq\lambda}{0.12}$$

$$\Rightarrow F = \frac{5kq}{0.12} \times \frac{Q}{L}$$

Substituting the given values,

$$k = 9 \times 10^9; q = 10^{-6}; Q = 24 \times 10^{-6}; L = 10 \text{ cm} = 0.10 \text{ m}$$

$$F = \frac{5 \times 9 \times 10^9 \times 1 \times 10^{-6} \times 24 \times 10^{-6}}{0.10 \times 0.12} \text{ N}$$

$$\Rightarrow F = \frac{216}{2.4}$$

$$\Rightarrow F = 90 \text{ N}$$

Therefore, the electrostatic force exerted by the finite wire on the point charge is 90 N .

Hence, the correct answer is 90 .

Question2

A three coulomb charge moves from the point $(0, -2, -5)$ to the point $(5, 1, 2)$ in an electric field expressed as

$\vec{E} = 2x\hat{i} + 3y^2\hat{j} + 4z\hat{k}$ N/C. The work done in moving the charge is _____ J.

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Answer: 186

Solution:

The work done (W) by an external agent in moving a charge (q) in an electric field ($\vec{E}$) is given by the change in potential energy, which is the negative of the work done by the field.

The electric force acting on the charge by electric field is $\vec{F} = q\vec{E}$

So, the work done is :

$$W = q \int \vec{E} \cdot d\vec{r}$$

Where $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ is the infinitesimal displacement vector.

$$W = q \int_{(0,-2,-5)}^{(5,1,2)} (2x\hat{i} + 3y^2\hat{j} + 4\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\Rightarrow W = q \left[\int_0^5 2x dx + \int_{-2}^1 3y^2 dy + \int_{-5}^2 4 dz \right]$$

$$\Rightarrow W = q \left[2 \left[\frac{x^2}{2} \right]_0^5 + 3 \left[\frac{y^3}{3} \right]_{-2}^1 + 4 [x]_{-5}^2 \right]$$

$$\Rightarrow W = 3 \left[[5^2 - 0^2] + [1^3 - (-2)^3] + 4[2 - (-5)] \right]$$

$$\Rightarrow W = 3 \times (25 + 9 + 28)$$

$$\Rightarrow W = 3 \times 62 = 186 \text{ J}$$

Therefore, the work done in moving the charge is 186 J.

Question3

A point charge of 10^{-8}C is placed at origin. The work done in moving a point charge $2\mu\text{C}$ from point $A(4, 4, 2)\text{m}$ to $B(2, 2, 1)\text{m}$ is _____ J. $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ in SI units} \right)$

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Options:

A.

$$30 \times 10^{-6}$$

B.

$$0$$

C.

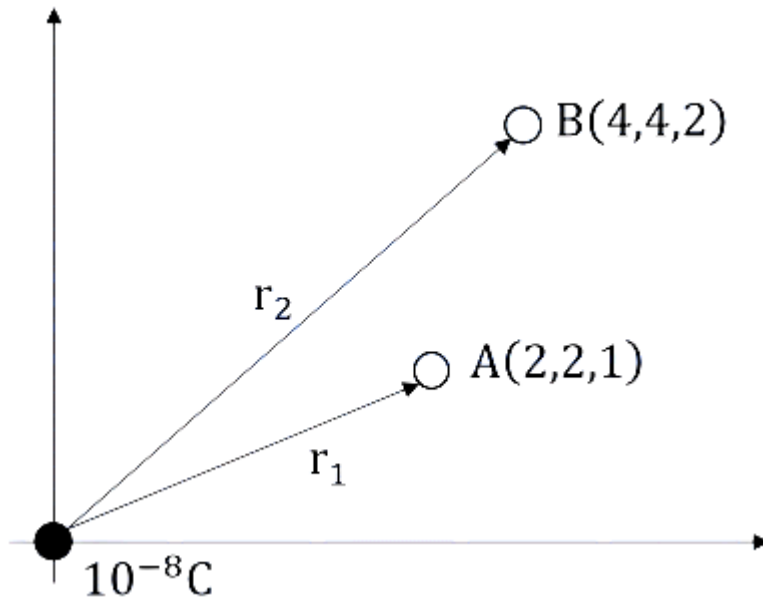
$$15 \times 10^{-6}$$

D.

$$45 \times 10^{-6}$$

Answer: A

Solution:



The work done (W) by an external agent in moving a charge q from point A to point B in the field of a source charge Q is equal to the change in electrostatic potential energy :

$$W = U_B - U_A = q (V_B - V_A)$$

Where the electrostatic potential V at a distance r from a source charge Q is :

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

The source charge at origin is, $Q = 10^{-8}\text{C}$

The test charge is $q = 2\mu\text{C} = 2 \times 10^{-6}\text{C}$

The distance r of a point (x, y, z) from the origin is $r = \sqrt{x^2 + y^2 + z^2}$

For Point A (4, 4, 2):

$$r_A = \sqrt{4^2 + 4^2 + 2^2} = \sqrt{16 + 16 + 4} = \sqrt{36} = 6 \text{ m}$$

For Point B (2, 2, 1):

$$r_B = \sqrt{2^2 + 2^2 + 1^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3 \text{ m}$$

So, the potential at A (V_A) is,

$$V_A = \frac{kQ}{r_A} = \frac{(9 \times 10^9) 10^{-8}}{6} = \frac{90}{6} = 15V$$

And the potential at B (V_B) is,

$$V_B = \frac{kQ}{r_B} = \frac{(9 \times 10^9)10^{-8}}{3} = \frac{90}{3} = 30V$$

So, the work done by external agent is,

$$W = q (V_B - V_A)$$

$$\Rightarrow W = (2 \times 10^{-6})(30 - 15)$$

$$\Rightarrow W = 30 \times 10^{-6} \text{ J}$$

Therefore, the work done to be $30 \times 10^{-6} \text{ J}$.

Hence, the correct option is (A).

Question4

Consider two identical metallic spheres of radius R each having charge Q and mass m . Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of u towards each other. The minimum value of u , so that they can just touch each other is:

(Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the Gravitational constant)

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Options:

A.

$$\sqrt{\frac{kQ^2}{4mR} \left(1 + \frac{Gm^2}{kQ^2}\right)}$$

B.

$$\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{2kQ^2}\right)}$$

C.

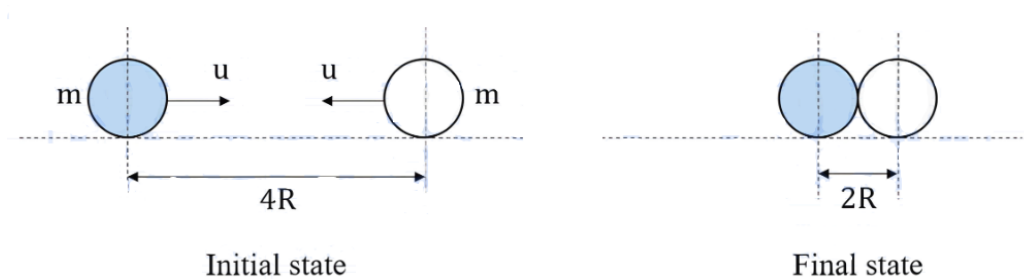
$$\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{kQ^2}\right)}$$

D.

$$\sqrt{\frac{kQ^2}{4mR} \left(1 - \frac{Gm^2}{kQ^2}\right)}$$

Answer: D

Solution:



At the start, the spheres are moving towards each other.

Since there are two spheres of mass m moving with speed u , so the initial kinetic energy is,

$$K_i = \frac{1}{2}mu^2 + \frac{1}{2}mu^2 = mu^2$$

Initial electrostatic potential energy is,

$$U_{ei} = \frac{kQ^2}{4R}$$

Initial gravitational potential energy is :

$$U_{gi} = -\frac{Gm^2}{4R}$$

When the spheres just touch, their centres are separated by a distance equal to the sum of their radii ($R + R = 2R$). To find the minimum u , we assume they come to a momentary rest at this point.

Total final kinetic energy, $K_f = 0$

Final electrostatic potential energy is,

$$U_{ef} = \frac{kQ^2}{2R}$$

Final gravitational potential energy is :

$$U_{gf} = -\frac{Gm^2}{2R}$$

The total energy remains constant :

$$K_i + U_{ei} + U_{gi} = K_f + U_{ef} + U_{gf}$$

$$\Rightarrow mu^2 + \frac{kQ^2}{4R} - \frac{Gm^2}{4R} = 0 + \frac{kQ^2}{2R} - \frac{Gm^2}{2R}$$

$$\Rightarrow mu^2 = \left(\frac{kQ^2}{2R} - \frac{kQ^2}{4R} \right) - \left(\frac{Gm^2}{2R} - \frac{Gm^2}{4R} \right)$$

$$\Rightarrow mu^2 = \frac{kQ^2}{4R} - \frac{Gm^2}{4R}$$

$$\Rightarrow mu^2 = \frac{kQ^2}{4R} \left(1 - \frac{Gm^2}{kQ^2} \right)$$

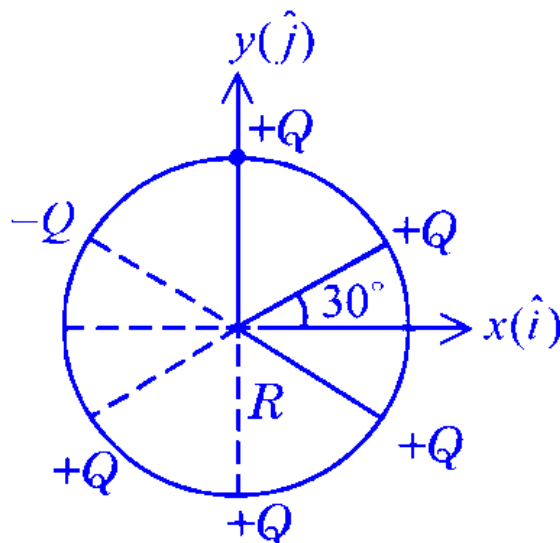
$$\Rightarrow u^2 = \frac{kQ^2}{4mR} \left(1 - \frac{Gm^2}{kQ^2} \right)$$

$$\Rightarrow u = \sqrt{\frac{kQ^2}{4mR} \left(1 - \frac{Gm^2}{kQ^2} \right)}$$

Question5

Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is ____ .

(ϵ_0 is permittivity of free space)



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Options:

A.

$$-\left(\frac{5Q}{8\pi\epsilon_0 R^2}\right)(\hat{i} - 3\hat{j})$$

B.

$$\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$$

C.

$$-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$$

D.

$$-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i} + \sqrt{3}\hat{j})$$

Answer: C

Solution:

The electric field ($\vec{E}$) created by a point charge q at a distance r is given by Coulomb's Law:

$$|\vec{E}| = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

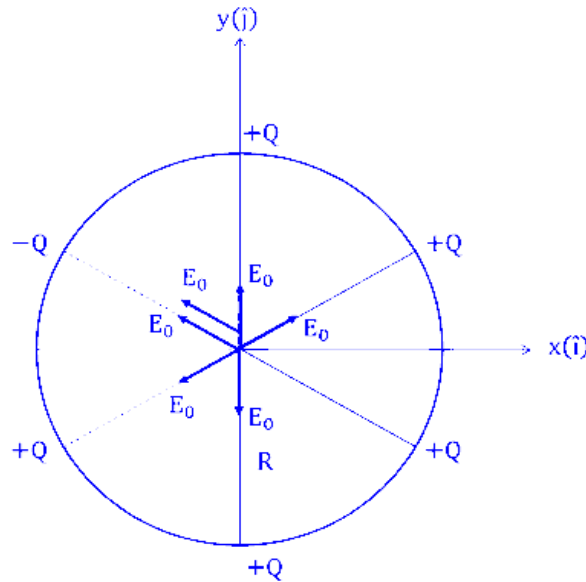
If the charge is positive ($+q$), the electric field points away from the charge. If the charge is negative ($-q$), the electric field points towards the charge.

When multiple charges are present, the net electric field at any point is simply the vector sum of the individual electric fields created by each charge.

Let the centre of the circle be the origin (0, 0). All six charges are at a distance R from the center.

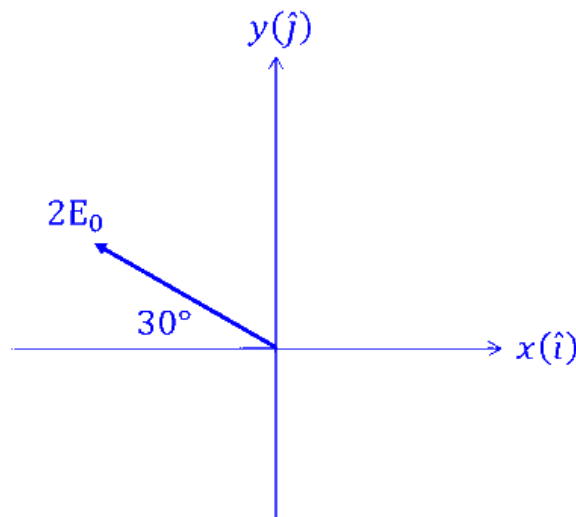
Let the magnitude of the electric field produced by a single charge of magnitude Q at the center as E_0 :

$$E_0 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$



The electric field due to charge $+Q$ charge at 90° is downward with magnitude E_0 and that for the $+Q$ at 270° it is upward with magnitude E_0 . These two vectors are equal and opposite, so they cancel each other out.

The electric field due to charge $+Q$ charge at 30° is with magnitude E_0 and opposite to that for the $+Q$ at 210° with magnitude E_0 . These two vectors are equal and opposite, so they cancel each other out.



$$\left| \vec{E}_{net} \right| = E_0 + E_0 = 2E_0$$

The direction of this net vector is 150° or 30° with negative x -axis.

So, the net electric field in vector form is,

$$\vec{E}_{\text{net}} = (2E_0) (\cos 150^\circ \hat{i} + \sin 150^\circ \hat{j})$$

$$\Rightarrow \vec{E}_{\text{net}} = (2E_0) \left(-\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j} \right)$$

$$\Rightarrow \vec{E}_{\text{net}} = -E_0(\sqrt{3}\hat{i} - \hat{j})$$

Putting the expression for E_0 ,

$$\vec{E}_{\text{net}} = -\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} (\sqrt{3}\hat{i} - \hat{j})$$

Therefore, the net electric field at the centre of circle is $-\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} (\sqrt{3}\hat{i} - \hat{j})$.

Hence, the correct option is (C).

Question6

A simple pendulum has a bob with mass m and charge q . The pendulum string has negligible mass. When a uniform and horizontal electric field $\vec{E}$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is _____ .

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Options:

A.

$$mg - qE$$

B.

$$\sqrt{m^2 g^2 + q^2 E^2}$$

C.

$$m g + qE$$

D.

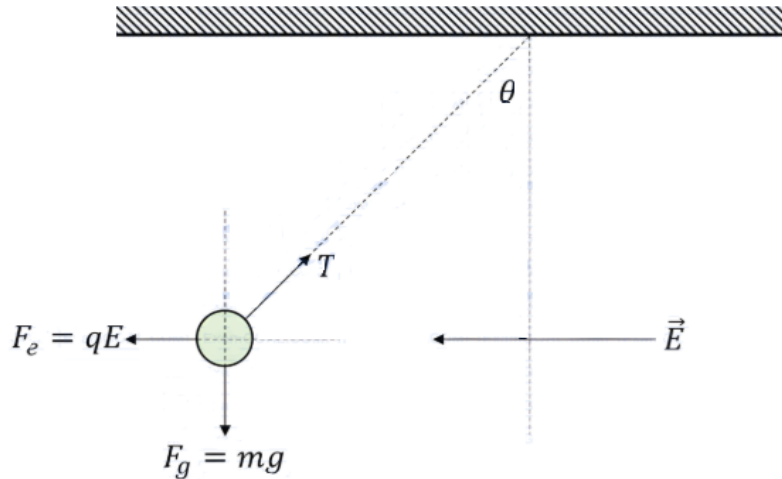
$$\sqrt{m^2 g^2 - q^2 E^2}$$

Answer: B

Solution:

When the bob is at its new equilibrium position, it is stationary. This means the net force acting on it is zero.

The free body diagram of bob is,



The gravitational force acting vertically downwards. Its magnitude is $F_g = mg$.

Because the bob has a charge q and is in a horizontal electric field $\vec{E}$, it experiences a horizontal force. Its magnitude is $F_e = qE$. Let's assume it points to the left.

The force exerted by the string on the bob, acting along the length of the string towards the pivot point with magnitude T .

Let the string make an angle θ with the vertical when it reaches this new equilibrium position.

On resolving the forces into vertical and horizontal components.

Along vertical direction (y - axis):

The upward component of the tension must balance the downward weight of the bob.

Therefore: $T \cos \theta = mg \dots (i)$

Along horizontal direction (x - axis): The horizontal component of the tension must balance the outward pull of the electric force.

Therefore : $T \sin \theta = qE \dots (ii)$

Squaring and adding both the equations,

$$T^2 \sin^2 \theta + T^2 \cos^2 \theta = q^2 E^2 + m^2 g^2$$

$$\Rightarrow T^2 (\sin^2 \theta + \cos^2 \theta) = q^2 E^2 + m^2 g^2$$

$$\Rightarrow T^2 = q^2 E^2 + m^2 g^2$$

$$\Rightarrow T = \sqrt{m^2 g^2 + q^2 E^2}$$

Therefore, the magnitude of the tension is $T = \sqrt{m^2 g^2 + q^2 E^2}$.

Hence, the correct option is (B).

Question7

Electric field in a region is given by $\vec{E} = Ax\hat{i} + By\hat{j}$, where $A = 10 \text{ V/m}^2$ and $B = 5 \text{ V/m}^2$. If the electric potential at a point $(10, 20)$ is 500 V , then the electric potential at origin is ____ V.

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Options:

A.

1000

B.

0

C.

2000

D.

500

Answer: C

Solution:

Electric potential difference between two points is defined as the negative of the work done by the electric field in moving a unit positive charge from the initial point to the final point.

The infinitesimal work done dW by an electric field $\vec{E}$ on a unit charge over a tiny displacement $d\vec{r}$ is $\vec{E} \cdot d\vec{r}$.

Therefore, the infinitesimal change in potential, dV , is :

$$dV = -\vec{E} \cdot d\vec{r}$$

To find the total potential difference between an initial position $\vec{r}_1$ and a final position $\vec{r}_2$, we integrate both sides :

$$\int_{V_1}^{V_2} dV = - \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

$$\Rightarrow V_2 - V_1 = - \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

The electric field is given as: $\vec{E} = Ax\hat{i} + By\hat{j}$

The general position displacement vector is: $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

The initial point (Point 1) is the origin: $(x_1, y_1) = (0, 0)$. Let the potential here be V_0 .

The final point (Point 2) is: $(x_2, y_2) = (10, 20)$. The potential here is $V = 500$ V.

$$V_2 - V_1 = - \int_{(0,0)}^{(10,20)} (Ax\hat{i} + By\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\Rightarrow V_2 - V_1 = - \int_{(0,0)}^{(10,20)} Axdx + Bydy$$

$$\Rightarrow V_2 - V_1 = - \left[\int_0^{10} Axdx + \int_0^{20} Bydy \right]$$

$$\Rightarrow V_2 - V_1 = - \left[\left[A \left(\frac{x^2}{2} \right) \right]_0^{10} + \left[B \left(\frac{y^2}{2} \right) \right]_0^{20} \right]$$

$$\Rightarrow V_2 - V_1 = - \left[A \left(\frac{10^2}{2} - \frac{0^2}{2} \right) + B \left(\frac{20^2}{2} - \frac{0^2}{2} \right) \right]$$

$$\Rightarrow V_2 - V_1 = - \left[A \left(\frac{100}{2} \right) + B \left(\frac{400}{2} \right) \right]$$

$$\Rightarrow V_2 - V_1 = -[50A + 200B]$$

We are given : $A = 10$, $B = 5$, $V_2 = 500$ V

$$500 - V_0 = -[50(10) + 200(5)]$$

$$\Rightarrow 500 - V_0 = -[500 + 1000]$$

$$\Rightarrow 500 - V_0 = -1500$$

$$\Rightarrow V_0 = 500 + 1500$$

$$\Rightarrow V_0 = 2000$$
 V

Question8

Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is :

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Options:

A.

$$1 : 3^{2/3}$$

B.

$$3^{2/3} : 1$$

C.

$$1 : 3^{1/3}$$

D.

$$1 : 2^{2/3}$$

Answer: A

Solution:

When small liquid bubbles (or drops) coalesce to form a larger one, two fundamental quantities are conserved: the total charge and the total volume.

Let the radius of each small initial bubble be r and the charge on each be q .

The electric potential (V_1) on the surface of one such small bubble is given by the formula for the potential of a charged sphere:

$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Three such identical bubbles coalesce to form one large bubble.

Volume of 3 small bubbles = Volume of 1 large bubble

$$3 \times \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3$$

$$\Rightarrow 3r^3 = R^3 \Rightarrow R = 3^{1/3}r$$

Using conservation of charge, the total charge (Q) on the new bigger bubble is simply the algebraic sum of the charges of the three initial bubbles.

$$Q = q + q + q = 3q$$

The electric potential (V_2) on the surface of the resultant bigger bubble is :

$$V_2 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

$$\Rightarrow V_2 = \frac{1}{4\pi\epsilon_0} \frac{3q}{3^{\frac{1}{3}}r} = 3^{\frac{2}{3}} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right)$$

$$\Rightarrow V_2 = 3^{\frac{2}{3}} V_1$$

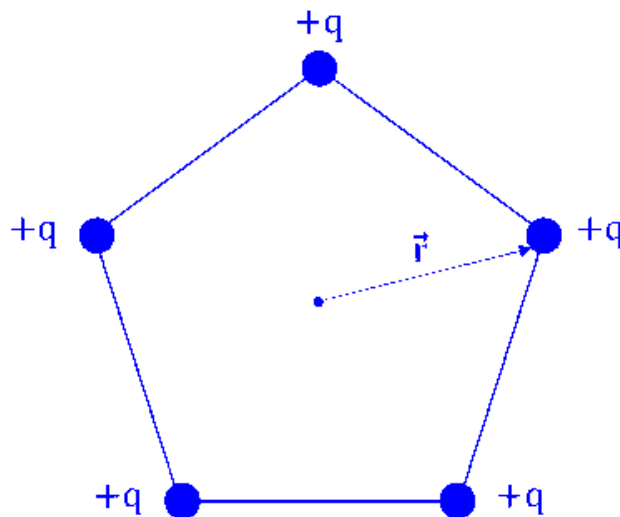
So, the ratio of the potential on one initial bubble (V_1) to that on the resultant bigger bubble (V_2),

$$\frac{V_1}{V_2} = \frac{V_1}{3^{\frac{2}{3}} V_1} = \frac{1}{3^{\frac{2}{3}}}$$

So, the ratio is $1 : 3^{2/3}$. Hence, the correct option is (A).

Question9

Five positive charges each having charge q are placed at the vertices of a pentagon as shown in the figure. The electric potential (V) and the electric field ($\vec{E}$) at the center O of the pentagon due to these five positive charges are :



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Options:

A.

$$V = \frac{5q}{4\pi\epsilon_0 r} \text{ and } \vec{E} = \frac{5\sqrt{3}q}{8\pi\epsilon_0 r^2} \hat{r}$$

B.

$$V = \frac{5q}{4\pi\epsilon_0 r} \text{ and } \vec{E} = 0$$

C.

$$V = 0 \text{ and } \vec{E} = 0$$

D.

$$V = \frac{5q}{4\pi\epsilon_0 r} \text{ and } \vec{E} = \frac{5q}{4\pi\epsilon_0 r^2} \hat{r}$$

Answer: B

Solution:

Electric potential is a scalar quantity. According to the principle of superposition the total electric potential at a point is the algebraic sum of the individual potentials created by each charge.

The potential V_i at the center (at distance r) due to a single point charge $+q$ at a vertex is :

$$V_i = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Since there are 5 identical positive charges situated at the exact same distance r from the center O , the total potential V is simply 5 times the potential of a single charge :

$$V = V_1 + V_2 + V_3 + V_4 + V_5$$

$$\Rightarrow V = 5 \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right)$$

$$V = \frac{5q}{4\pi\epsilon_0 r}$$

Electric field is a vector quantity, which means both magnitude and direction must be considered.

The net electric field at the centre is the vector sum of the electric fields produced by each individual charge :

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \vec{E}_5$$

$$\Rightarrow \vec{E}_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}_1 + \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}_2 + \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}_3 + \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}_4 + \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}_5$$

$$\Rightarrow \vec{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} (\hat{r}_1 + \hat{r}_2 + \hat{r}_3 + \hat{r}_4 + \hat{r}_5)$$

$$\Rightarrow \vec{E}_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{\vec{r}_1}{r} + \frac{\vec{r}_2}{r} + \frac{\vec{r}_3}{r} + \frac{\vec{r}_4}{r} + \frac{\vec{r}_5}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} (\vec{r}_1 + \vec{r}_2 + \vec{r}_3 + \vec{r}_4 + \vec{r}_5)$$

Using closed polygon rule, $\vec{r}_1 + \vec{r}_2 + \vec{r}_3 + \vec{r}_4 + \vec{r}_5 = \vec{0}$

Therefore, $\vec{E}_{\text{net}} = 0$

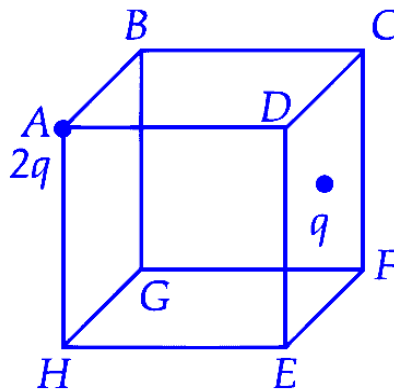
So, at the centre of regular polygon,

The electric potential is $V = \frac{5q}{4\pi\epsilon_0 r}$ and the electric field is $\vec{E} = 0$.

Hence, the correct option is (B).

Question10

Two point charges $2q$ and q are placed at vertex A and centre of face $CDEF$ of the cube as shown in figure. The electric flux passing through the cube is:



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Options:

A.

$$\frac{3q}{2\epsilon_0}$$

B.

$$\frac{3q}{4\epsilon_0}$$

C.

$$\frac{q}{\epsilon_0}$$

D.

$$\frac{3q}{\epsilon_0}$$

Answer: B

Solution:

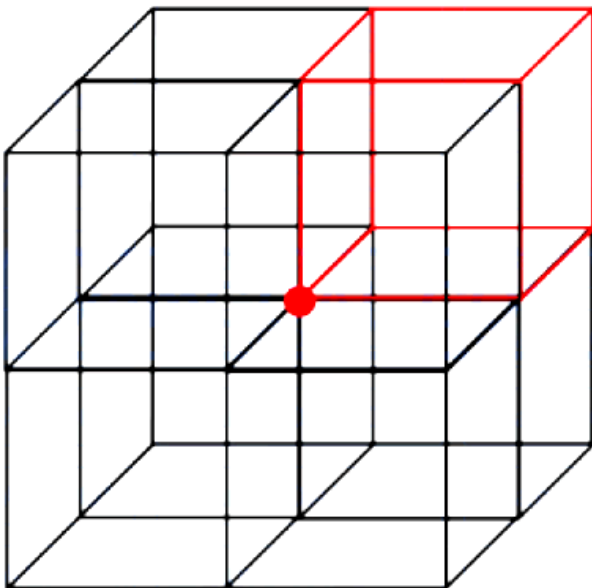
According to Gauss's Law, the total electric flux Φ through a closed surface is

$$\Phi = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

For the flux contribution from charge at vertex A ($2q$)

As the charge $2q$ is placed at a corner (vertex) of the cube.

A corner of a cube is a meeting point for 8 identical cubes stacked together

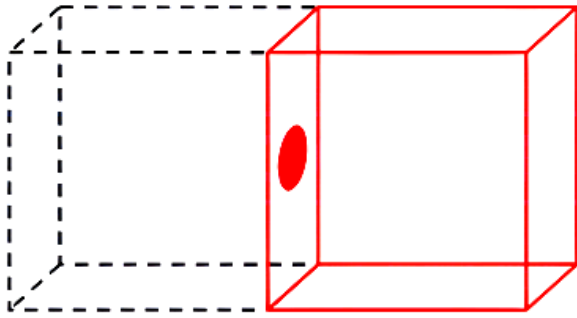


Since the charge is shared equally among 8 cubes, the portion of the flux entering the specific cube is $\frac{1}{8}$ of the total flux produced by the charge.

$$\Phi_A = \frac{1}{8} \times \frac{2q}{\epsilon_0} = \frac{2q}{8\epsilon_0} = \frac{q}{4\epsilon_0}$$

For the flux contribution from charge at the centre of face CDEF (q)

The charge q is placed at the center of the face CDEF and a face of a cube is shared by exactly 2 identical cubes placed side-by-side.



Since the charge is shared equally between 2 cubes, the portion of the flux entering the specific cube is $\frac{1}{2}$ of the total flux produced by the charge.

$$\Phi_{\text{face}} = \frac{1}{2} \times \frac{q}{\epsilon_0} = \frac{q}{2\epsilon_0}$$

The total flux passing through the cube is the sum of the contributions from both charges.

$$\Phi_{\text{total}} = \Phi_A + \Phi_{\text{face}}$$

$$\Rightarrow \Phi_{\text{total}} = \frac{q}{4\epsilon_0} + \frac{q}{2\epsilon_0}$$

$$\Rightarrow \Phi_{\text{total}} = \frac{q+2q}{4\epsilon_0} = \frac{3q}{4\epsilon_0}$$

Therefore, the total electric flux passing through the cube is $\frac{3q}{4\epsilon_0}$.

Hence, the correct option is (B).

Question11

Two charges $7\mu\text{C}$ and $-2\mu\text{C}$ are placed at $(-9, 0, 0)\text{cm}$ and $(9, 0, 0)\text{cm}$ respectively in an external field $E = \frac{A}{r^2}\hat{r}$, where $A = 9 \times 10^5 \text{ N/C} \cdot \text{m}^2$. Considering the potential at infinity is 0 , the electrostatic energy of the configuration is _____ J.

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Options:

A.

49.3

B.

1.4

C.

24.3

D.

-90.7

Answer: A

Solution:

The total electrostatic energy (U) of this system consists of two parts :

1. The energy of each individual charge due to the external electric field.
2. The interaction energy between the two charges.

The external electric field is given as $E = \frac{A}{r^2}$.

The relation between electric field and potential (V) is:

$$V(r) = - \int E \cdot dr = - \int \frac{A}{r^2} dr = \frac{A}{r}$$

The energy of a charge q at distance r is $U = qV(r) = \frac{qA}{r}$.

For $q_1 = 7\mu\text{C} = 7 \times 10^{-6}\text{C}$:

Position $r_1 = 9 \text{ cm} = 0.09 \text{ m}$.

$$U_1 = \frac{q_1 A}{r_1} = \frac{(7 \times 10^{-6})(9 \times 10^5)}{0.09} = \frac{6.3}{0.09} = 70 \text{ J}$$

For $q_2 = -2\mu\text{C} = -2 \times 10^{-6}\text{C}$:

Position $r_2 = 9 \text{ cm} = 0.09 \text{ m}$.

$$U_2 = \frac{q_2 A}{r_2} = \frac{(-2 \times 10^{-6})(9 \times 10^5)}{0.09} = -\frac{1.8}{0.09} = -20 \text{ J}$$

The energy due to the interaction between the two point charges is given by :

$$U_{\text{int}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{d}$$

The distance (d) between (-9, 0, 0) and (9, 0, 0) is $18 \text{ cm} = 0.18 \text{ m}$.

Using $k = 9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$:

$$U_{\text{int}} = \frac{(9 \times 10^9)(7 \times 10^{-6})(-2 \times 10^{-6})}{0.18}$$

$$\Rightarrow U_{\text{int}} = -\frac{0.126}{0.18} = -0.7 \text{ J}$$

So, total electrostatic energy (U_{total}),

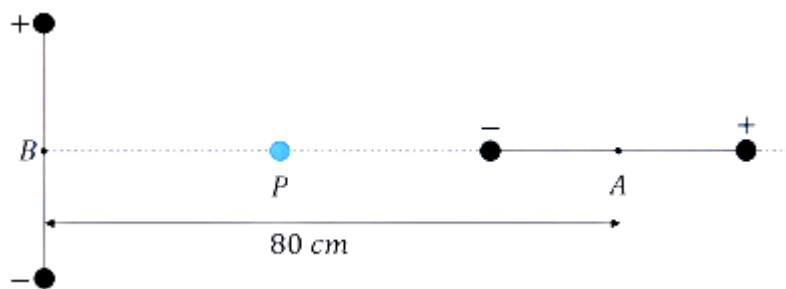
$$U_{\text{total}} = U_1 + U_2 + U_{\text{int}}$$

$$\Rightarrow U_{\text{total}} = 70 + (-20) + (-0.7)$$

$$\Rightarrow U_{\text{total}} = 50 - 0.7 = 49.3 \text{ J}$$

Question12

Two short dipoles (A , B), A having charges $\pm 2\mu\text{C}$ and length 1 cm and B having charges $\pm 4\mu\text{C}$ and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P , equi-distant from the centres of both dipoles is _____ N/C.



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Options:

A.

$$\frac{9}{16}\sqrt{2} \times 10^5$$

B.

$$\frac{9}{16}\sqrt{2} \times 10^4$$

C.

$$9\sqrt{2} \times 10^4$$

D.

$$4.5\sqrt{2} \times 10^4$$

Answer: B

Solution:

The dipole moment is given by $p = q \times 2l$.

For Dipole A : $q_A = 2 \times 10^{-6} \text{C}$, $2l = 1 \text{ cm} = 0.01 \text{ m}$.

$$p_A = (2 \times 10^{-6})(0.01) = 2 \times 10^{-8} \text{C} \cdot \text{m}$$

For Dipole B : $q_B = 4 \times 10^{-6} \text{C}$, $2l = 1 \text{ cm} = 0.01 \text{ m}$.

$$p_B = (4 \times 10^{-6})(0.01) = 4 \times 10^{-8} \text{C} \cdot \text{m}$$

The distance between the centres is 80 cm .

Point P is equi-distant from both, so it is at the midpoint : $r = 40 \text{ cm} = 0.4 \text{ m}$.

Point P is on the axial line of dipole A and on the equatorial line of dipole B .

Field due to A (Axial) :

$$E_A = \frac{1}{4\pi\epsilon_0} \frac{2p_A}{r^3} = \frac{(9 \times 10^9) 2(2 \times 10^{-8})}{(0.4)^3} = \frac{360}{0.064} = 5625 \text{ N/C}$$

It is along the axis of dipole A (horizontal)

Field due to B (Equatorial) :

$$E_B = \frac{1}{4\pi\epsilon_0} \frac{p_B}{r^3} = \frac{(9 \times 10^9) 4 \times 10^{-8}}{(0.4)^3} = \frac{360}{0.064} = 5625 \text{ N/C}$$

It acts opposite to the direction of dipole B (vertical).

Since the vectors are perpendicular and equal in magnitude ($E_A = E_B = E$) :

$$E_{\text{net}} = \sqrt{E_A^2 + E_B^2} = E\sqrt{2}$$

$$E_{\text{net}} = 5625\sqrt{2} \frac{\text{N}}{\text{C}} = 0.5625\sqrt{2} \times 10^4 = \frac{9}{16}\sqrt{2} \times 10^4 \frac{\text{N}}{\text{C}}$$

So, net electric field at point P is, $E_{\text{net}} = \frac{9}{16}\sqrt{2} \times 10^4 \text{ N/C}$.

Hence, the correct option is (B).

Question13

Three charges $+2q$, $+3q$ and $-4q$ are situated at $(0, -3a)$, $(2a, 0)$ and $(-2a, 0)$ respectively in the xy plane. The resultant dipole moment about origin is _____ .

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Options:

A.

$$2qa(7\hat{i} - 3\hat{j})$$

B.

$$2qa(3\hat{j} - 7\hat{i})$$

C.

$$2qa(3\hat{i} - 7\hat{j})$$

D.

$$2qa(3\hat{j} - \hat{i})$$

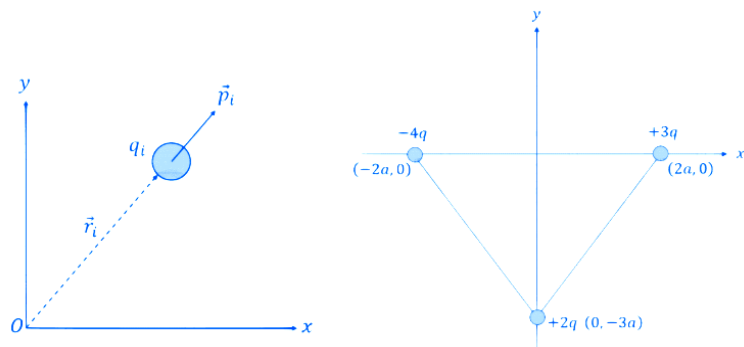
Answer: A

Solution:

For a system of discrete point charges q_i located at positions $\vec{r}_i$, the electric dipole moment $\vec{P}$ with respect to the origin is defined as the vector sum of the product of each charge and its position vector.

$$\vec{P} = \sum_i q_i \vec{r}_i$$

$$\Rightarrow \vec{P} = q_1 \vec{r}_1 + q_2 \vec{r}_2 + q_3 \vec{r}_3 + \dots$$



For charge 1 (q_1) :

$q_1 = +2q$ and it is at coordinates $(0, -3a)$

So, position vector with respect to origin is, $\vec{r}_1 = 0\hat{i} - 3a\hat{j} = -3a\hat{j}$

So, electric dipole moment is, $\vec{p}_1 = 2q(-3a\hat{j}) = -6aq\hat{j}$

For charge 2 (q_2) :

$q_2 = +3q$ and it is at coordinates $(2a, 0)$

So, position vector with respect to origin is, $\vec{r}_2 = 2a\hat{i} - 0\hat{j} = 2a\hat{i}$

So, electric dipole moment is, $\vec{p}_2 = 3q(2a\hat{i}) = 6aq\hat{i}$

For charge 3 (q_3) :

$q_3 = -4q$ and it is at coordinates $(-2a, 0)$

So, position vector with respect to origin is, $\vec{r}_3 = -2a\hat{i} - 0\hat{j} = -2a\hat{i}$

So, electric dipole moment is, $\vec{p}_3 = -4q(-2a\hat{i}) = 8aq\hat{i}$

So, the net electric dipole moment is the sum of the individual moments :

$$\vec{P} = \vec{P}_1 + \vec{P}_2 + \vec{P}_3$$

$$\Rightarrow \vec{P} = (-6aq\hat{j}) + (6aq\hat{i}) + (8aq\hat{i})$$

$$\Rightarrow \vec{P} = 14aq\hat{i} - 6aq\hat{j}$$

$$\Rightarrow \vec{P} = 2qa(7\hat{i} - 3\hat{j})$$

Hence, the correct option is (A).

Question14

There are three co-centric conducting spherical shells A , B and C of radii a , b and c respectively ($c > b > a$) and they are charged with charge q_1 , q_2 and q_3 respectively. The potentials of the spheres A , B and C respectively, are :

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Options:

A.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{c} \right)$$

B.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{c} \right)$$

C.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right)$$

D.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{c} \right)$$

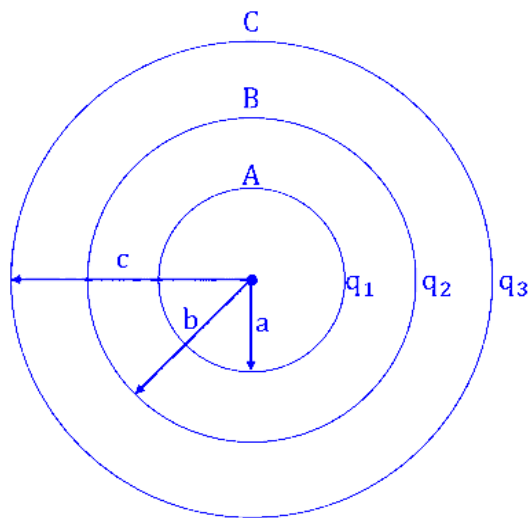
Answer: D

Solution:

The electric potential V due to a charged spherical shell of radius R and charge Q is given by :

$$V = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{Q}{R}, & r < R \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{R}, & r = R \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, & r > R \end{cases}$$

Point A is on the surface of shell A, inside shell B, and inside shell C. The total potential is the sum of potentials due to each shell (Superposition Principle).



So, the potential at A ;

- 1. Due to A (Surface, $r = a$) $\Rightarrow V_{AA} = \frac{kq_1}{a}$
- 2. Due to B (Inside, $r < b$) $\Rightarrow V_{AB} = \frac{kq_2}{b}$
- 3. Due to C (Inside, $r < c$) $\Rightarrow V_{AC} = \frac{kq_3}{c}$

Hence, total potential at point A is,

$$V_A = \left(\frac{kq_1}{a} + \frac{kq_2}{b} + \frac{kq_3}{c} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right)$$

For the point B which is outside shell A , on the surface of shell B , and inside shell C . So, the potential at point B,

- 1. Due to A (Outside, distance b) $\Rightarrow V_{BA} = \frac{kq_1}{b}$
- 2. Due to B (Surface, r = b) $\Rightarrow V_{BB} = \frac{kq_2}{b}$
- 3. Due to C (Inside, r < c) $\Rightarrow V_{BC} = \frac{kq_3}{c}$

Hence, total potential at point B is:

$$V_B = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{b} + \frac{q_2}{b} + \frac{q_3}{c} \right)$$

$$\Rightarrow V_B = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2}{b} + \frac{q_3}{c} \right)$$

Point C is outside shell A , outside shell B , and on the surface of shell C . So, potential at point C is,

- 1. Due to A (Outside, distance c) $\Rightarrow V_{CA} = \frac{kq_1}{c}$
- 2. Due to B (Outside, distance c) $\Rightarrow V_{CB} = \frac{kq_2}{c}$
- 3. Due to C (Surface, r = c) $\Rightarrow V_{CC} = \frac{kq_3}{c}$

Hence, total potential at point C is :

$$V_C = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{c} + \frac{q_2}{c} + \frac{q_3}{c} \right)$$

$$\Rightarrow V_C = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{c} \right)$$

$$\text{So, } V_A = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), V_B = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2}{b} + \frac{q_3}{c} \right), V_C = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1+q_2+q_3}{c} \right).$$

Hence, the correct option is (D).

Question15

The electrostatic potential in a charged spherical region of radius r varies as $V = ar^3 + b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi a \epsilon_0$. The value of α is ____ .

(permittivity of vacuum is ϵ_0)

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Options:

A.

-12

B.

-8

C.

-9

D.

-6

Answer: A

Solution:

The electric field $\vec{E}$ is related to the electric potential V by $E = -\frac{dV}{dr}$.

Since the potential V depends only on the radial distance r , the electric field is radial.

Given the potential function :

$$V = ar^3 + b$$

Then the electric field is,

$$E = -\frac{dV}{dr} = -\left[\frac{d}{dr}(ar^3 + b)\right] = -3ar^2$$

We need to find the total charge enclosed in a sphere of unit radius ($R = 1$).

The electric flux passing through the surface of this sphere is given by $\Phi_E = \oint \vec{E} \cdot d\vec{A}$

At $R = 1$, the electric field magnitude is :

$$E(1) = -3a(1)^2 = -3a$$

The surface area of the sphere is $4\pi R^2 = 4\pi(1)^2 = 4\pi$.

Thus, the flux is :

$$\Phi_E = E \times \text{Area} = (-3a) \times (4\pi) = -12\pi a$$

According to Gauss's Law,

$$\Phi_E = \frac{Q_{\text{enclosed}}}{\epsilon_0} \Rightarrow Q_{\text{enclosed}} = \Phi_E \times \epsilon_0$$

$$\Rightarrow Q_{\text{enclosed}} = (-12\pi a) \times \epsilon_0$$

$$\Rightarrow Q_{\text{enclosed}} = -12\pi a \epsilon_0 = \alpha \times \pi a \epsilon_0 \Rightarrow \alpha = -12$$

Question 16

Two point charges of 1 nC and 2 nC are placed at the two corners of an equilateral triangle of side 3 cm. The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is _____ μJ .

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

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Options:

A.

3.3

B.

27

C.

2.7

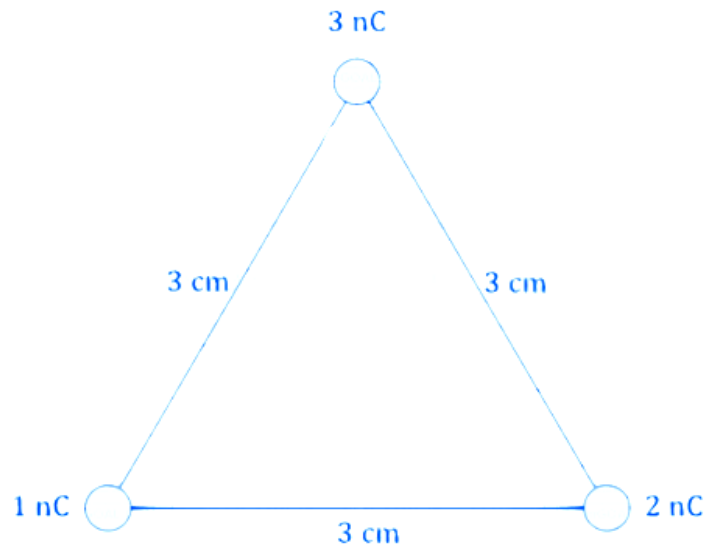
D.

5.4

Answer: C

Solution:

When 3 nC is at infinity then electric potential due to charges 1 nC and 2 nC is zero. So the initial potential is V_i



The total potential at the third corner due to $q_1(1\text{nC})$ and $q_2(2\text{nC})$ is the sum of their individual potentials:

$$V_f = \frac{kq_1}{r} + \frac{kq_2}{r} = \frac{k(q_1+q_2)}{r}$$

The work done is given by $W = q_3 \times \Delta V$, where ΔV = potential difference between initial and final position.

$$W = q_3 \times \left[\frac{k(q_1+q_2)}{r} - 0 \right]$$

$$\Rightarrow W = \frac{k \times q_3 \times (q_1 + q_2)}{r}$$

Substituting the values,

$$W = \frac{(9 \times 10^9) \times (3 \times 10^{-9}) \times (1 \times 10^{-9} + 2 \times 10^{-9})}{0.03}$$

$$\Rightarrow W = \frac{(27) \times (3 \times 10^{-9})}{0.03} = 27 \times 10^{-7} \text{ J}$$

$$\text{As } 1 \text{ J} = 10^6 \mu \text{ J}$$

$$W = 27 \times 10^{-7} \times 10^6 \mu \text{ J}$$

$$\Rightarrow W = 27 \times 10^{-1} \mu \text{ J} = 2.7 \mu \text{ J}$$

So, the work done in bring a charge of 3 nC from infinity to the third corner of the triangle is $2.7 \mu \text{ J}$.

Hence, the correct option is (C).

Question17

Identify the correct statements :

A. Electrostatic field lines form closed loops.

B. The electric field lines point radially outward when charge is greater than zero.

C. The Gauss - Law is valid only for inverse-square force.

D. The work done in moving a charged particle in a static electric field around a closed path is zero.

E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the correct answer from the options given below :

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Options:

A.

A, C, E Only

B.

A, B, D, E Only

C.

B, C, D, E Only

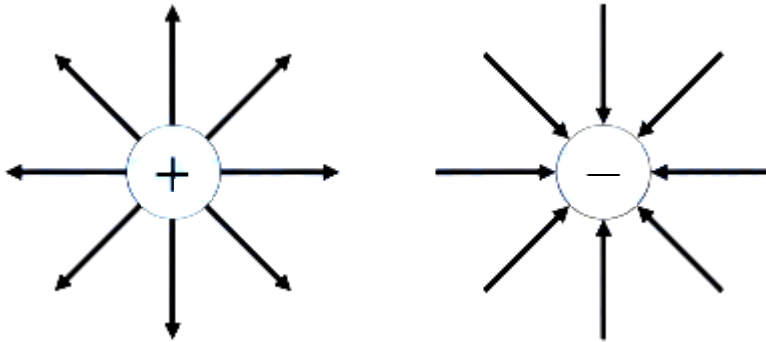
D.

A, B, C, D Only

Answer: C

Solution:

Electrostatic field lines originate from positive charges and terminate on negative charges. Unlike magnetic field lines, they do not form closed loops because the electrostatic force is conservative and a charge cannot return to its starting point along a continuous line of force without a change in potential.



Hence, statement (A) is incorrect.

By convention, electric field lines represent the direction of force on a positive test charge. If a charge is positive ($q > 0$), the field lines point radially outward from the charge.

Hence, the statement B is correct.

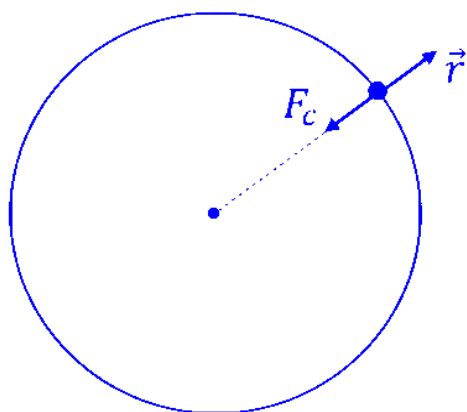
Gauss's Law is a fundamental law of electromagnetism that relates electric flux to enclosed charge. It is mathematically derived based on the inverse-square nature of Coulomb's Law ($F \propto \frac{1}{r^2}$). So, Gauss-Law is valid only for inverse square force (like gravitational force too).

Hence, the statement C is correct.

The electrostatic field is a conservative field. In a conservative field the work done in moving a charge between two points is independent of the path taken, and the total work done over any closed loop is exactly zero.

Hence, the statement D is Correct.

Coulomb's force is a central force, i.e., it always acts along the line joining the centres of two charges. The torque on the particle ($\vec{\tau} = \vec{r} \times \vec{F}$) is zero because the central force ($\vec{F}$) is anti-parallel to the position vector, so to make the angular momentum ($\vec{\tau} = \frac{d\vec{L}}{dt}$) will be constant. For the constant angular momentum, the motion is in a plane. Therefore, the motion of a particle under Coulomb's force must take in a plane.



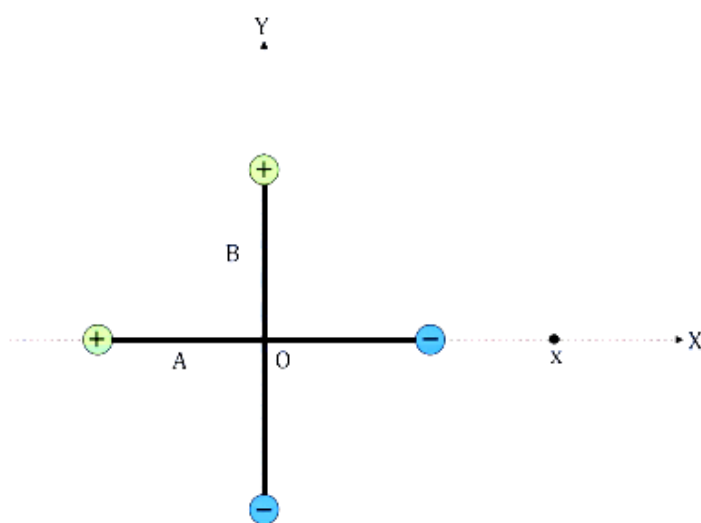
Hence, the statement D is correct.

Therefore, statements B, C, D, and E are correct, while statement A is false.

Hence, the correct option is (C).

Question18

Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular as shown in the figure. The resultant electric field at a point x is making an angle of 60° with the line joining points O and x . The ratio of the dipole moments p_2/p_1 is _____.



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Options:

A.

$$\frac{\sqrt{3}}{2}$$

B.

$$2\sqrt{3}$$

C.

$$\frac{1}{\sqrt{3}}$$

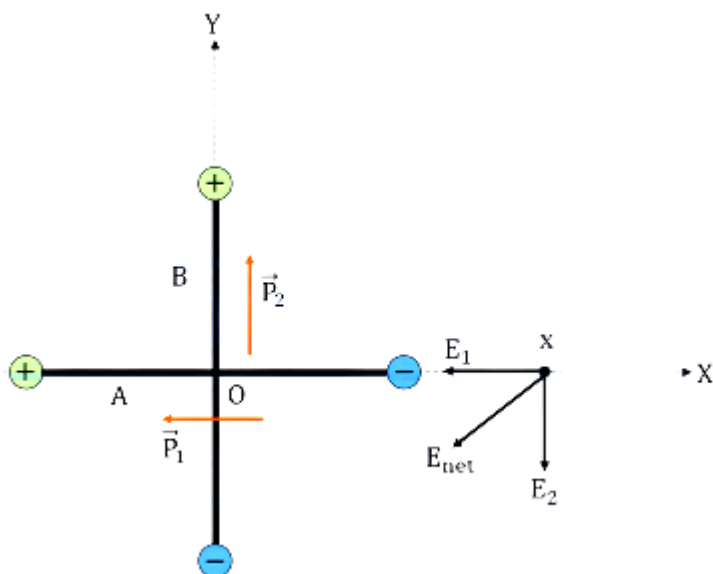
D.

$$\sqrt{3}$$

Answer: B

Solution:

The direction of electric dipole is from negative charge to positive charge. So the direction of dipole moment for A ($\vec{P}_1$) is along $-\hat{i}$; and that of dipole moment for B ($\vec{P}_2$) is along $\hat{j}$.



Dipole A (p_1) is oriented along the line OX. So, the point X lies on the axial line of dipole A. For a short dipole, the electric field at a distance r on the axial line is:

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2p_1}{r^3}$$

This field E_1 acts along the direction of the dipole moment,

$$\vec{E}_1 = - \left(\frac{1}{4\pi\epsilon_0} \frac{2p_1}{r^3} \right) \hat{i}$$

Dipole B (p_2) is oriented perpendicular to the line OX. So, point X lies on the equatorial line of dipole B. The electric field at a distance r on the equatorial line is:

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{p_2}{r^3}$$

This field E_2 acts in a direction opposite to the dipole moment p_2 (perpendicular to OX),

$$\vec{E}_2 = - \left(\frac{1}{4\pi\epsilon_0} \frac{p_2}{r^3} \right) \hat{j}$$

The resultant electric field makes an angle $\theta = 60^\circ$ with the line OX.

$$\tan \theta = \frac{E_2}{E_1}$$

Substituting the expressions for E_1 and E_2 :

$$\tan 60^\circ = \frac{\frac{1}{4\pi\epsilon_0} \frac{p_2}{r^3}}{\frac{1}{4\pi\epsilon_0} \frac{2p_1}{r^3}}$$

$$\tan 60^\circ = \frac{p_2}{2p_1}$$

$$\sqrt{3} = \frac{p_2}{2p_1}$$

$$\Rightarrow \frac{p_2}{p_1} = 2\sqrt{3}$$

Therefore, the ratio of the dipole moments is $2\sqrt{3}$. Thus, the correct option is (B).

Question19

Two charged conducting spheres S_1 and S_2 of radii 8 cm and 18 cm are connected to each other by a wire. After equilibrium is established, the ratio of electric fields on S_1 and S_2 spheres are E_{S1} and E_{S2} respectively. The value of $\frac{E_{S1}}{E_{S2}}$ is _____.

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Options:

A.

$$\frac{3}{2}$$

B.

$$\frac{2}{3}$$

C.

$$\frac{4}{9}$$

D.

$$\frac{9}{4}$$

Answer: D

Solution:

When two conducting spheres are connected by a wire, charge flows from one to the other until they reach the same electrical potential.

At equilibrium, the potential (V) of both spheres must be equal.

$$V_1 = V_2$$

The potential at the surface of a conducting sphere of radius R and charge Q is given by:

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

Therefore, for the two spheres:

$$V_1 = V_2 \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Q_1}{R_1} = \frac{1}{4\pi\epsilon_0} \frac{Q_2}{R_2}$$

$$\frac{Q_1}{R_1} = \frac{Q_2}{R_2} \Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$$

The electric field (E) at the surface of a conducting sphere is given by:

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

So, the ratio $\frac{E_{S1}}{E_{S2}}$:

$$\frac{E_{S1}}{E_{S2}} = \frac{\frac{1}{4\pi\epsilon_0} \frac{Q_1}{R_1^2}}{\frac{1}{4\pi\epsilon_0} \frac{Q_2}{R_2^2}}$$

$$\frac{E_{S1}}{E_{S2}} = \left(\frac{Q_1}{Q_2} \right) \times \left(\frac{R_2}{R_1} \right)^2$$

Substituting the equilibrium condition $\frac{Q_1}{Q_2} = \frac{R_1}{R_2}$ into the electric field ratio:

$$\frac{E_{S1}}{E_{S2}} = \left(\frac{R_1}{R_2}\right) \times \left(\frac{R_2}{R_1}\right)^2 = \frac{R_2}{R_1}$$

The radii are given as $R_1 = 8$ cm and $R_2 = 18$ cm.

$$\frac{E_{S1}}{E_{S2}} = \frac{18}{8}$$

$$\frac{E_{S1}}{E_{S2}} = \frac{9}{4}$$

Therefore, the value of the ratio is $9/4$. Thus, the correct option is (D).

Question20

Two point charges $8 \mu C$ and $-2 \mu C$ are located at $x = 2$ cm and $x = 4$ cm, respectively on the x -axis. The ratio of electric flux due to these charges through two spheres of radii 3 cm and 5 cm with their centers at the origin is _____.

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Options:

A.

4 : 1

B.

3 : 4

C.

4 : 3

D.

4 : 5

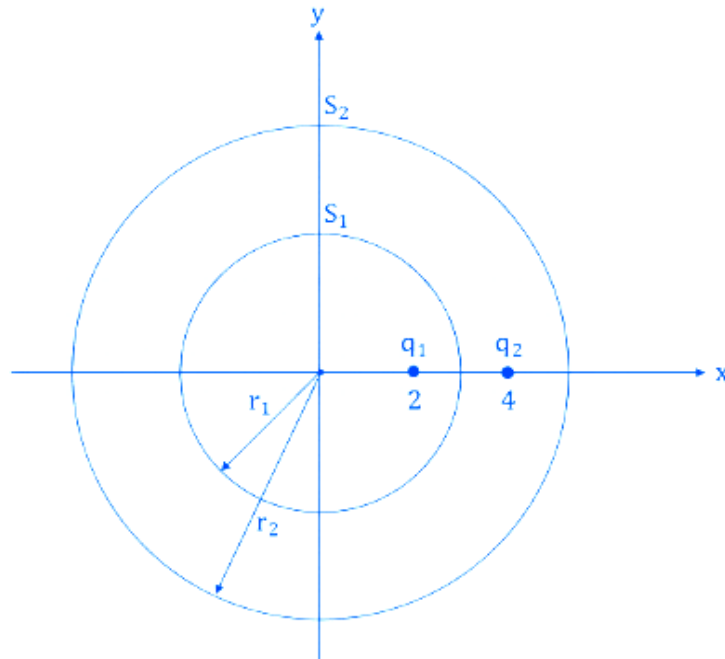
Answer: C

Solution:

Gauss's law states that the total electric flux (ϕ) through any closed surface is given by :

$$\phi = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

Where q_{enclosed} is the net charge inside the surface and ϵ_0 is the permittivity of free space. Any charge located outside the closed surface does not contribute to the net flux through it.



The first sphere has a radius $R_1 = 3$ cm centred at the origin.

Charge $q_1 = 8\mu\text{C}$ is at $x = 2$ cm. So, charge q_1 is inside S_1 .

Charge $q_2 = -2\mu\text{C}$ is at $x = 4$ cm. So, this charge is outside S_1 .

Thus, total charge enclosed by S_1 is,

$$q_{\text{enc1}} = 8\mu\text{C}$$

Using Gauss law, the electric flux through sphere 1 is,

$$\phi_1 = \frac{8\mu\text{C}}{\epsilon_0}$$

The second sphere has a radius $R_2 = 5$ cm centred at the origin.

Charge $q_1 = 8\mu\text{C}$ is at $x = 2$ cm. So, is inside S_2 .

Charge $q_2 = -2\mu\text{C}$ is at $x = 4$ cm. It is also inside S_2 .

Thus, total charge enclosed by S_2 is,

$$q_{\text{enc } 2} = 8\mu\text{C} + (-2\mu\text{C}) = 6\mu\text{C}$$

Using Gauss law, the electric flux through sphere 2 is,

$$\phi_2 = \frac{6\mu\text{C}}{\epsilon_0}$$

So, the ratio of electric flux through sphere 1 to sphere 2 is,

$$\frac{\phi_1}{\phi_2} = \frac{8/\epsilon_0}{6/\epsilon_0} = \frac{8}{6} = \frac{4}{3}$$

Therefore, the ratio of the electric flux is 4 : 3.

Hence, the correct option is (C).

Question21

Two metal plates (A, B) are kept horizontally with separation of $\left(\frac{12}{\pi}\right)$ cm, with plate A on the top.

An atomizer jet sprays oil (density 1.5 g/cm^3) droplets of radius 1 mm horizontally. All oil droplets carry a charge 5 nC. The potentials V_A and V_B are required on plates A and B respectively in order to ensure the droplets do not descend. The values of V_A and V_B are _____.

(Neglect the air resistance to the droplets and take $g = 10 \text{ m/s}^2$)

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

100 V and 580 V

B.

580 V and 100 V

C.

60 V and 400 V

D.

0 V and -200 V

Answer: A

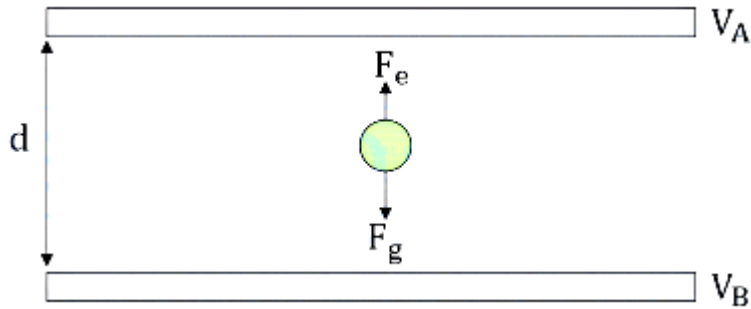
Solution:

The mass of a spherical droplet is $m = V \times \rho$, where V is the volume and ρ is the density.

The radius of droplet is $r = 1 \text{ mm} = 10^{-3} \text{ m} = 0.1 \text{ cm}$ and the density is $\rho = 1.5 \text{ g/cm}^3 = 1500 \text{ kg/m}^3$

So, the mass of spherical droplet is $m = \frac{4}{3}\pi r^3 \rho = \frac{4}{3}\pi (10^{-3})^3 (1500)$

$$m = \frac{4}{3}\pi (10^{-9})(1500) = 2\pi \times 10^{-6} \text{ kg}$$



So, the gravitational force acting on it is vertically downward,

$$F_g = mg = (2\pi \times 10^{-6}) \times 10 = 2\pi \times 10^{-5} \text{ N}$$

The upward force is due to electric force $F_e = qE$.

For the droplet to be at rest, the net force acting on it should be zero.

$$F_e = F_g$$

$$\Rightarrow qE = mg \Rightarrow E = \frac{mg}{q}$$

Given charge on the droplet as $q = 5 \text{ nC} = 5 \times 10^{-9} \text{ C}$:

$$E = \frac{2\pi \times 10^{-5}}{5 \times 10^{-9}} = \frac{2\pi}{5} \times 10^4 = 0.4\pi \times 10^4 = 4000\pi \text{ V/m}$$

The electric field between plates with separation $d = 12\pi \text{ cm} = 0.12\pi \text{ m}$ is :

$$\Delta V = E \times d = (4000\pi) \times (0.12\pi)$$

$$\Rightarrow \Delta V = 4000 \times 0.12 = 480 \text{ V}$$

Since the charge is positive (5 nC) and the force must be upward, the electric field must point upward. This means the bottom plate (B) must be at a higher potential than the top plate (A).

$$V_B - V_A = 480 \text{ V}$$

So, if $V_B = 580 \text{ V}$ and $V_A = 100 \text{ V}$ then the potential difference between the plates will be 480 V and the direction of electric field will be upward.

For, $V_B = 100$ and $V_A = 580 \text{ V}$ the potential difference between the plates is still 480 V but the direction of electric field will be downward, which doesn't satisfy the condition.

Therefore, the correct option (A).

Question22

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : In electrostatics, a conductor does not store any net charge inside.

Reason R : Inside the capacitor (with no dielectric medium), the free charge carriers, if placed between the plates of capacitor, experience force and drift.

Choose the correct answer from the options given below

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

Both **A** and **R** are true and **R** is the correct explanation of **A**

B.

Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

C.

A is true but **R** is false

D.

A is false but **R** is true

Answer: B

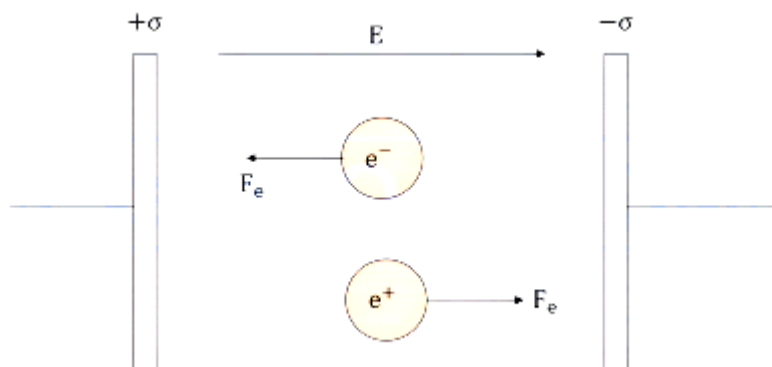
Solution:

In electrostatic equilibrium, the electric field inside a conductor is zero ($E = 0$).

According to Gauss's Law $\left(\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0} \right)$, if the electric field is zero everywhere inside, the net charge enclosed (q_{in}) by any Gaussian surface inside the conductor must also be zero. Therefore, any excess charge resides entirely on the outer surface.

Therefore, assertion A is true.

Inside a charged capacitor, there is a uniform electric field between the plates.



If free charge carriers (like electrons or ions) are placed in this region, they will experience an electric force ($\mathbf{F} = q\mathbf{E}$). This force causes the charges to accelerate or drift toward the plate of opposite polarity.

Therefore, reason R is true.

While both statements are scientifically true, reason R describes the behaviour of charges in an external field between two conductors (the plates). It does not explain why the net charge inside a solid conductor is zero (which is due to mutual repulsion and the resulting zero internal field).

So, both A and R are true, but R is NOT the correct explanation of A.

Thus, the correct option is (B).

Question23

A rigid dipole undergoes a simple harmonic motion about its centre in the presence of an electric field $\vec{E}_1 = E_0 \hat{x}$. If another electric field $\vec{E}_2 = 2E_0(\hat{y} + \hat{z})$ is introduced to the system, what will be the percentage change in the frequency of the oscillation (approximate)?

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

73%

B.

63%

C.

83%

D.

53%

Answer: A

Solution:

When a dipole with dipole moment $\vec{p}$ is placed in an electric field $\vec{E}$, the restoring torque is $\vec{\tau} = -\vec{p} \times \vec{E}$

The magnitude of restoring torque is $\tau = -pE \sin \theta$.

For small θ ,

$$\tau \approx -pE\theta$$

The equation of motion is $I \frac{d^2\theta}{dt^2} = -pE\theta$, where I is the moment of inertia.

The angular frequency is $\omega = \sqrt{\frac{pE}{I}}$, so the frequency f is:

$$f = \frac{\omega}{2\pi}$$

$$\Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{pE}{I}}$$

$$\Rightarrow f \propto \sqrt{E}$$

Where E is the magnitude of the net electric field.

Initial electric field is $\vec{E}_1 = E_0 \hat{x}$.

Magnitude of initial electric field is $E_i = E_0$.

The additional electric field is $\vec{E}_2 = 2E_0 \hat{y} + 2E_0 \hat{z}$.

So, the net electric field is

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 = E_0 \hat{x} + 2E_0 \hat{y} + 2E_0 \hat{z}$$

Magnitude of the net electric field is,

$$E_{\text{net}} = \sqrt{(E_0)^2 + (2E_0)^2 + (2E_0)^2} = \sqrt{E_0^2 + 4E_0^2 + 4E_0^2}$$

$$\Rightarrow E_{\text{net}} = \sqrt{9E_0^2} = 3E_0$$

Let f_1 be the initial frequency and f_2 be the final frequency.

$$\frac{f_2}{f_1} = \sqrt{\frac{E_f}{E_i}} = \sqrt{\frac{3E_0}{E_0}} = \sqrt{3} \approx 1.732$$

So, the percentage change is

$$\Delta\% = \left(\frac{f_2 - f_1}{f_1} \right) \times 100$$

$$\Rightarrow \Delta\% = (\sqrt{3} - 1) \times 100$$

$$\Rightarrow \Delta\% = (1.732 - 1) \times 100 = 0.732 \times 100 = 73.2\%$$

Therefore, the approximate percentage change in the frequency of oscillation is 73%.

Thus, the correct option is (A).

Question24

A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is 100 V/m, then the value of Q is ____ nC .

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2/\text{Nm}^2 \text{ and } \pi = 3.14)$$

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

2.14

B.

2.44

C.

3.25

D.

0.7

Answer: A

Solution:

For a uniformly charged semicircular ring with linear charge density λ and radius R , the electric field E at the center is:

$$E = \frac{2k\lambda}{R} = \frac{2\lambda}{4\pi\epsilon_0 R} = \frac{\lambda}{2\pi\epsilon_0 R}$$

$$\text{Where } \lambda = \frac{\text{Total Charge}}{\text{Length}} = \frac{Q}{\pi R}.$$

Substituting λ into the field formula:

$$E = \frac{Q/\pi R}{2\pi\epsilon_0 R} = \frac{Q}{2\pi^2\epsilon_0 R^2}$$

$$\Rightarrow Q = E \cdot 2\pi^2\epsilon_0 R^2$$

The magnitude of electric field at the centre is $E = 100 \text{ V/m}$

The radius of the semi-circle is $R = 35 \text{ cm} = 0.35 \text{ m}$

The value of $\epsilon_0 = 8.85 \times 10^{-12}$ and $\pi = 3.14$

$$Q = 100 \times 2 \times (3.14)^2 \times 8.85 \times 10^{-12} \times (0.35)^2$$

$$\Rightarrow Q = 100 \times 2 \times 9.8596 \times 8.85 \times 10^{-12} \times 0.1225$$

$$\Rightarrow Q \approx 2.14 \times 10^{-9} \text{ C}$$

$$\Rightarrow Q = 2.14 \text{ nC}$$

Therefore, the value of Q is 2.14 nC . Thus, the correct option is (A).

Question25

The electric potential as a function of x, y is given by

$V = 5(x^2 - y^2) \text{ V}$. The electric field at a point $(2, 3) \text{ m}$ is _____ V/m .

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

$$(-20\hat{i} + 30\hat{j})$$

B.

$$(20\hat{i} - 30\hat{j})$$

C.

$$(20\hat{i} + 45\hat{j})$$

D.

$$(-4\hat{i} + 6\hat{j})$$

Answer: A

Solution:

The electric field $\vec{E}$ is the negative gradient of the electric potential V :

$$\vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

The electric potential is given as $V = 5(x^2 - y^2) = 5x^2 - 5y^2$.

The x -component of electric field is,

$$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x}(5x^2 - 5y^2) = -(10x) = -10x$$

The y -component of electric field is,

$$E_y = -\frac{\partial V}{\partial y} = -\frac{\partial}{\partial y}(5x^2 - 5y^2) = (10y) = 10x$$

At Point (2, 3)

$$E_x = -10(2) = -20 \text{ V/m}$$

$$\Rightarrow E_y = 10(3) = 30 \text{ V/m}$$

So, the resulting electric field vector is :

$$\vec{E} = -20\hat{i} + 30\hat{j}$$

Therefore, the correct option is (A).

Question26

Two point charges $q_1 = 3\mu C$ and $q_2 = -4\mu C$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. Force on charge q_2 is

_____ N. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI Units)

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$$(12\hat{i} + 24\hat{j} + 24\hat{k}) \times 10^{-3}$$

B.

$$(4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3}$$

C.

$$(3\hat{i} + 6\hat{j} + 6\hat{k}) \times 10^{-3}$$

D.

$$(-4\hat{i} - 8\hat{j} - 8\hat{k}) \times 10^{-3}$$

Answer: B

Solution:

Coulomb's Law states that the force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them.

In vector form, the force $\vec{F}_{21}$ exerted on charge q_2 by charge q_1 is:

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}|^2} \hat{r}$$

Where,

$\vec{r} = \vec{r}_2 - \vec{r}_1$ is the position vector from q_1 to q_2 , $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$ is the unit vector and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$

Substituting the unit vector, we get:

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}|^3} \vec{r}$$

Let $\vec{r}_1$ be the position of q_1 and $\vec{r}_2$ be the position of q_2 :

$$\vec{r}_1 = 2\hat{i} + 3\hat{j} + 3\hat{k}$$

$$\vec{r}_2 = 1\hat{i} + 1\hat{j} + 1\hat{k}$$

The vector from q_1 to q_2 is :

$$\vec{r} = \vec{r}_2 - \vec{r}_1 = (1 - 2)\hat{i} + (1 - 3)\hat{j} + (1 - 3)\hat{k}$$

$$\Rightarrow \vec{r} = -\hat{i} - 2\hat{j} - 2\hat{k}$$

So, the magnitude of position vector is,

$$|\vec{r}| = \sqrt{(-1)^2 + (-2)^2 + (-2)^2}$$

$$\Rightarrow |\vec{r}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \text{ m}$$

Substituting the values into the vector formula :

$$\vec{F} = (9 \times 10^9) \frac{(3 \times 10^{-6})(-4 \times 10^{-6})}{3^3} (-\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\Rightarrow \vec{F} = \frac{9 \times 3 \times (-4) \times 10^{-3}}{27} (-\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\Rightarrow \vec{F} = \frac{-108 \times 10^{-3}}{27} (-\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\Rightarrow \vec{F} = -4 \times 10^{-3} (-\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\Rightarrow \vec{F} = (4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3} \text{ N}$$

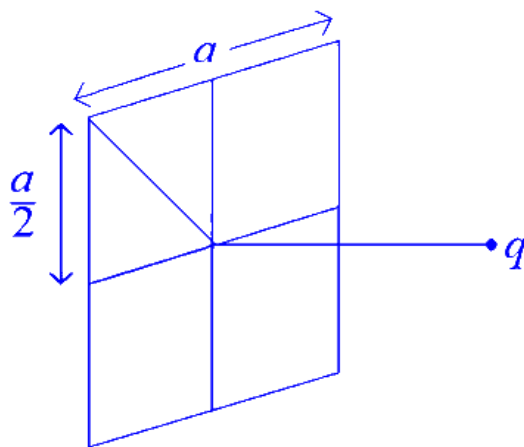
Therefore, the force on charge q_2 is $(4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3} \text{ N}$.

Hence, the correct option is (B).

Question27

A square loop of sides $a = 1 \text{ m}$ is held normally in front of a point charge $q = 1\text{C}$ at a distance $\frac{a}{2}$. The flux of the electric field through

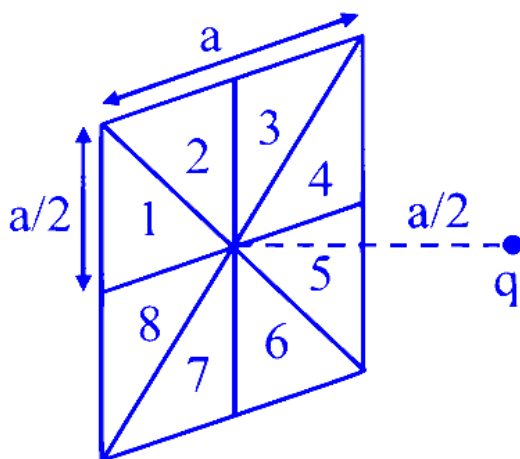
the shaded region is $\frac{5}{p} \times \frac{1}{\epsilon_0} \frac{Nm^2}{C}$, where the value of p is _____ .



JEE Main 2025 (Online) 24th January Morning Shift

Answer: 48

Solution:



Total flux through square = $\frac{q}{\epsilon_0} \left(\frac{1}{6} \right)$

Lets divide square is 8 equal parts.

Flux is same for each part.

∴ Flux through shaded portion is $\frac{5}{8}$ (Total flux)

$$= \frac{5}{8} \times \frac{q}{\epsilon_0} \frac{1}{6} = \frac{5}{48} \frac{1}{\epsilon_0}$$

∴ Required Ans. is 48

Question28

An electric dipole of dipole moment $6 \times 10^{-6} \text{ Cm}$ is placed in uniform electric field of magnitude 10^6 V/m . Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J.

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 12

Solution:

$$p = 6 \times 10^{-6} \text{ Cm}$$

$$E = 10^6 \text{ V/m}$$

$$W = \Delta U = -pE (\cos \theta_f - \cos \theta_i)$$

$$W = 2pE = 12 \text{ J}$$

Question29

The electric field in a region is given by

$\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3 \text{ N/C}$. The flux of the field through a rectangular surface parallel to $x - z$ plane is $6.0 \text{ Nm}^2 \text{ C}^{-1}$. The area of the surface is _____ cm^2 .

JEE Main 2025 (Online) 7th April Evening Shift

Answer: 15

Solution:

To calculate the area of the surface through which the electric flux is determined, we use the given electric field and the specified surface orientation. The electric field is expressed as $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3 \text{ N/C}$.

Since the surface is parallel to the $x - z$ plane, its area vector $\vec{A}$ is directed along the y -axis, making it $A\hat{j}$.

The electric flux ϕ through the surface is given by:

$$\phi = \vec{E} \cdot \vec{A} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3 \cdot A\hat{j}$$

Calculating the dot product, the only component contributing to the flux is the y -component:

$$\phi = (4 \times 10^3) A$$

Given that the electric flux ϕ is $6.0 \text{ Nm}^2/\text{C}$, we can equate and solve for A :

$$6 = 4 \times 10^3 A$$

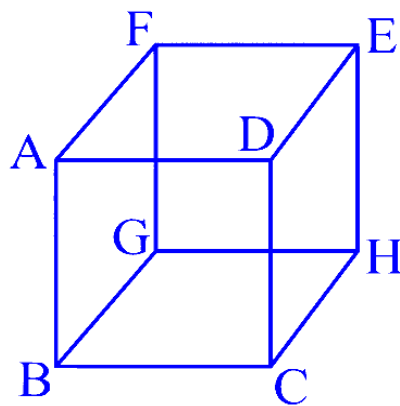
$$A = \frac{6}{4 \times 10^3} = 1.5 \times 10^{-3} \text{ m}^2$$

Converting this area from square meters to square centimeters:

$$A = 1.5 \times 10^{-3} \text{ m}^2 = 15 \text{ cm}^2$$

Question30

A line charge of length $\frac{a}{2}$ is kept at the center of an edge BC of a cube $ABCDEFGH$ having edge length ' a ' as shown in the figure. If the density of line charge is $\lambda \text{ C per unit length}$, then the total electric flux through all the faces of the cube will be _____. (Take, ϵ_0 as the free space permittivity)



JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $\frac{\lambda a}{2\epsilon_0}$

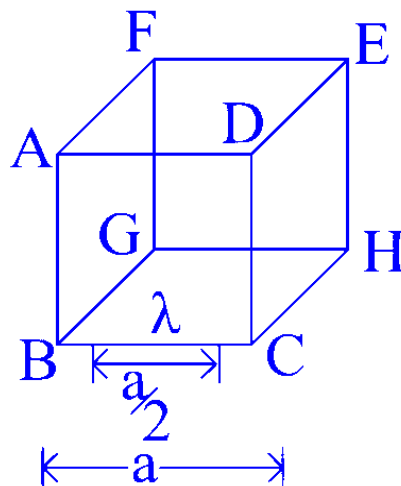
B. $\frac{\lambda a}{8\epsilon_0}$

C. $\frac{\lambda a}{4\epsilon_0}$

D. $\frac{\lambda a}{16\epsilon_0}$

Answer: B

Solution:



$$\phi = \frac{q_{enc}}{\epsilon_0}, q_{enc} = \text{enclosed charge}$$

here, we need three other identical cubes to enclose the charge fully.

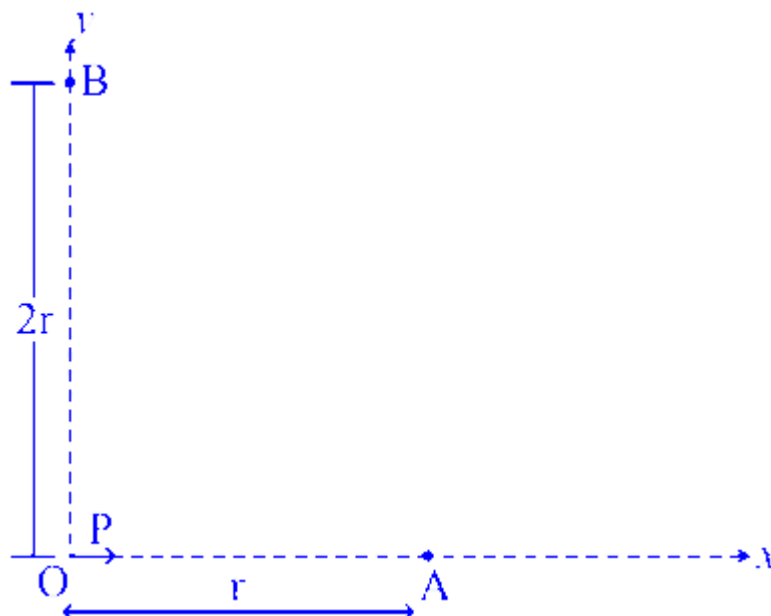
So, total four cubes enclose the charge

$$\text{Hence, } \phi_{cube} = \frac{\phi}{4}$$

$$= \frac{q_{enc}}{4\epsilon_0} = \frac{\lambda(\frac{a}{2})}{4\epsilon_0} \Rightarrow \phi_{cube} = \frac{\lambda a}{8\epsilon_0}$$

Question31

For a short dipole placed at origin O , the dipole moment P is along x -axis, as shown in the figure. If the electric potential and electric field at A are V_0 and E_0 , respectively, then the correct combination of the electric potential and electric field, respectively, at point B on the y -axis is given by



JEE Main 2025 (Online) 22nd January Evening Shift

Options:

- A. $\frac{V_0}{2}$ and $\frac{E_0}{16}$
- B. zero and $\frac{E_0}{8}$
- C. V_0 and $\frac{E_0}{4}$
- D. zero and $\frac{E_0}{16}$

Answer: D

Solution:

$$E_A = \frac{2kP}{r^3} = E_0 \& V_A = \frac{kP}{r^2} = v_0$$
$$E_B = \frac{kP}{(2r)^3} = \frac{E_0}{16} \& V_B = \frac{k\vec{p} \cdot \hat{r}}{r^2} = 0$$

Question32

The electric flux is $\phi = \alpha\sigma + \beta\lambda$ where λ and σ are linear and surface charge density, respectively. $\left(\frac{\alpha}{\beta}\right)$ represents

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

- A. displacement
- B. charge

C. electric field

D. area

Answer: A

Solution:

Let's analyze the given expression step by step.

We are given the electric flux:

$$\phi = \alpha\sigma + \beta\lambda$$

where:

σ is a surface charge density with units of charge per unit area (C/m^2).

λ is a linear charge density with units of charge per unit length (C/m).

For the equation to be dimensionally consistent, the two terms on the right must have the same units as the flux ϕ .

For the term $\alpha\sigma$:

Since σ has units C/m^2 , the constant α must have units that convert σ into the same units as the flux.

For the term $\beta\lambda$:

Since λ has units C/m , the constant β carries its own units to ensure the term matches the flux's dimensions.

Since both terms add to give the flux, they must share the same overall dimension. Thus, the dimensions of $\alpha\sigma$ and $\beta\lambda$ must be equal:

$$[\alpha] \cdot \frac{Q}{L^2} = [\beta] \cdot \frac{Q}{L}$$

Here, Q represents the unit of charge and L represents a unit of length. Canceling the common factors, we find:

$$\frac{[\alpha]}{[\beta]} = L$$

That is, the ratio $\frac{\alpha}{\beta}$ has the dimension of length.

Now, let's match this with the given options:

Option A: Displacement (commonly measured as a length)

Option B: Charge (has the unit Coulomb)

Option C: Electric Field (has units of V/m or N/C)

Option D: Area (has units of length squared, L^2)

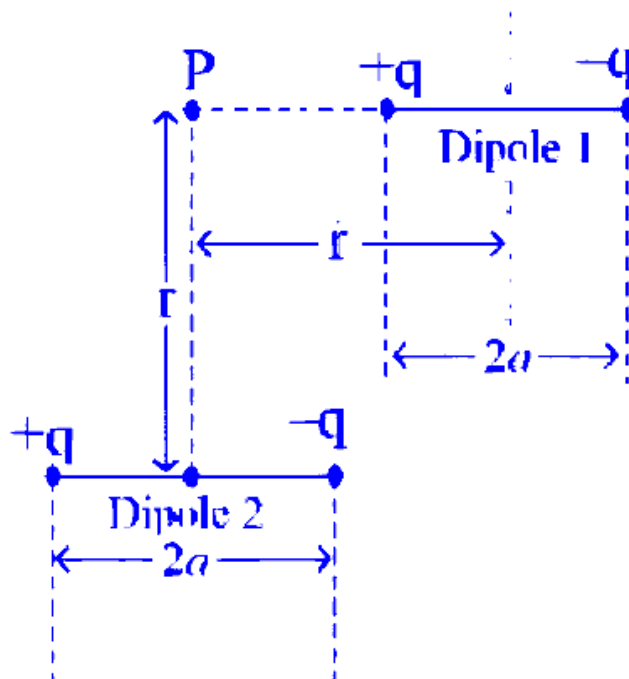
Since $\frac{\alpha}{\beta}$ has the dimensions of a length, it logically corresponds to a displacement.

Thus, the correct answer is:

Option A – displacement.

Question33

A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r . The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?



JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $\frac{a}{r} \sim 0.5$

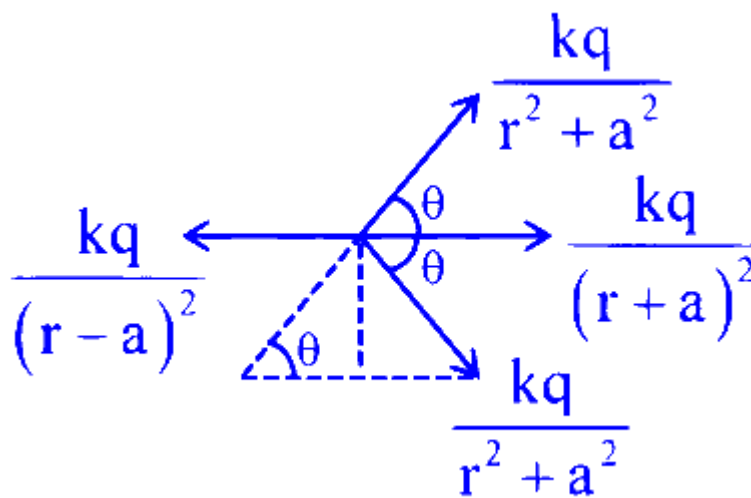
B. $\frac{a}{r} \sim 3$

C. $\frac{a}{r} \sim 10$

D. $\frac{a}{r} \sim 20$

Answer: B

Solution:



$$\frac{kq}{(r-a)^2} = \frac{kq}{(r+a)^2} + \frac{2kq}{(r^2+a^2)} \cos \theta$$

$$\frac{1}{(r-a)^2} = \frac{1}{(r+a)^2} + \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{4ra}{(r^2-a^2)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{2r}{(r^2 - a^2)^2} = \frac{1}{(r^2 + a^2)^{\frac{3}{2}}}$$

$$\frac{4r^2}{(r^2 - a^2)^4} = \frac{1}{(r^2 + a^2)^3}$$

$$\Rightarrow 4r^2(r^2 + a^2)^3 = (r^2 - a^2)^4$$

$$4r^8 \left(1 + \frac{a^2}{r^2}\right)^3 = r^8 \left(1 - \frac{a^2}{r^2}\right)^4$$

$$4 \left(1 + \frac{a^2}{r^2}\right)^3 = \left(1 - \frac{a^2}{r^2}\right)^4$$

Exact value cannot be solved in exam for this equation to be true

$$\left| \frac{a}{r} \right| > 1 \Rightarrow a > r$$

But point charge Q lies between charges of dipole

1 hence electric field cannot be zero.

There for it should be bonus.

But by solving from mathematical software we are getting $a/r \approx 3$.

Question34

Two charges $7\mu\text{C}$ and $-4\mu\text{C}$ are placed at $(-7\text{ cm}, 0, 0)$ and $(7\text{ cm}, 0, 0)$ respectively. Given, $\epsilon_0 = 8.85 \times 10^{-12}\text{C}^2 \text{N}^{-1} \text{m}^{-2}$, the electrostatic potential energy of the charge configuration is :

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Options:

A. -2.0 J

B. -1.5 J

C. -1.2 J

D. -1.8 J

Answer: D

Solution:

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

Here,

$$q_1 = 7 \times 10^{-6} \text{ C}$$

$$q_2 = -4 \times 10^{-6} \text{ C}$$

The separation between the charges is along the x-axis from -7 cm to 7 cm , so

$$r = 0.14 \text{ m}$$

The Coulomb constant is approximated as

$$\frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

Substitute these values into the equation:

$$U = 9 \times 10^9 \times \frac{(7 \times 10^{-6})(-4 \times 10^{-6})}{0.14}$$

Calculate the product of the charges:

$$(7 \times 10^{-6})(-4 \times 10^{-6}) = -28 \times 10^{-12} \text{ C}^2$$

Now, divide by the distance:

$$\frac{-28 \times 10^{-12}}{0.14} = -2 \times 10^{-10} \text{ C}^2/\text{m}$$

Finally, multiply by the Coulomb constant:

$$U = 9 \times 10^9 \times (-2 \times 10^{-10}) = -1.8 \text{ J}$$

Thus, the electrostatic potential energy of the configuration is:

$$-1.8 \text{ J}$$

This corresponds to Option D.

Question 35

Two point charges $-4\mu\text{C}$ and $4\mu\text{C}$, constituting an electric dipole, are placed at $(-9, 0, 0)\text{cm}$ and $(9, 0, 0)\text{cm}$ in a uniform electric field of strength 10^4 NC^{-1} . The work done on the dipole in rotating it from the equilibrium through 180° is :

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Options:

A. 18.4 mJ

B. 12.4 mJ

C. 16.4 mJ

D. 14.4 mJ

Answer: D

Solution:

The work done on an electric dipole in a uniform electric field when it is rotated from an angle θ_1 to an angle θ_2 is given by:

$$W = -pE(\cos \theta_2 - \cos \theta_1)$$

where:

$p = q \cdot d$ is the dipole moment,

E is the electric field strength,

θ_1 and θ_2 are the initial and final angles between the dipole moment and the electric field.

Given:

$$\text{Charge } q = 4 \mu\text{C} = 4 \times 10^{-6} \text{ C}$$

$$\text{Distance } d = 18 \text{ cm} = 0.18 \text{ m}$$

$$\text{Electric field } E = 10^4 \text{ N/C}$$

Initial angle $\theta_1 = 0^\circ$ (aligned with the field, equilibrium position)

Final angle $\theta_2 = 180^\circ$

First, calculate the dipole moment p :

$$p = q \cdot d = 4 \times 10^{-6} \text{ C} \cdot 0.18 \text{ m} = 7.2 \times 10^{-7} \text{ Cm}$$

Calculate the work done W :

$$W = -pE(\cos 180^\circ - \cos 0^\circ)$$

$$W = -7.2 \times 10^{-7} \text{ Cm} \cdot 10^4 \text{ N/C} \cdot (-1 - 1)$$

$$W = 7.2 \times 10^{-7} \text{ Cm} \cdot 10^4 \text{ N/C} \cdot 2$$

$$W = 1.44 \times 10^{-2} \text{ J} = 14.4 \text{ mJ}$$

Therefore, the work done on the dipole in rotating it from the equilibrium position through 180° is **14.4 mJ** (Option D).

Question36

Consider a parallel plate capacitor of area A (of each plate) and separation ' d ' between the plates. If E is the electric field and ϵ_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

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Options:

A. $\epsilon_0 E^2 A d$

B. $\frac{3}{4} \epsilon_0 E^2 A d$

C. $\frac{1}{4} \epsilon_0 E^2 A d$

D. $\frac{1}{2} \epsilon_0 E^2 A d$

Answer: D

Solution:

$$u = \frac{1}{2} \epsilon_0 E^2$$

The energy density of an electric field in free space is given by the expression above. For a parallel plate capacitor with plate area A and separation d , the volume between the plates is

$$V = A d.$$

Multiplying the energy density by the volume gives the total energy stored:

$$U = u \times V = \frac{1}{2} \epsilon_0 E^2 \times A d = \frac{1}{2} \epsilon_0 E^2 A d.$$

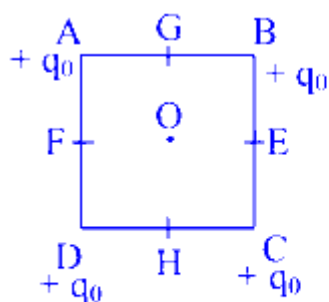
Thus, the potential energy stored in the capacitor is

$$\frac{1}{2} \epsilon_0 E^2 A d.$$

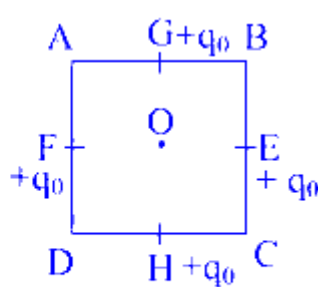
This corresponds to the option:

Option D: $\frac{1}{2} \epsilon_0 E^2 A d.$

Question37



Configuration (1)



Configuration (2)

In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A, B, C and D of square of side length ' a '. In the second configuration (2), the same charges are shifted to mid points G , E , H and F , of the square. If $K = \frac{1}{4\pi\epsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by :

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Options:

A. $\frac{Kq_0^2}{a}(4 - 2\sqrt{2})$

B. $\frac{Kq_0^2}{a}(4\sqrt{2} - 2)$

C. $\frac{Kq_0^2}{a}(3 - \sqrt{2})$

D. $\frac{Kq_0^2}{a}(3\sqrt{2} - 2)$

Answer: D

Solution:

$$U_1 = \frac{4Kq_0^2}{a} + \frac{2Kq_0^2}{\sqrt{2}a} = \frac{Kq_0^2}{a}(4 + \sqrt{2})$$

$$U_2 = \frac{Kq_0^2}{\left(\frac{a}{\sqrt{2}}\right)}(4 + \sqrt{2}) = \frac{Kq_0^2}{a}(4\sqrt{2} + 2)$$

$$U_2 - U_1 = \frac{Kq_0^2}{a}(3\sqrt{2} - 2)$$

Question38

A small uncharged conducting sphere is placed in contact with an identical sphere but having $4 \times 10^{-8} \text{ C}$ charge and then removed to a distance such that the force of repulsion between them is $9 \times 10^{-3} \text{ N}$. The distance between them is (Take $\frac{1}{4\pi\epsilon_0}$ as 9×10^9 in SI units)

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Options:

A. 1 cm

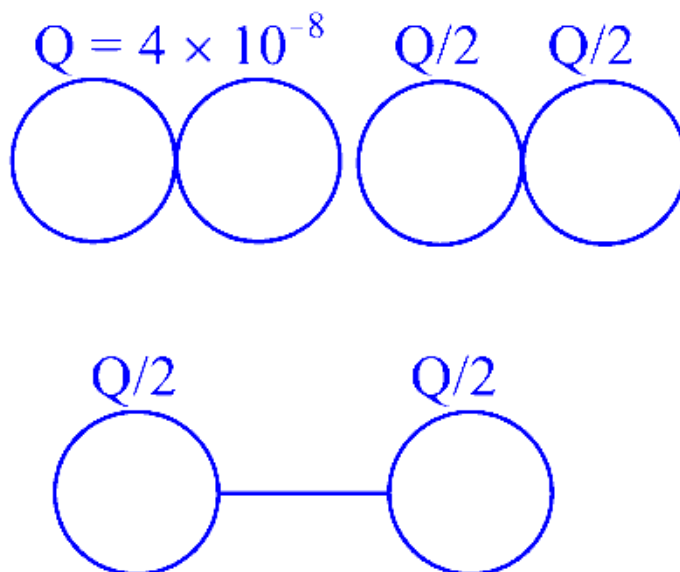
B. 2 cm

C. 4 cm

D. 3 cm

Answer: B

Solution:



$$F = \frac{k \left(\frac{\theta}{2}\right) \left(\frac{\theta}{2}\right)}{r^2}$$

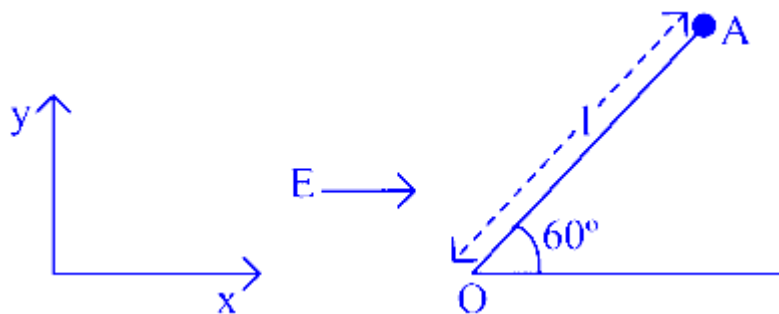
$$9 \times 10^{-3} = \frac{9 \times 10^9 \times (4 \times 10^{-8}) \times 4 \times 10^{-8}}{4 \times r^2}$$

$$r^2 = \frac{9 \times 10^9 \times 16 \times 10^{-16}}{4 \times 9 \times 10^{-3}} = 4 \times 10^{-4}$$

$$r = 2 \times 10^{-2} \text{ m} \Rightarrow 2 \text{ cm}$$

Question39

A particle of mass ' m ' and charge ' q ' is fastened to one end ' A ' of a massless string having equilibrium length l , whose other end is fixed at point ' O '. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x -axis is



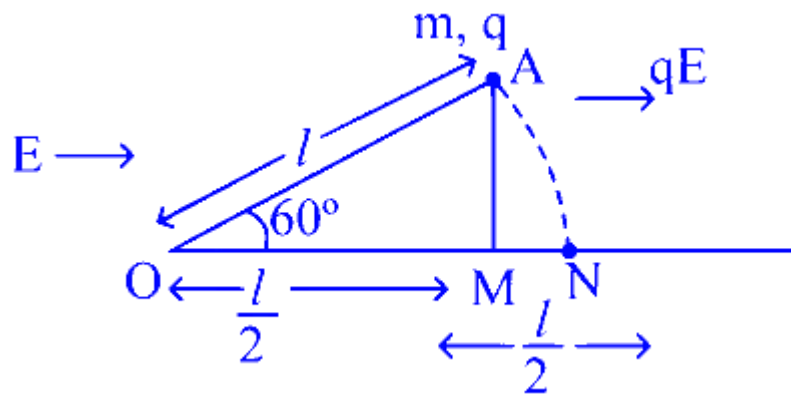
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Options:

- A. $\sqrt{\frac{qEl}{2m}}$
- B. $\sqrt{\frac{qEl}{4m}}$
- C. $\sqrt{\frac{qEl}{m}}$
- D. $\sqrt{\frac{2qEl}{m}}$

Answer: C

Solution:



$$\text{In } \triangle OMA, \frac{OM}{OA} = \cos 60^\circ = \frac{1}{2}$$

$$\Rightarrow OM = \frac{l}{2} \text{ (as } OA = l)$$

$$MN = ON - OM = l - \frac{l}{2} = \frac{l}{2}$$

Using work – energy theorem,

$$W_{all} = \Delta k$$

$$\Rightarrow W_c = k_f - k_i$$

$$\Rightarrow qE \frac{l}{2} = \frac{1}{2}mv^2 - 0 \text{ (As } v_i = 0 \text{ and } w = FS)$$

$$\Rightarrow v^2 = \frac{qEl}{m}$$

$$\Rightarrow v = \sqrt{\frac{qEl}{m}}$$

Question40

Three infinitely long wires with linear charge density λ are placed along the x – axis, y – axis and z – axis respectively. Which of the following denotes an equipotential surface?

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Options:

A. $(x^2 + y^2) (y^2 + z^2) (z^2 + x^2) = \text{constant}$

B. $xyz = \text{constant}$

C. $xy + yz + zx = \text{constant}$

D. $(x + y)(y + z)(z + x) = \text{constant}$

Answer: A

Solution:

Electric field due to infinitely long wire

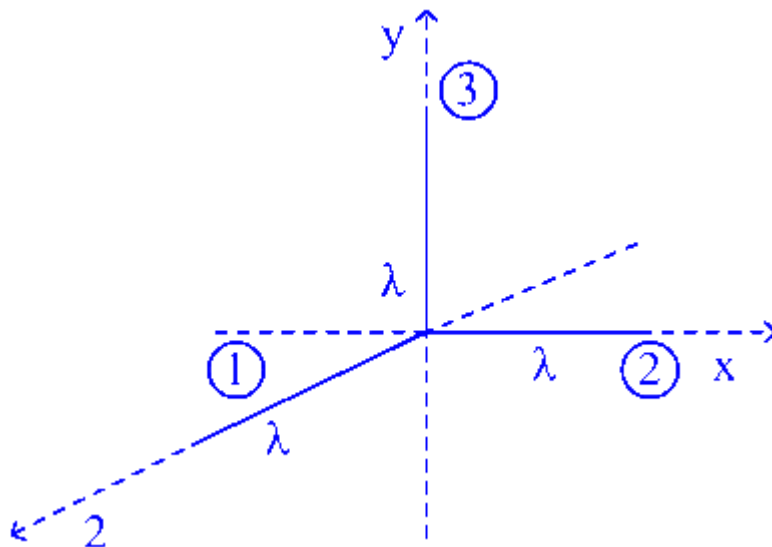
$$\vec{E} = -\frac{2k\lambda}{r} \hat{r}$$

We know, $V = -\int \vec{E} \cdot \vec{dr}$

$$= -\int \frac{-2k\lambda}{r} \hat{r} \cdot \vec{dr}$$

$$v = \int \frac{2k\lambda}{r} dr = 2k\lambda \ln r + c$$

Net potential due to all wires,



$$V = v_1 + v_2 + v_3$$

$$= 2k\lambda \ln \left(\sqrt{x^2 + y^2} \right) + 2k\lambda \ln \left(\sqrt{y^2 + z^2} \right) + 2k\lambda \ln \left(\sqrt{z^2 + x^2} \right) + c$$

for equipotential surface, $v = c$

$$\Rightarrow 2k\lambda \left[\ln \left(\sqrt{x^2 + y^2} \cdot \sqrt{y^2 + z^2} \cdot \sqrt{z^2 + x^2} \right) \right] = c$$

$$\Rightarrow \sqrt{(x^2 + y^2) \cdot (y^2 + z^2) \cdot (z^2 + x^2)} = c$$

$$\Rightarrow (x^2 + y^2)(y^2 + z^2)(z^2 + x^2) = c$$

where, $c = \text{constant}$.

Question41

Match List - I with List - II.

List - I	List - II
(A) Electric field inside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	(I) σ/ϵ_0
(B) Electric field at distance $r > 0$ from a uniformly charged infinite plane sheet with surface charge density σ .	(II) $\sigma/2\epsilon_0$
(C) Electric field outside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	(III) 0
(D) Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ .	(IV) $\frac{\sigma}{\epsilon_0 r^2}$

Choose the correct answer from the options given below :

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Options:

A.

(A)-(IV), (B)-(I), (C)-(III), (D)-(II)

B.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

C.

(A)-(III), (B)-(II), (C)-(IV), (D)-(I)

D.

(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Answer: C

Solution:

(A) $\rightarrow$ 0 (III)

$$(B) \rightarrow \frac{\sigma}{2\epsilon_0}$$

$$(C) \rightarrow \frac{\sigma R^2}{\epsilon_0 r^2} \text{ (No row matching)}$$

$$(D) \rightarrow \frac{\sigma}{\epsilon_0} (I)$$

Question42

An electric dipole of mass m , charge q , and length l is placed in a uniform electric field $\vec{E} = E_0 \hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be :

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Options:

A.

$$\frac{1}{2\pi} \sqrt{\frac{ml}{2qE_0}}$$

B.

$$2\pi \sqrt{\frac{ml}{2qE_0}}$$

C.

$$2\pi \sqrt{\frac{ml}{qE_0}}$$

D.

$$\frac{1}{2\pi} \sqrt{\frac{2ml}{qE_0}}$$

Answer: B

Solution:

$$I\omega 2\theta = q\ell E_0\theta$$

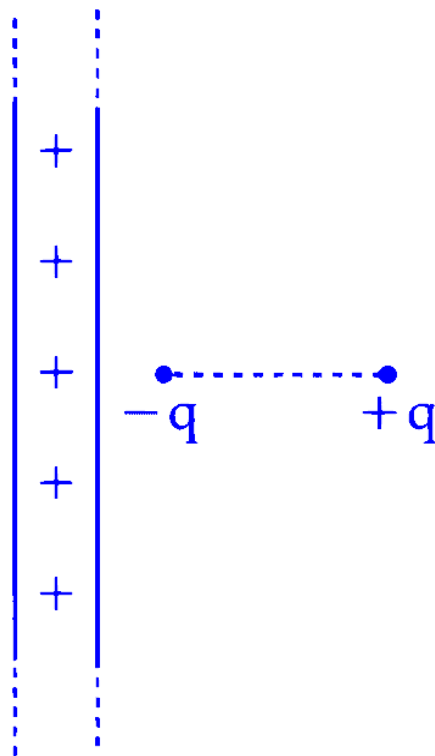
$$2m\left(\frac{\ell}{2}\right)^2\omega^2 = q\ell E_0$$

$$\omega^2 = \frac{2qE_0}{m\ell}$$

$$T = 2\pi\sqrt{\frac{m\ell}{2qE_0}}$$

Question43

An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0 . Choose the correct option from the following.



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Options:

A.

Torque on dipole is zero and net force is directed away from the sheet.

B.

Potential energy of dipole is minimum and torque is zero.

C.

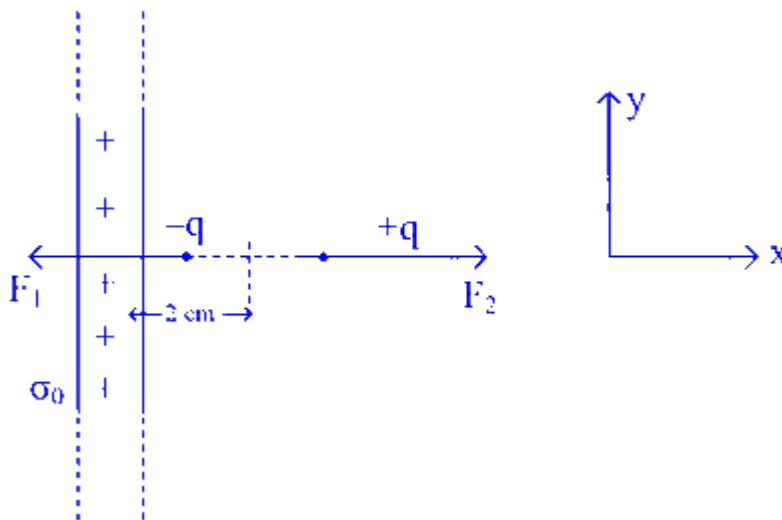
Potential energy and torque both are maximum.

D.

Torque on dipole is zero and net force acts towards the sheet.

Answer: B

Solution:



We know, electric field due to uniformly charged infinite plane sheet,

$$E = \frac{\sigma_0}{2\epsilon_0}$$

perpendicularly directed away from the sheet for positive charge

$$\text{So, } \vec{E} = \frac{\sigma_0}{2\epsilon_0} \hat{i}$$

note : E is constant

Let length of dipole = d

so dipole moment, $\vec{p} = qd \hat{i}$

We know,

$$\text{Torque, } \vec{\tau} = \vec{p} \times \vec{E}$$

$$\vec{\tau} = qd\hat{i} \times \frac{\sigma_0}{2\epsilon_0}\hat{i}$$

$$\vec{\tau} = 0 \text{ (as } \hat{i} \times \hat{i} = 0)$$

$$\text{force, on charge } -q, \vec{F}_1 = -qE\hat{i}$$

$$\text{on charge } +q, \vec{F}_2 = qE\hat{i}$$

$$\text{So net force on dipole, } \vec{F} = \vec{F}_1 + \vec{F}_2$$

$$= -qE\hat{i} + qE\hat{i}$$

$$\vec{F} = 0$$

We know, potential energy of dipole

$$u(O) = -pE \cos \theta$$

here, $\theta = 0^\circ$ (given)

$$\text{So, } u_{\min} = -pE$$

Hence, option B is correct.

Question44

A point charge causes an electric flux of $-2 \times 10^4 \text{ Nm}^2 \text{ C}^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is :

(Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

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Options:

A. $17.7 \times 10^{-8} \text{ C}$

B. $-17.7 \times 10^{-8} \text{C}$

C. $15.7 \times 10^{-8} \text{C}$

D. $-15.7 \times 10^{-8} \text{C}$

Answer: B

Solution:

Given an electric flux $\phi = -2 \times 10^4 \text{ Nm}^2\text{C}^{-1}$ through a spherical Gaussian surface with a radius of 8.0 cm ($r = 0.08 \text{ m}$), we are to find the value of the point charge Q . According to Gauss's Law:

$$\phi = \frac{Q}{\epsilon_0}$$

Where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$.

Rearranging the equation to solve for Q , we have:

$$Q = \phi \epsilon_0$$

Substituting the given values:

$$Q = -2 \times 10^4 \times 8.85 \times 10^{-12}$$

Calculating this yields:

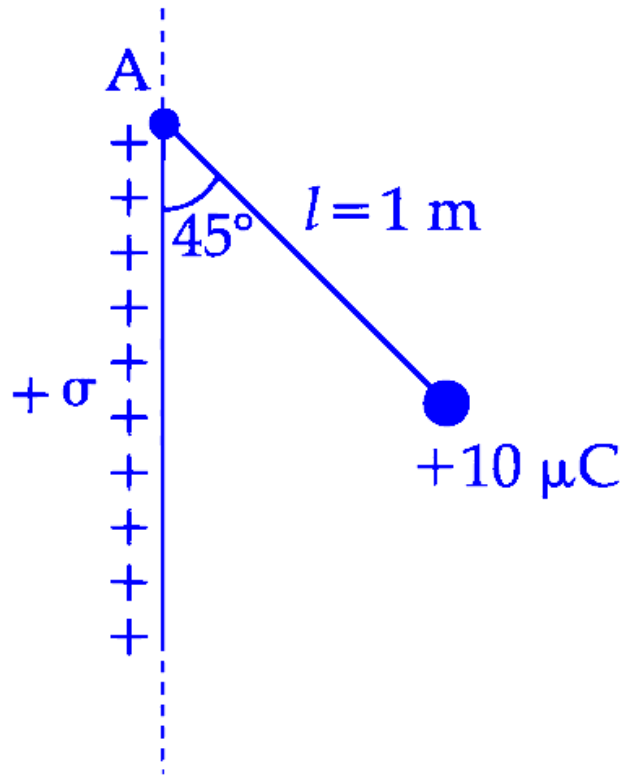
$$Q = -17.7 \times 10^{-8} \text{ C}$$

Thus, the value of the point charge is $Q = -17.7 \times 10^{-8} \text{ C}$.

Question45

A small bob of mass 100 mg and charge $+10\mu\text{C}$ is connected to an insulating string of length 1 m . It is brought near to an infinitely long non-conducting sheet of charge density ' σ ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be.

(Given, $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{F}}{\text{m}}$ and acceleration due to gravity, $g = 10 \frac{\text{m}}{\text{s}^2}$)



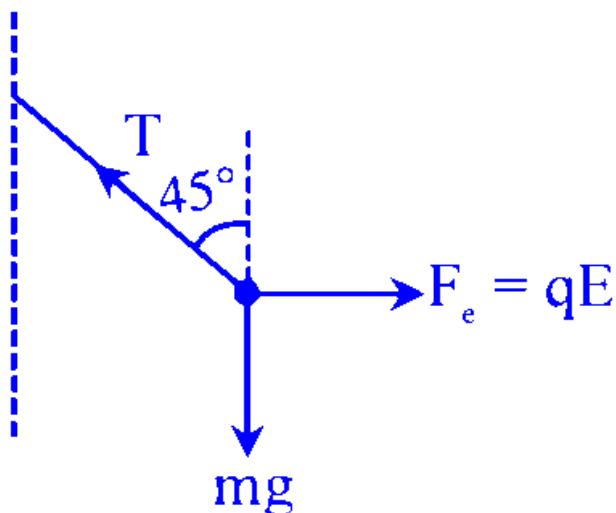
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Options:

- A. $1.77\ \text{nC/m}^2$
- B. $0.885\ \text{nC/m}^2$
- C. $885\ \text{nC/m}^2$
- D. $17.7\ \text{nC/m}^2$

Answer: A

Solution:



$$qE = mg$$

$$q \left[\frac{\sigma}{2\epsilon_0} \right] = mg$$

$$\sigma = \frac{2\epsilon_0 mg}{q}$$

$$\sigma = \frac{2 \times 8.85 \times 10^{-12} \times 100 \times 10^{-6} \times 10}{10 \times 10^{-6}}$$

$$\sigma = 17.7 \times 10^{-10} \text{ C/m}^2$$

$$\sigma = 1.77 \text{ nC/m}^2$$

Question 46

A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

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Options:

A. $(3d/4, 0, 0)$

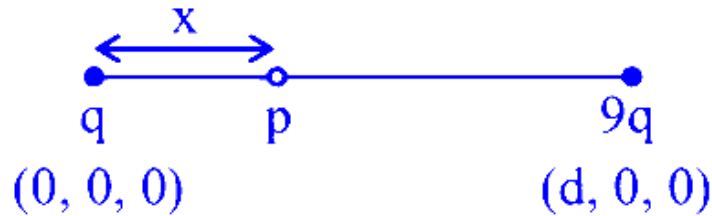
B. $(d/4, 0, 0)$

C. $(4d/3, 0, 0)$

D. $(d/3, 0, 0)$

Answer: B

Solution:



Let $E_p = 0$

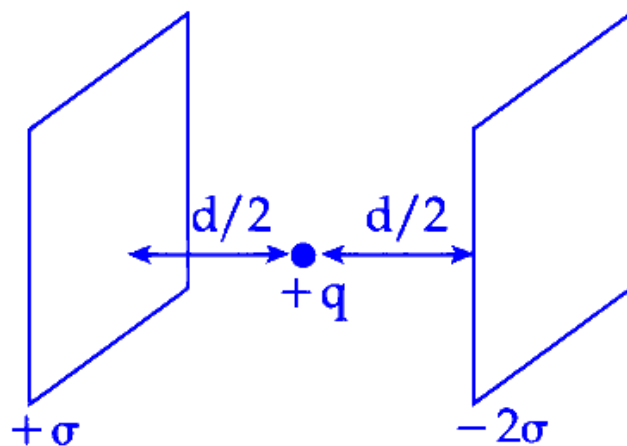
$$\therefore \frac{kq}{x^2} = \frac{k9q}{(d-x)^2}$$

$$\Rightarrow \frac{d-x}{x} = 3 \Rightarrow x = \frac{d}{4}$$

$\therefore$ co-ordinate of P is $(\frac{d}{4}, 0, 0)$

Question47

Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ . The force experienced by a point charge $+q$ placed at the mid point between two plates will be:



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Options:

A. $\frac{3\sigma q}{4\epsilon_0}$

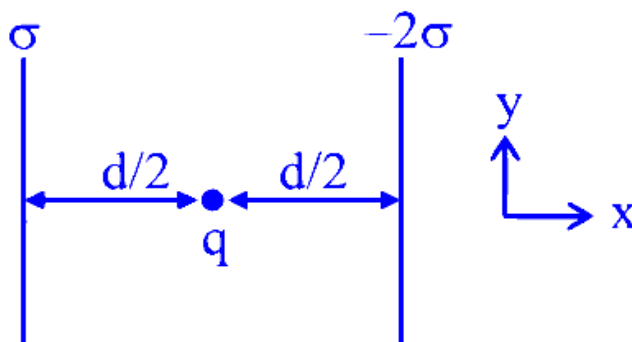
B. $\frac{3\sigma q}{2\epsilon_0}$

C. $\frac{\sigma q}{4\epsilon_0}$

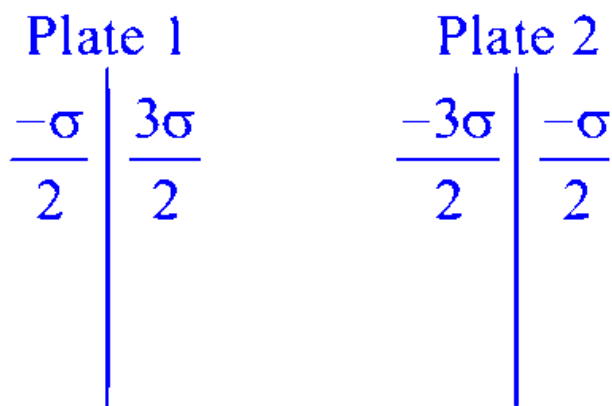
D. $\frac{\sigma q}{2\epsilon_0}$

Answer: B

Solution:



Final charge distribution will be



$$\therefore F_{\text{net}} = \frac{3\sigma}{2\epsilon_0} q$$

Question48

Consider a circular loop that is uniformly charged and has a radius $a\sqrt{2}$. Find the position along the positive z -axis of the cartesian coordinate system where the electric field is maximum if the ring was assumed to be placed in xy plane at the origin :

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Options:

A. a

B. $a/2$

C. 0

D.

$a/\sqrt{2}$

Answer: A

Solution:

To determine the position along the positive z -axis where the electric field due to a uniformly charged circular loop is at its maximum, we follow these steps:

Expression for Electric Field (E):

The electric field E at a point along the z -axis for a charged circular loop can be expressed as:

$$E = \frac{KQr}{(x^2 + R^2)^{3/2}}$$

Here, K is the Coulomb's constant, Q is the total charge, r is a constant involving charge distribution, x is the distance along the z -axis, and R is the radius of the loop.

Maximizing the Electric Field:

To find where E is maximum, we take the derivative of E with respect to x and set it to zero:

$$\frac{dE}{dx} = 0$$

Solve for x :

Solving the equation from the derivative, we find:

$$x = \frac{R}{\sqrt{2}}$$

Substitute Given Radius:

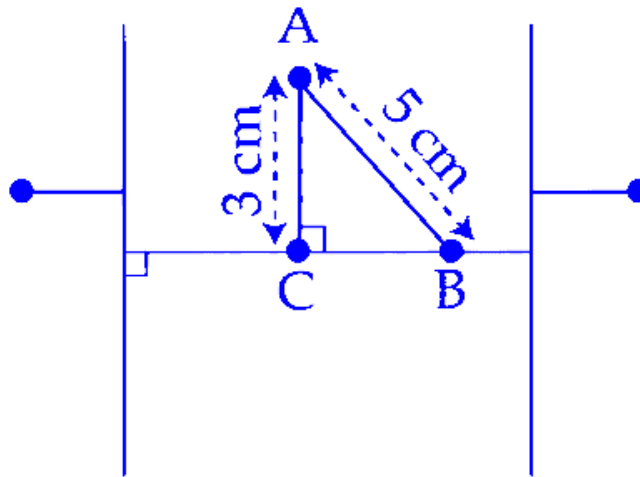
Given the radius $R = a\sqrt{2}$, substituting into the expression for x :

$$x = \frac{\sqrt{2}a}{\sqrt{2}} = a$$

Thus, the position along the positive z -axis where the electric field is maximum is at $x = a$.

Question49

Two large plane parallel conducting plates are kept 10 cm apart as shown in figure. The potential difference between them is V . The potential difference between the points A and B (shown in the figure) is :



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Options:

A. 1 V

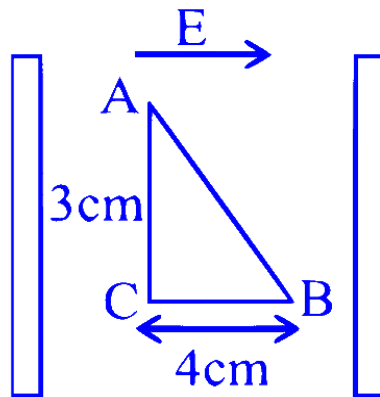
B. $\frac{1}{4}$ V

C. $\frac{3}{4}$ V

D. $\frac{2}{5}$ V

Answer: D

Solution:



Using $\Delta V = E(\Delta d)$

$$V = E(10)$$

$$V_{AB} = E \cdot 4 = \frac{V}{10} \times 4 = \frac{2V}{5}$$

Question 50

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.

Reason (R) : In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A) 2.
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is not correct but (R) is correct
- D. (A) is correct but (R) is not correct

Answer: C

Solution:

Assertion (A): The net dipole moment of a polar linear isotropic dielectric substance is not zero, even when there is no external electric field.

Reason (R): In the absence of an external electric field, the different permanent dipoles in a polar dielectric substance are oriented in random directions.

Explanation:

Assertion (A): The statement is incorrect. In a polar linear isotropic dielectric substance, the net dipole moment is zero when an external electric field is absent. This is because the dipoles within the material are randomly oriented and cancel each other out.

Reason (R): The statement is correct. In the absence of an external electric field, the dipoles in a polar dielectric substance remain inherently polarized but are oriented randomly. This randomness in orientation leads to a cancellation of the net dipole moment.

Therefore, while the reason is correct, it does not support the validity of the assertion as stated.

Question51

The electrostatic potential on the surface of uniformly charged spherical shell of radius $R = 10$ cm is 120 V . The potential at the centre of shell, at a distance $r = 5$ cm from centre, and at a distance $r = 15$ cm from the centre of the shell respectively, are:

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Options:

- A. 0 V, 120 V, 40 V
- B. 120 V, 120 V, 80 V
- C. 40 V, 40 V, 80 V
- D. 0 V, 0 V, 80 V

Answer: B

Solution:

The potential inside a uniformly charged spherical shell is equal to the potential on its surface. This means:

$$V_{\text{in}} = V_{\text{surface}} = \frac{kQ}{R} = 120 \text{ V}$$

This maintains the potential at both the center of the shell and any point inside it up to the surface. Therefore, the potential at the center of the shell and at any point inside the shell (like at a distance $r = 5 \text{ cm}$) is also 120 V.

For a point outside the shell, at a distance $r = 15 \text{ cm}$, the potential is given by the formula:

$$V = \frac{kQ}{r} = \frac{120 \times 10}{15} = 80 \text{ V}$$

Question52

Two small spherical balls of mass 10 g each with charges $-2\mu\text{C}$ and $2\mu\text{C}$, are attached to two ends of very light rigid rod of length 20 cm . The arrangement is now placed near an infinite nonconducting charge sheet with uniform charge density of $100\mu\text{C}/\text{m}^2$ such that length of rod makes an angle of 30° with electric field generated by charge sheet. Net torque acting on the rod is:(Take $\epsilon_0 : 8.85 \times 10^{-12} \text{C}^2/\text{Nm}^2$)

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Options:

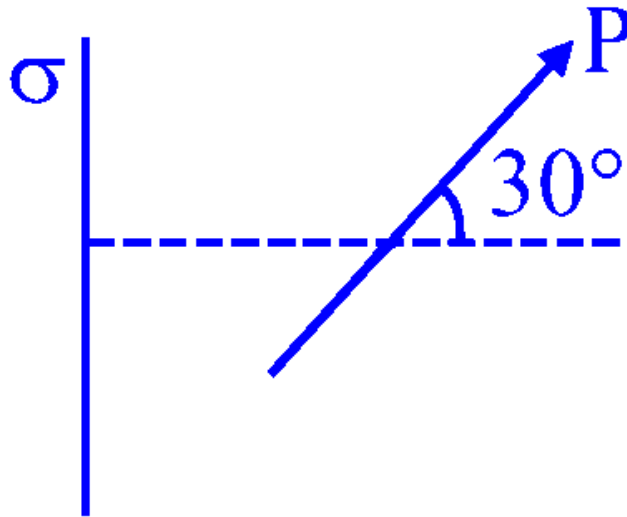
- A. 1.12 Nm
- B. 2.24 Nm

C. 11.2 Nm

D. 112 Nm

Answer: A

Solution:



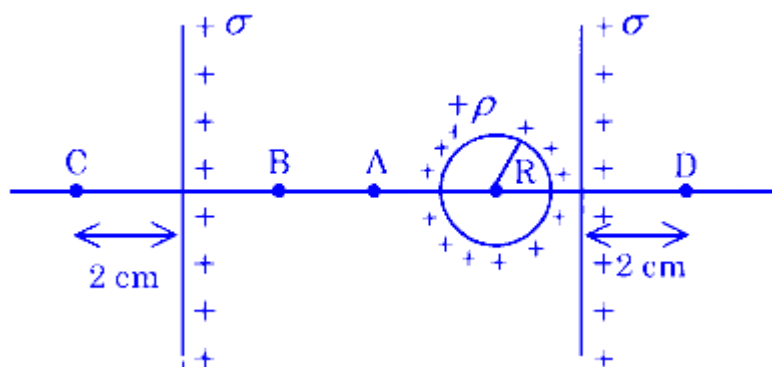
$$E = \frac{\sigma}{2\epsilon_0}$$

$$\tau = PE \sin \theta$$

$$= \left[(2 \times 10^{-6}) \left(\frac{2}{10} \right) \right] \left[\frac{100 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} \right] \left(\frac{1}{2} \right)$$
$$= \frac{10}{8.85} = 1.12 \text{ Nm}$$

Question53

Two infinite identical charged sheets and a charged spherical body of charge density ' ρ ' are arranged as shown in figure. Then the correct relation between the electrical fields at A, B, C and D points is:



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Options:

A. $|\vec{E}_A| = |\vec{E}_B|; \vec{E}_C > \vec{E}_D$

B. $\vec{E}_A = \vec{E}_B; \vec{E}_C = \vec{E}_D$

C. $\vec{E}_C \neq \vec{E}_D; \vec{E}_A > \vec{E}_B$

D. $\vec{E}_A > \vec{E}_B; \vec{E}_C = \vec{E}_D$

Answer: C

Solution:

In this scenario, two infinite identical charged sheets and a charged spherical body with charge density ' ρ ' are positioned as shown in the figure. We aim to determine the relationship between the electrical fields at points A, B, C, and D.

When considering the electric fields:

Electric Field Due to Charged Sheets:

The infinite charged sheets produce a uniform electric field. The electric field due to a single infinite sheet of charge is constant on either side of the sheet and is directed perpendicularly away from the sheet.

Electric Field Due to a Spherical Charge:

Outside the charged sphere, the electric field behaves as if the entire charge were concentrated at the center of the sphere. The electric field decreases with the square of the distance from the center, following the inverse square law.

Given these principles:

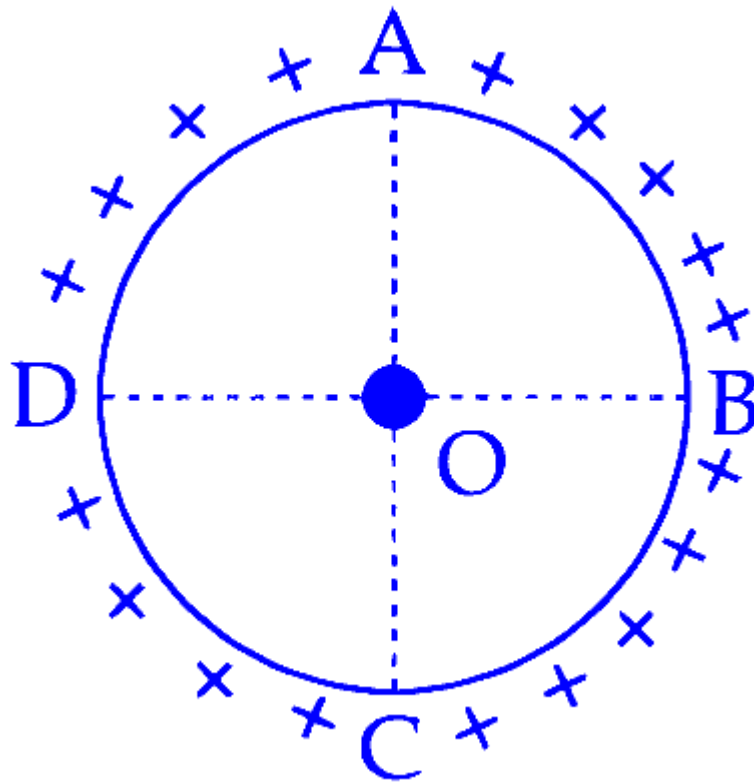
At Points C and D: These are located in different regions regarding the charged sheets and spherical body. Due to the inverse square nature of the electric field from the spherical charge and the uniform field of the sheets, the field strength will vary, resulting in $\vec{E}_C \neq \vec{E}_D$.

At Points A and B: Both points are influenced by the uniform field of the charged sheets. However, if the distance or position concerning the sphere also affects these points, this could lead to differences in field strength, thus $\vec{E}_A > \vec{E}_B$.

This analysis leads us to conclude that the electric field relations are $E_C \neq E_D$ and $E_A > E_B$.

Question54

A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB at 'O' is 'E' in magnitude. What would be the magnitude of electric field at 'O' due to arc ABC ?



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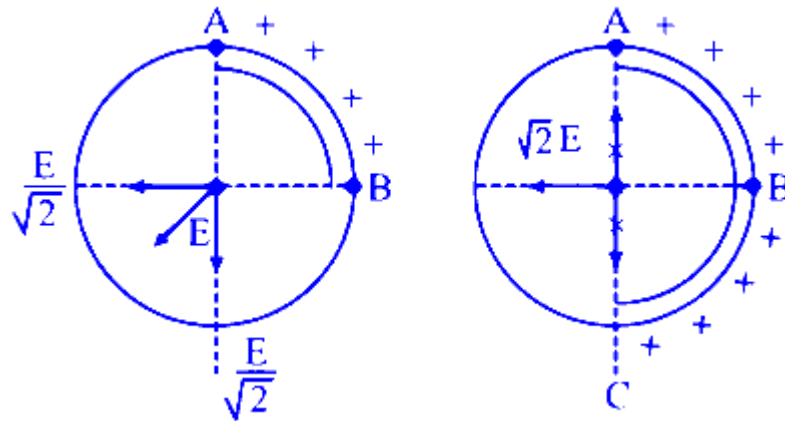
Options:

- A. $2E$
- B. Zero
- C. $E/2$

D. $\sqrt{2}E$

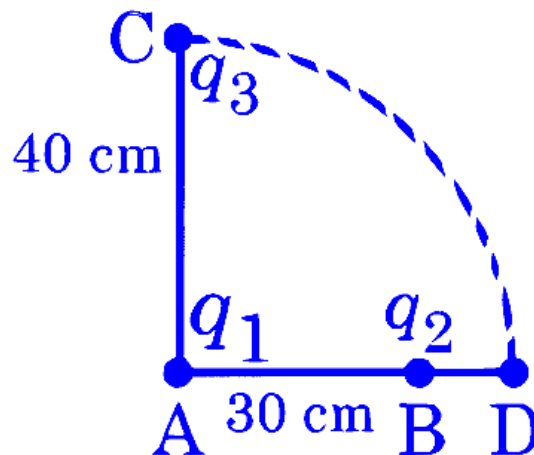
Answer: D

Solution:



Question55

Two charges q_1 and q_2 are separated by a distance of 30 cm . A third charge q_3 initially at ' C ' as shown in the figure, is moved along the circular path of radius 40 cm from C to D . If the difference in potential energy due to movement of q_3 from C to D is given by $\frac{q_3 K}{4\pi\epsilon_0}$, the value of K is :



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Options:

A. $6q_2$

B. $6q_1$

C. $8q_1$

D. $8q_2$

Answer: D

Solution:

To determine the value of K for the given electrical potential scenario, we start by calculating the potential at points C and D.

Potential at Point C:

The potential at point C, V_C , is given by the sum of the potentials due to charges q_1 and q_2 :

$$V_C = \frac{kq_1}{0.4} + \frac{kq_2}{0.5}$$

Potential at Point D:

Similarly, the potential at point D, V_D , is:

$$V_D = \frac{kq_1}{0.4} + \frac{kq_2}{0.1}$$

Change in Potential Energy:

The change in potential energy ΔU when q_3 moves from C to D is:

$$\Delta U = (V_D - V_C) \times q_3$$

Substituting the expressions for V_D and V_C , we get:

$$\Delta U = \left(\frac{kq_2}{0.1} - \frac{kq_2}{0.5} \right) \times q_3$$

Simplifying the Expression:

By simplifying the expression, we find:

$$\Delta U = \left(\frac{5kq_2}{0.5} \right) \times q_3 - \left(\frac{kq_2}{0.5} \right) \times q_3$$

$$\Delta U = \frac{4kq_2}{0.5} \times q_3 = 8kq_2q_3$$

$$\Delta U = \frac{8q_2q_3}{4\pi\epsilon_0}$$

Thus, the value of K is $8q_2$.

Question 56

If ϵ_0 denotes the permittivity of free space and Φ_E is the flux of the electric field through the area bounded by the closed surface, then dimensions of $\left(\epsilon_0 \frac{d\Phi_E}{dt}\right)$ are that of :

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Options:

- A. electric charge
- B. electric field
- C. electric current
- D. electric potential

Answer: C

Solution:

We can use Gauss's law to analyze the dimensions. Gauss's law states that

$$\Phi_E = \frac{q}{\epsilon_0},$$

where

Φ_E is the electric flux,

q is the electric charge,

ϵ_0 is the permittivity of free space.

To find the dimensions of $\epsilon_0 \frac{d\Phi_E}{dt}$, follow these steps:

Differentiate Gauss's law with respect to time:

$$\frac{d\Phi_E}{dt} = \frac{1}{\epsilon_0} \frac{dq}{dt}.$$

Multiply the equation by ϵ_0 :

$$\epsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt}.$$

Recognize that the time rate of change of charge, $\frac{dq}{dt}$, is by definition the electric current.

Thus, the dimensions of $\epsilon_0 \frac{d\Phi_E}{dt}$ are those of electric current.

The correct answer is: Option C, electric current.

Question 57

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The outer body of an aircraft is made of metal which protects persons sitting inside from lightning strikes.

Reason (R) : The electric field inside the cavity enclosed by a conductor is zero.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

Both (A) and (R) are correct but (R) is not the correct explanation of (A)

B.

(A) is correct but (R) is not correct

C.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

D.

(A) is not correct but (R) is correct

Answer: C

Solution:

The assertion suggests that the outer body of an aircraft is made of metal to protect passengers from lightning strikes. The reason provided is that the electric field inside a cavity enclosed by a conductor is zero.

This reasoning is based on the property of metallic conductors, where the electric field within a conductor's cavity is indeed zero. This phenomenon occurs because the charges on a conductor redistribute themselves to cancel any external electric fields within the conductor. Thus, if lightning were to strike an aircraft, the metal body would direct the electric charges along its surface, effectively shielding the interior where passengers are located.

Therefore, the assertion and the reason are both correct, and the reason correctly explains why the metal body of an aircraft shields passengers from lightning. This aligns with the principles of electromagnetic shielding. The key point is that the conductive nature of the aircraft's metal body ensures no electric field permeates the interior, keeping passengers safe.

Question58

A dipole with two electric charges of $2\ \mu\text{C}$ magnitude each, with separation distance $0.5\ \mu\text{m}$, is placed between the plates of a capacitor such that its axis is parallel to an electric field established between the plates when a potential difference of $5\ \text{V}$ is applied. Separation between the plates is $0.5\ \text{mm}$. If the dipole is rotated by 30° from the axis, it tends to realign in the direction due to a torque. The value of torque is:

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Options:

A.

$$2.5 \times 10^{-9}\ \text{Nm}$$

B.

$$2.5 \times 10^{-12}\ \text{Nm}$$

C.

$$5 \times 10^{-3}\ \text{Nm}$$

D.

$$5 \times 10^{-9} \text{ Nm}$$

Answer: D

Solution:

Given:

Potential difference $V = 5 \text{ V}$

Separation between the plates $d = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$

The electric field E is calculated as:

$$E = \frac{V}{d} = \frac{5}{0.5 \times 10^{-3}} = 10^4 \text{ V/m}$$

The torque τ experienced by the dipole in the electric field is given by:

$$\tau = PE \sin \theta$$

Where:

P is the dipole moment,

E is the electric field,

$\theta = 30^\circ$ is the angle of rotation.

The dipole moment P is calculated by:

$$P = q \times a = (2 \times 10^{-6} \text{ C}) \times (0.5 \times 10^{-6} \text{ m}) = 1 \times 10^{-12} \text{ C} \cdot \text{m}$$

Substituting the values into the torque equation:

$$\tau = (1 \times 10^{-12} \text{ C} \cdot \text{m}) \times 10^4 \text{ V/m} \times \sin 30^\circ$$

Since $\sin 30^\circ = 0.5$, the torque simplifies to:

$$\tau = 1 \times 10^{-12} \times 10^4 \times 0.5 = 5 \times 10^{-9} \text{ N} \cdot \text{m}$$

Thus, the torque is $5 \times 10^{-9} \text{ N} \cdot \text{m}$.

Question 59

An infinitely long wire has uniform linear charge density $\lambda = 2 \text{ nC/m}$. The net flux through a Gaussian cube of side length $\sqrt{3} \text{ cm}$, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be $x \text{ Nm}^2\text{C}^{-1}$, where x is:

[Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI units]

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Options:

A.

$$6.48\pi$$

B.

$$0.72\pi$$

C.

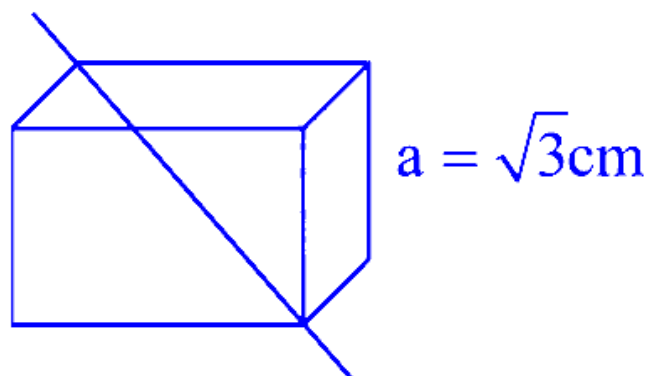
$$1.44\pi$$

D.

$$2.16\pi$$

Answer: D

Solution:



To find the net flux through a Gaussian cube when an infinitely long wire with a uniform linear charge density $\lambda = 2$ nC/m passes through the cube's diagonal corners, we use Gauss's Law. The relevant expression is:

$$\phi = \frac{q_{\text{enc}}}{\varepsilon_0}$$

Here, q_{enc} is the charge enclosed by the Gaussian surface, and ε_0 is the permittivity of free space.

Given:

The wire passes through two opposite corners of the cube.

The side length of the cube, $a = \sqrt{3} \text{ cm} = \sqrt{3} \times 10^{-2} \text{ m}$.

The length of the wire inside the cube is equal to its diagonal, which is $\sqrt{3} \times a$.

Thus, the charge enclosed q_{enc} is:

$$q_{\text{enc}} = \lambda \times (\text{length of wire inside the cube}) = \lambda \times \sqrt{3} \times a$$

Substitute the given values:

$$\lambda = 2 \times 10^{-9} \text{ C/m}, \quad a = \sqrt{3} \times 10^{-2} \text{ m}$$

Calculating:

$$q_{\text{enc}} = 2 \times 10^{-9} \times \sqrt{3} \times \sqrt{3} \times 10^{-2}$$

Now apply Gauss's Law for net flux:

$$\phi = \frac{\lambda \cdot \sqrt{3} \cdot a}{\varepsilon_0}$$

$$\text{Substitute } \varepsilon_0 = \frac{1}{4\pi \times 9 \times 10^9}:$$

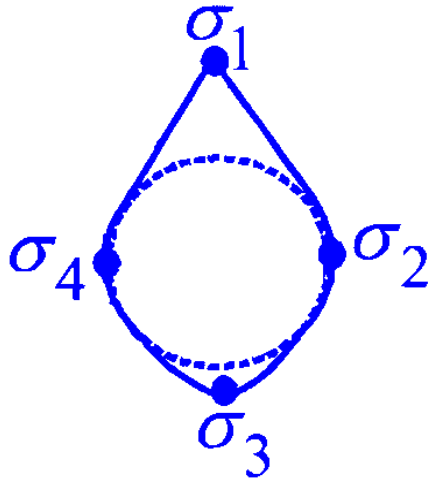
$$\phi = 2 \times 10^{-9} \times \sqrt{3} \times \sqrt{3} \times 10^{-2} \times 36\pi \times 10^9$$

$$\phi = 2.16\pi \text{ Nm}^2\text{C}^{-1}$$

Thus, the net flux through the Gaussian cube is $2.16\pi \text{ Nm}^2\text{C}^{-1}$.

Question60

Electric charge is transferred to an irregular metallic disk as shown in the figure. If σ_1 , σ_2 , σ_3 and σ_4 are charge densities at given points then, choose the correct answer from the options given below:



- A. $\sigma_1 > \sigma_3; \sigma_2 = \sigma_4$
- B. $\sigma_1 > \sigma_2; \sigma_3 > \sigma_4$
- C. $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$
- D. $\sigma_1 < \sigma_3 < \sigma_2 = \sigma_4$
- E. $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$

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Options:

- A. B and C Only
- B. A and C Only
- C. D and E Only
- D. A, B and C Only

Answer: D

Solution:

Therefore, the relationship between charge density and ROC at different points can be expressed as:

$$\sigma \propto \frac{1}{\text{ROC}}$$

From this, the relationship between the charge densities at the specified points becomes clear:

$$(\text{ROC})_1 < (\text{ROC})_3 < (\text{ROC})_2 = (\text{ROC})_4$$

This implies:

$$\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$$

Thus, the charge density is greatest at the point with the smallest ROC and decreases as the ROC increases, keeping in mind that σ_2 and σ_4 have equal ROC, hence equal charge densities.

Question61

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Work done in moving a test charge between two points inside a uniformly charged spherical shell is zero, no matter which path is chosen.

Reason R : Electrostatic potential inside a uniformly charged spherical shell is constant and is same as that on the surface of the shell.

In the light of the above statements, choose the correct answer from the options given below.

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Options:

A.

A is true but R is false

B.

Both A and R are true and R is the correct explanation of A

C.

Both A and R are true but R is NOT the correct explanation of A

D.

A is false but R is true

Answer: B

Solution:

Assertion A:

For a uniformly charged spherical shell, Gauss's law tells us that the electric field inside the shell is zero.

Since the electric field $\vec{E}$ is zero everywhere inside, the potential V must be constant.

Work done in moving a charge from one point to another in an electric field is given by the difference in potential energy, which is $q(V_B - V_A)$.

If the potential difference $V_B - V_A$ is zero (because the potential is constant), then the work done is zero, regardless of the path taken.

Therefore, Assertion A is true.

Reason R:

It states that the electrostatic potential inside a uniformly charged spherical shell is constant and equal to that on its surface.

This is correct because the electric field inside is zero, ensuring that the potential remains uniform inside.

Thus, Reason R is also true.

Relationship between A and R:

The work done on a test charge moving inside the shell being zero is a direct consequence of the fact that the potential is constant inside.

Hence, Reason R correctly explains why the work done (Assertion A) is zero.

Given this analysis, the correct answer is:

Option B

"Both A and R are true and R is the correct explanation of A."

Question62

Two metal spheres of radius R and $3R$ have same surface charge density σ . If they are brought in contact and then separated, the surface charge density on smaller and bigger sphere becomes σ_1 and σ_2 , respectively. The ratio $\frac{\sigma_1}{\sigma_2}$ is

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Options:

A.

$$\frac{1}{3}$$

B.

$$\frac{1}{9}$$

C.

$$9$$

D.

$$3$$

Answer: D

Solution:

When two conducting spheres are in contact, they achieve the same electric potential. Consider two metal spheres with radii R and $3R$ that have the same surface charge density σ .

For a conductive sphere, the potential V of a sphere can be expressed as $V = \frac{\sigma r}{\epsilon_0}$, where r is the radius of the sphere and ϵ_0 is the permittivity of free space.

When the two spheres are brought into contact and then separated, they equalize their potentials:

$$V_1 = V_2$$

Thus, the equations become:

$$\sigma_1 r_1 = \sigma_2 r_2$$

Substituting for the radii:

$$\sigma_1 R = \sigma_2 (3R)$$

From this, we can solve for the ratio of the surface charge densities:

$$\frac{\sigma_1}{\sigma_2} = \frac{3R}{R} = 3$$

Therefore, the ratio of the surface charge densities on the smaller sphere to the larger sphere after contact is 3.

Question63

An electric charge $10^{-6}\mu\text{C}$ is placed at origin (0,0) m of X – Y co-ordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})\text{m}$ and $(\sqrt{6}, 0)\text{m}$ respectively. The potential difference between the points P and Q will be :

[27-Jan-2024 Shift 1]

Options:

A.

$$\sqrt{3}\text{V}$$

B.

$$\sqrt{6}\text{V}$$

C.

$$0\text{V}$$

D.

$$3\text{V}$$

Answer: C

Solution:

$$\text{Potential difference} = \frac{KQ}{r_1} - \frac{KQ}{r_2}$$

$$r_1 = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2}$$

$$r_2 = \sqrt{(\sqrt{6})^2 + 0}$$

$$\text{As } r_1 = r_2 = \sqrt{6}\text{m}$$

$$\text{So potential difference} = 0$$

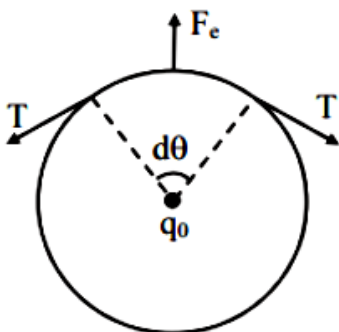
Question64

A thin metallic wire having cross sectional area of 10^{-4}m^2 is used to make a ring of radius 30cm. A positive charge of $2\pi\text{C}$ is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is ____N ; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$)

[27-Jan-2024 Shift 1]

Answer: 3

Solution:



$$2T \sin \frac{d\theta}{2} = \frac{kq_0}{R^2} \cdot \lambda R d\theta$$

$$\left[\lambda = \frac{Q}{2\pi R} \right]$$

$$\Rightarrow T = \frac{Kq_0 Q}{(R^2) \times 2\pi}$$

$$= \frac{(9 \times 10^9)(2\pi \times 30 \times 10^{-12})}{(0.30)^2 \times 2\pi}$$

$$= \frac{9 \times 10^{-3} \times 30}{9 \times 10^{-2}} = 3N$$

Question 65

Given below are two statements : one is labelled a

Assertion (A) and the other is labelled as Reason(R)

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below :

[27-Jan-2024 Shift 2]

Options:

A.

Both (A) and (R) are correct but (R) is not the correct explanation of (A)

B.

((A) is correct but (R) is not correct

C.

(A) is not correct but (R) is correct

D.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

Answer: D

Solution:

Electric line of force are always perpendicular to equipotential surface so angle between force and displacement will always be 90° . So work done equal to 0.

Question66

The electric potential at the surface of an atomic nucleus ($z = 50$) of radius 9×10^{-13} cm is _____ $\times 10^6$ V

[27-Jan-2024 Shift 2]

Answer: 8

Solution:

$$\begin{aligned}\text{Potential} &= \frac{kQ}{R} = \frac{k \cdot Z e}{R} \\ &= \frac{9 \times 10^9 \times 50 \times 1.6 \times 10^{-19}}{9 \times 10^{-13} \times 10^{-2}} \\ &= 8 \times 10^6 \text{ V}\end{aligned}$$

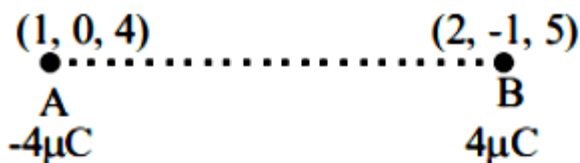
Question67

Two charges of $-4\mu\text{C}$ and $+4\mu\text{C}$ are placed at the points $A(1, 0, 4)\text{m}$ and $B(2, -1, 5)\text{m}$ located in an electric field $\vec{E} = 0.20\hat{i}\text{V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}\text{Nm}$, Where $\alpha = \underline{\hspace{1cm}}$

[27-Jan-2024 Shift 2]

Answer: 2

Solution:



$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\vec{p} = q \vec{\ell}$$

$$\vec{E} = 0.2 \frac{\text{V}}{\text{cm}} = 20 \frac{\text{V}}{\text{m}}$$

$$\vec{p} = 4 \times (\hat{i} - \hat{j} + \hat{k})$$

$$= (4\hat{i} - 4\hat{j} + 4\hat{k}) \mu\text{C} - \text{m}$$

$$\vec{\tau} = (4\hat{i} - 4\hat{j} + 4\hat{k}) \times (20\hat{i}) \times 10^{-6} \text{Nm}$$

$$= (8\hat{k} + 8\hat{j}) \times 10^{-5} = 8\sqrt{2} \times 10^{-5}$$

$$\alpha = 2$$

Question68

Two charges of $5Q$ and $-2Q$ are situated at the points $(3a,0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius ' $4a$ ' having center at origin is :

[29-Jan-2024 Shift 1]

Options:

A.

$$\frac{2Q}{\epsilon_0}$$

B.

$$\frac{5Q}{\epsilon_0}$$

C.

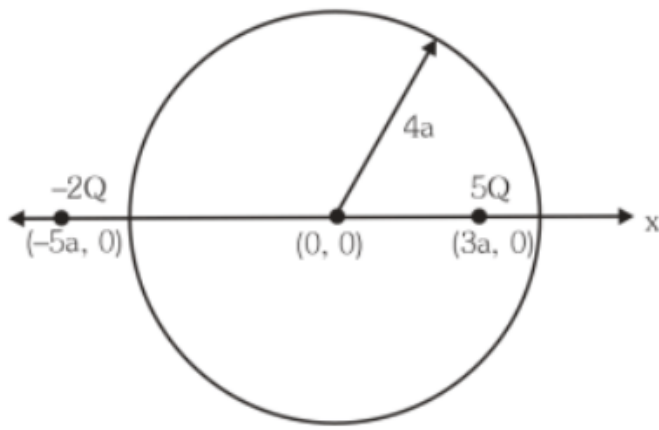
$$\frac{7Q}{\epsilon_0}$$

D.

$$\frac{3Q}{\epsilon_0}$$

Answer: B

Solution:



5Q charge is inside the spherical region flux through sphere = $5Q/\epsilon_0$

Question69

Match List I with List II

	List I		List II
A.	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$	I.	Gauss law for electricity
B.	$\oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$	II.	Gauss law for magnetism
C.	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$	III.	Faraday law
D.	$\oint \vec{B} \cdot d\vec{A} = 0$	IV.	Ampere -Maxwell law

Choose the correct answer from the options given below

[29-Jan-2024 Shift 1]

Options:

A.

A-IV, B-I, C-III, D-II

B.

A-II, B-III, C-I, D-IV

C.

A-IV, B-III, C-I, D-II

D.

A-I, B-II, C-III, D-IV

Answer: C

Solution:

Ampere-Maxwell law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

$$\text{Faraday law } \oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$$

$$\text{Gauss' law for electricity } \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$\text{Gauss' law for magnetism } \oint \vec{B} \cdot d\vec{A} = 0$$

Question 70

An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of 1m from S and has a speed of 1m/s. The maximum value of σ if the electron strikes

S at $t = 1$ s is $\alpha \left[\frac{m\epsilon_0}{e} \right] \frac{C}{m^2}$ the value of α is ____

[29-Jan-2024 Shift 1]

Answer: 8

Solution:

$$u = 1 \text{ m/s}; a = -\frac{\sigma e}{2\epsilon_0 m}$$

$$t = 1 \text{ s}$$

$$S = -1 \text{ m}$$

$$\text{Using } S = ut + \frac{1}{2}at^2$$

$$-1 = 1 \times 1 - \frac{1}{2} \times \frac{\sigma e}{2\epsilon_0 m} \times (1)^2$$

$$\therefore \sigma = 8 \frac{\epsilon_0 m}{e}$$

$$\therefore \alpha = 8$$

Question 71

An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$. The electric flux through a surface area $30\hat{i} \text{ m}^2$ lying in YZ-plane (in SI unit) is :

[29-Jan-2024 Shift 2]

Options:

A.

90

B.

150

C.

180

D.

60

Answer: C

Solution:

$$\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$$

$$\vec{A} = 30\hat{i}$$

$$\phi = \vec{E} \cdot \vec{A}$$

$$\phi = (6\hat{i} + 5\hat{j} + 3\hat{k}) \cdot (30\hat{i})$$

$$\phi = 6 \times 30 = 180$$

Question 72

The electrostatic potential due to an electric dipole at a distance ' r ' varies as :

[30-Jan-2024 Shift 1]

Options:

A.

r

B.

$1/r^2$

C.

$1/r^3$

D.

$1/r$

Answer: B

Solution:

$$V = \frac{kP \cos \theta}{r^2}$$

& can also be checked dimensionally

Question 73

A particle of charge ' $-q$ ' and mass ' m ' moves in a circle of radius ' r ' around an infinitely long line charge of linear density ' $+\lambda$ '. Then time period will be given as:

(Consider k as Coulomb's constant)

[30-Jan-2024 Shift 2]

Options:

A.

$$T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$$

B.

$$T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$$

C.

$$T = \frac{1}{2\pi r} \sqrt{\frac{m}{2k\lambda q}}$$

D.

$$T = \frac{1}{2\pi} \sqrt{\frac{2k\lambda q}{m}}$$

Answer: B

Solution:

$$\frac{2k\lambda q}{r} = m\omega^2 r$$

$$\omega^2 = \frac{2k\lambda q}{mr^2}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{2k\lambda q}{mr^2}$$

$$T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$$

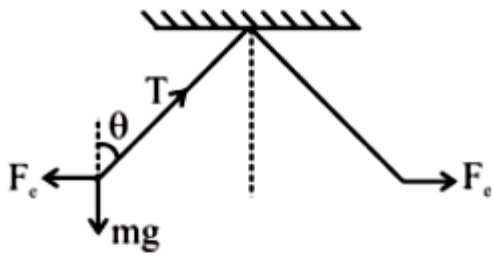
Question74

Two identical charged spheres are suspended by string of equal lengths. The string make an angle of 37° with each other. When suspended in a liquid of density 0.7g/cm^3 , the angle remains same. If density of material of the sphere is 1.4g/cm^3 , the dielectric constant of the liquid is _____ ($\tan 37^\circ = 3/4$).

[30-Jan-2024 Shift 2]

Answer: 2

Solution:



$$T \cos \theta = mg$$

$$T \sin \theta = F_e$$

$$\tan \theta = \frac{F_e}{mg}$$

$$\tan \theta = \frac{F_e}{\rho_B V g} \dots\dots\dots(i)$$

$$\tan \theta = \frac{\frac{F_e}{k}}{(\rho_B - \rho_L) V g} \dots\dots\dots(ii)$$

From Eq. (i) & (ii)

$$\rho_B V g = (\rho_B - \rho_L) k V g$$

$$1.4 = 0.7k$$

$$k = 2$$

Question75

Two charges q and $3q$ are separated by a distance ' r ' in air. At a distance x from charge q , the resultant electric field is zero. The value of x is :

[31-Jan-2024 Shift 1]

Options:

A.

$$\frac{(1 + \sqrt{3})}{r}$$

B.

$$\frac{r}{3(1 + \sqrt{3})}$$

C.

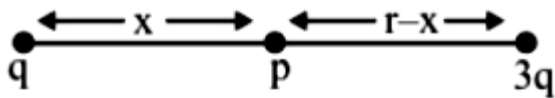
$$\frac{r}{(1 + \sqrt{3})}$$

D.

$$r(1 + \sqrt{3})$$

Answer: C

Solution:



$$\left(\vec{E}_{\text{net}} \right)_P = 0$$

$$\frac{kq}{x^2} = \frac{k \cdot 3q}{(r-x)^2}$$

$$(r-x)^2 = 3x^2$$

$$r-x = \sqrt{3}x$$

$$x = \frac{r}{\sqrt{3} + 1}$$

Question76

Force between two point charges q_1 and q_2 placed in vacuum at ' r ' cm apart is F . Force between them when placed in a medium having dielectric $K = 5$ at ' $r/5$ ' cm apart will be:

[31-Jan-2024 Shift 2]

Options:

A.

$F/25$

B.

$5F$

C.

$F/5$

D.

$25F$

Answer: B

Solution:

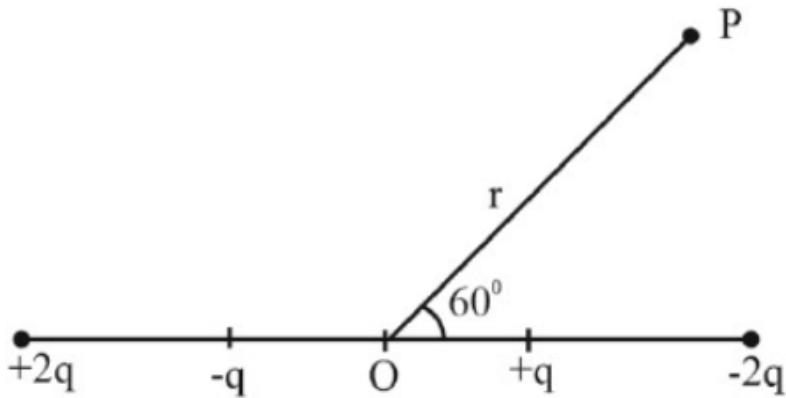
$$\text{In air } F = \frac{1}{4\pi E_0} \frac{q_1 q_2}{r^2}$$

In medium

$$F' = \frac{1}{4\pi(K E_0)} \frac{q_1 q_2}{(r')^2} = \frac{25}{4\pi(5 E_0)} \frac{q_1 q_2}{(r)^2} = 5F$$

Question 77

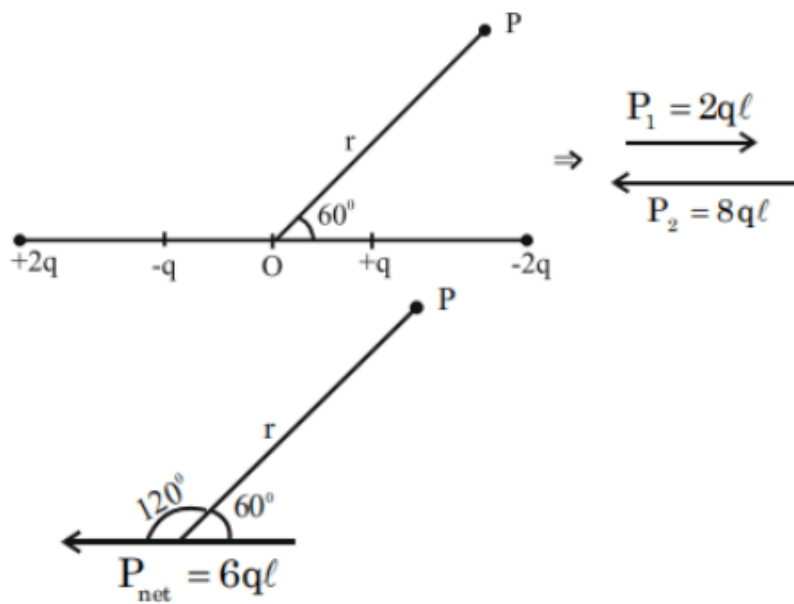
The distance between charges $+q$ and $-q$ is $2l$ and between $+2q$ and $-2q$ is $4l$. The electrostatic potential at point P at a distance r from centre O is $-\alpha \left[\frac{ql}{r^2} \right] \times 10^9 \text{V}$, where the value of α is (Use $1/4\pi\epsilon_0 = 9 \times 10^9 \text{NM}^2\text{C}^{-2}$)



[31-Jan-2024 Shift 2]

Answer: 27

Solution:



$$V = \frac{K \vec{p} \cdot \vec{r}}{r^3} = \frac{9 \times 10^9 (6q\ell)}{r^2} \cos(120^\circ)$$

$$= -(27) \left(\frac{q\ell}{r^2} \right) \times 10^9 \text{Nm}^2\text{c}^{-2}$$

$$\Rightarrow \alpha = 27$$

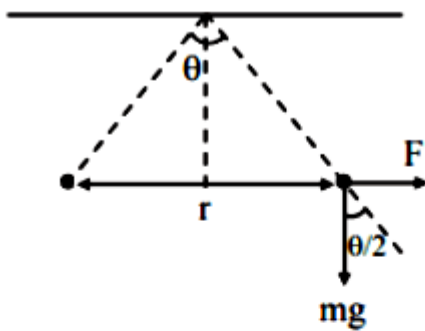
Question78

Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5g/cc , the dielectric constant of water will be _____ (Take density of water = 1g/cc)

[1-Feb-2024 Shift 1]

Answer: 3

Solution:



$$\text{In air } \tan \frac{\theta}{2} = \frac{F}{mg} = \frac{q^2}{4\pi\epsilon_0 r^2 mg}$$

$$\text{In water } \tan \frac{\theta}{2} = \frac{F'}{mg'} = \frac{q^2}{4\pi\epsilon_0\epsilon_r r^2 mg_{\text{eff}}}$$

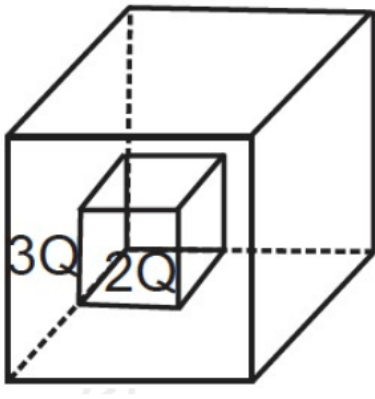
Equate both equations

$$\epsilon_0 g = \epsilon_0 \epsilon_r g \left[1 - \frac{1}{1.5} \right]$$

$$\epsilon_r = 3$$

Question 79

C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is :



[1-Feb-2024 Shift 2]

Options:

A.

2 : 5

B.

5 : 2

C.

2 : 3

D.

3 : 2

Answer: A

Solution:

$$\phi_{\text{smaller cube}} = \frac{2Q}{\epsilon_0}$$

$$\phi_{\text{bigger cube}} = \frac{5Q}{\epsilon_0}$$

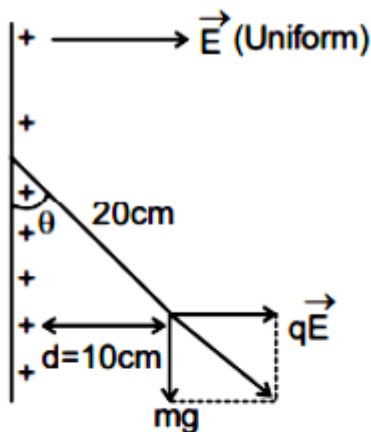
$$\frac{\phi_{\text{smaller cube}}}{\phi_{\text{bigger cube}}} = \frac{2}{5}$$

Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4 \text{ N/C}$ normally. A charged particle of mass 2g being suspended through a silk thread of length 20cm and remain stayed at a distance of 10cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}} \mu\text{C}$ where $x = \dots$ [use $g = 10 \text{ m/s}^2$]

[1-Feb-2024 Shift 2]

Answer: 3

Solution:



$$\sin \theta = \frac{10}{20} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\tan \theta = \frac{qE}{mg}$$

$$\tan 30^\circ = \frac{q \times 2 \times 10^4}{1 \times 10^{-3} \times 10}$$

$$\frac{1}{\sqrt{3}} = q \times 10^6$$

$$q = \frac{1}{\sqrt{3}} \times 10^{-6} \text{ C}$$

$$x = 3$$

Question81

Two small spheres each of mass 10mg are suspended from a point by threads 0.5m long. They are equally charged and repel each other to a distance of 0.20m. The charge on each of the sphere is $\frac{a}{21} \times 10^{-8} \text{C}$.

The value of a will be

[Given, $g = 10 \text{ms}^{-2}$]

[25 Feb 2021 Shift 2]

Answer: 630

Solution:

Given, mass of each spheres, $m = 10 \text{mg} = 10 \times 10^{-3} \text{g}$

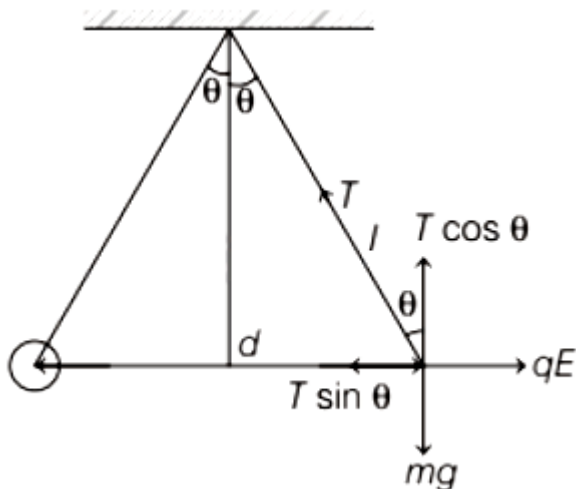
Length of thread (l) = 0.5m

Separation between charges, $d = 0.2 \text{m}$

Charge of each sphere, $q = \frac{a}{21} \times 10^{-8} \text{C}$

Acceleration due to gravity (g) = 10ms^{-2}

The situation can be shown as below,



Taking component of tension (T)

$$T \cos \theta = mg \dots (i)$$

$$T \sin \theta = qE = \frac{k |q|^2}{d^2} \dots (ii)$$

$$\sin \theta = \frac{d/2}{l} = \frac{0.1}{0.5} = \frac{1}{5}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - 1/25} = \frac{\sqrt{24}}{5}$$

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{kq^2}{d^2 mg} \quad [\text{using Eqs. (i) and (ii)}]$$

$$\Rightarrow \frac{1/5}{\sqrt{24}/5} = \frac{kq^2}{d^2 mg}$$

$$\Rightarrow \frac{1}{\sqrt{24}} = \frac{9 \times 10^9 \times q^2}{(0.2)^2 \times 10 \times 10^{-3} \times 10}$$

$$\Rightarrow q = \sqrt{\frac{(0.2)^2 \times 10^{-1}}{\sqrt{24} \times 9 \times 10^9}} \Rightarrow q = 3 \times 10^{-7} \text{C}$$

$$\Rightarrow \frac{a}{21} \times 10^{-8} = 30 \times 10^{-8}$$

$$a = 630$$

Question82

Two identical conducting spheres with negligible volume have 2.1nC and −0.1nC charges, respectively. They are brought into contact and then separated by a distance of 0.5m. The electrostatic force acting between the spheres is _____ $\times 10^{-9} \text{N}$.

[Given, $4\pi\epsilon_0 = \frac{1}{9 \times 10^9} \text{ SI unit}$]

[25 Feb 2021 Shift 2]

Answer: 36

Solution:

According to Coulomb's law, the force of attraction or repulsion is directly proportional to the product of distance between them and inversely proportional to the square of the distance between them.

$$F = K \frac{q_1 q_2}{r^2}$$

Here we have, q_1 is the charge of the first mass, q_2 is the charge of the second mass, r is the distance and K is the proportionally constant.

Given: Charge of first sphere $q_1 = 2.1 \text{ nc}$

and Charge of second sphere $q_2 = -0.1 \text{ NC}$

when they connect with each other the charge becomes, $q = \frac{q_2 - q_1}{2}$

$$\Rightarrow q = \frac{2.1 - 0.1}{2}$$

$$\Rightarrow q = 1 \text{ nC}$$

$$\sigma$$

According to Coulomb's law, we have;

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

Now, on putting the given values we have;

$$F = 9 \times 10^9 \times \frac{1 \times 10^{-9} \times 1 \times 10^{-9}}{0.5^2}$$

$$F = 9 \times 10^9 \times \frac{10^{-18}}{0.25}$$

$$\Rightarrow F = 36 \times 10^{-9} \text{ N}$$

$$\text{Hence, } F = 36 \times 10^{-9} \text{ N}$$

Question83

Two electrons each are fixed at a distance $2d$. A third charge proton placed at the mid-point is displaced slightly by a distance x ($x \ll d$) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency? (m = mass of charged particle)

[24 Feb 2021 Shift 2]

Options:

A. $\left(\frac{2q^2}{\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

B. $\left(\frac{\pi\epsilon_0 m d^3}{2q^2} \right)^{\frac{1}{2}}$

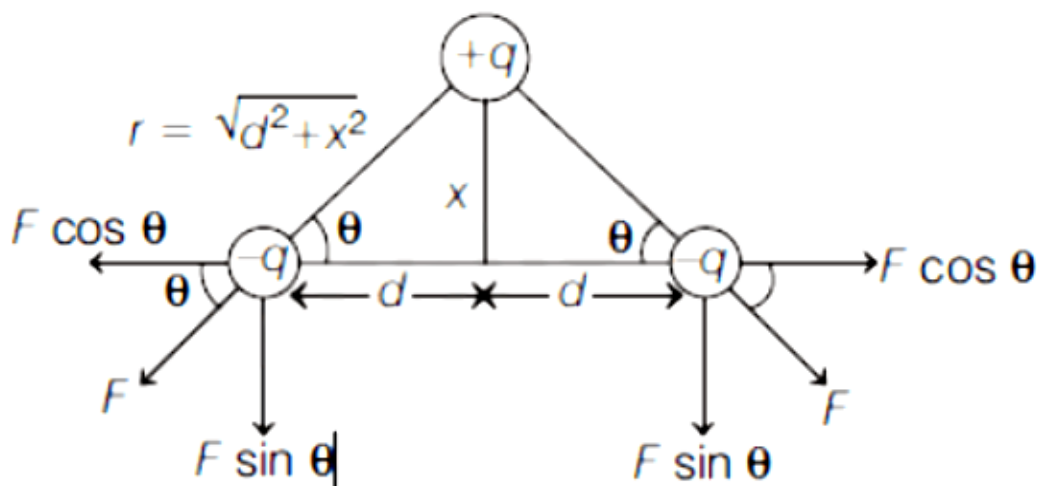
C. $\left(\frac{q^2}{2\pi\epsilon_0 m d^3} \right)^{\frac{1}{2}}$

D. $\left(\frac{2\pi\epsilon_0 m d^3}{q^2} \right)^{\frac{1}{2}}$

Answer: C

Solution:

The arrangement of charges is shown below



As we know that,

Coulomb's force between two charges. i.e. q_1 and q_2 ,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{(d^2 + x^2)} \dots (i)$$

Here, $q_1 = q_2 = q$

Force in SHM, $F = m\omega^2 x \dots (ii)$

Since, in order to have SHM $+q$ should move downwards and force responsible for this will be only

$$F' = F \sin \theta + F \sin \theta = 2F \sin \theta \dots (iii)$$

Using Eqs. (ii) and (iii), we get

$$2F \sin \theta = m\omega^2 x$$

$$\Rightarrow \frac{2}{4\pi\epsilon_0} \frac{q^2}{(d^2 + x^2)} \sin \theta = m\omega^2 x$$

$$\Rightarrow \frac{2}{4\pi\epsilon_0} \frac{q^2}{(d^2 + x^2)} \cdot \frac{x}{(d^2 + x^2)^{1/2}} = m\omega^2 x$$

$$\Rightarrow \omega = \left(\frac{1}{2\pi\epsilon_0} \frac{q^2}{(d^2 + x^2)^{3/2} m} \right)^{1/2}$$

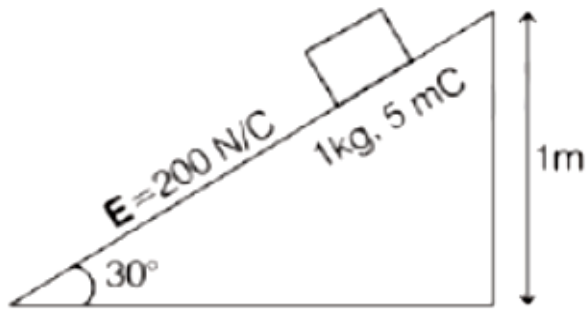
As, $x \ll d$

$$\therefore \omega = \left(\frac{1}{2\pi\epsilon_0} \frac{q^2}{md^3} \right)^{1/2}$$

Question84

An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field 200N/C as shown in the figure. A body of mass 1kg and charge 5mC is allowed to slide down from rest at a height of 1m . If the coefficient of friction is 0.2 , find the time taken by the body to reach the bottom.

[Take, $g = 9.8 \text{ m/s}^2$, $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$]



[26 Feb 2021 Shift 2]

Options:

A. 0.92 s

B. 0.46 s

C. 2.3 s

D. 1.3 s

Answer: D

Solution:

Given, mass of block, $m = 1 \text{ kg}$

Acceleration due to gravity, $g = 9.8 \text{ ms}^{-2}$

Inclination, $\theta = 30^\circ$

Electric field, $E = 200 \text{ N/C}$

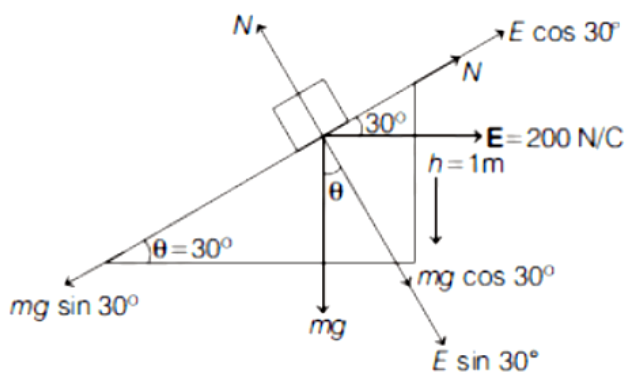
Coefficient of friction, $\mu = 0.2$

Charge, $q = 5 \text{ mC} = 5 \times 10^{-3} \text{ C}$

Let friction force, $f = \mu N$

where, N be the normal reaction.

Since, net force is zero along the perpendicular direction of incline.



Therefore, force along Y-axis will be zero.

$$\Rightarrow N = mg \cos 30^\circ + qE \sin 30^\circ$$

$$\Rightarrow N = 1 \times 9.8 \times \sqrt{3}/2 + 5 \times 10^{-3} \times 200 \times 1/2$$

$$= 8.49 + 0.5$$

$$= 8.99\text{N} \sim \text{eq } 9\text{N}$$

$$\therefore f = \mu N = \frac{2}{10} \times 9 = \frac{18}{10} = 1.8\text{N}$$

Now, total force along the plane of incline,

$$mg \sin 30^\circ - f - qE \cos 30^\circ = ma$$

$$\Rightarrow 1 \times 9.8 \times 1/2 - 1.8 - 5 \times 10^{-3} \times 200 \sqrt{3}/2 = a$$

$$\Rightarrow 5 - 1.8 - 1.732/2 = a$$

$$\Rightarrow a = 2.34\text{ms}^{-2}$$

Since, initial velocity of body, $u = 0\text{ms}^{-1}$ and distance along incline, $s = h/\sin 30^\circ = 1/\sin 30^\circ = 2$

By using second equation of motion,

$$s = ut + 1/2 at^2$$

$$\Rightarrow 2 = 0 + 1/2 \times 2.34 \times t^2$$

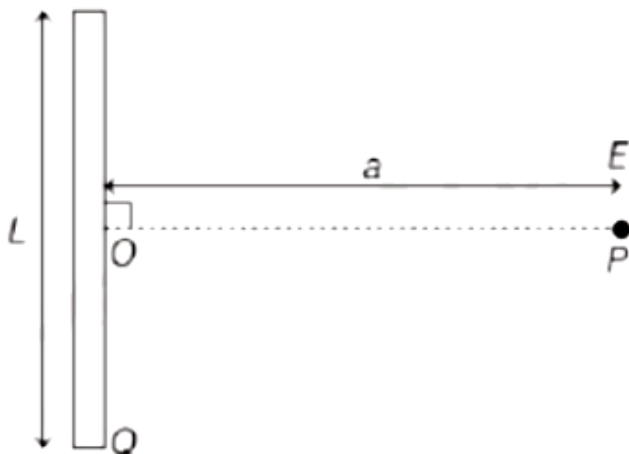
$$\Rightarrow t^2 = \frac{4}{2.34}$$

$$\Rightarrow t = \frac{2}{\sqrt{2.34}}$$

Question85

Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is

$$a = \frac{\sqrt{3}}{2}L$$



[26 Feb 2021 Shift 1]

Options:

A. $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$

B. $\frac{Q}{3\pi\epsilon_0 L^2}$

C. $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$

D. $\frac{Q}{4\pi\epsilon_0 L^2}$

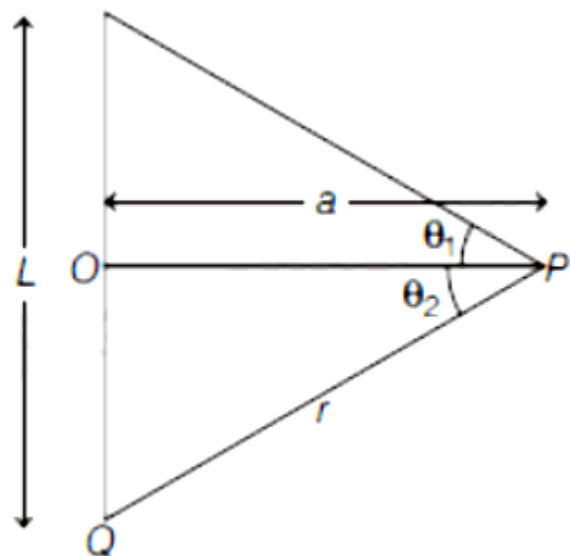
Answer: C

Solution:

Given, length of conductor = L

Charge on conductor = Q

According to figure, $OP = a = \frac{\sqrt{3}}{2}L$, $OQ = \frac{L}{2}$



Let $PQ = r = \sqrt{OP^2 + OQ^2}$

$$\Rightarrow PQ = \sqrt{\left(\frac{\sqrt{3}}{2}L\right)^2 + \left(\frac{L}{2}\right)^2} = \sqrt{\frac{3}{4}L^2 + \frac{L^2}{4}} = L \text{ and } E \text{ be the electric field at point P.}$$

Since, E (due to finite wire) = $\frac{k\lambda}{a}(\sin \phi_1 + \sin \phi_2)$. . . (i)

where, k = Coulomb's constant = $\frac{1}{4\pi\epsilon_0}$

λ = linear charge density = $\frac{Q}{L}$ and $\sin \phi = \sin \phi_2 = \frac{L/2}{L} = \frac{1}{2}$

Substituting the above value in Eq. (i), we get

$$E = \frac{k\lambda}{a} = \frac{1}{4\pi\epsilon_0} \frac{Q}{L \cdot \frac{\sqrt{3}L}{2}} = \frac{1}{2\sqrt{3}\pi\epsilon_0 L^2}$$

Question86

Given below are two statements:

Statement I An electric dipole is placed at the centre of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement II If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius $r (< R)$ is zero but the electric flux passing through this closed spherical surface of radius r is not zero.

In the light of the above statements, choose the correct answer from the options given below.

[26 Feb 2021 Shift 2]

Options:

- A. Both Statement I and Statement II are true.
- B. Statement I is true but Statement II is false.
- C. Both Statement I and Statement II are false.
- D. Statement I is false but Statement II is true.

Answer: B

Solution:

Net charge on electric dipole $= +q - q = 0$

Hence, according to Gauss's law,

$$\text{Electric flux, } \phi = \frac{q_{\text{net}}}{\epsilon_0} = \frac{0}{\epsilon_0} = 0$$

Electric field due to electric dipole is non-zero and varies at point to point.

Hence, statement I is true.

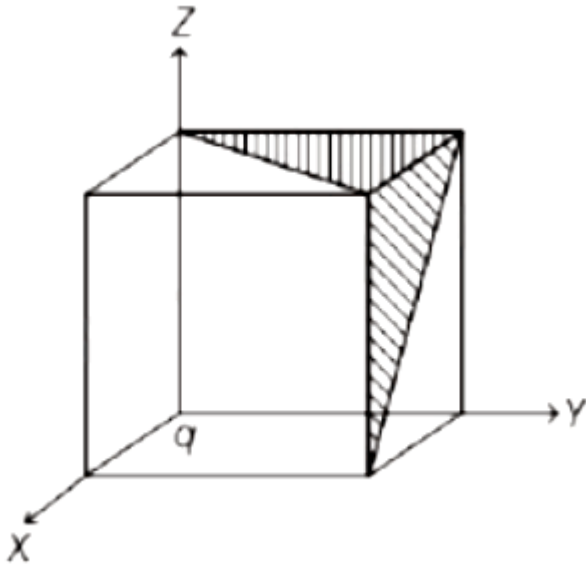
Electric field due to charged solid sphere at a distance r from centre.

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Qr}{R^3} \left[\begin{array}{l} \text{when } r < R \\ R \rightarrow \text{radius} \end{array} \right]$$

which is non-zero. Hence, statement II is false. Hence, option (b) is the correct.

Question 87

A charge q is placed at one corner of a cube as shown in figure. The flux of electrostatic field E through the shaded area is



[25 Feb 2021 Shift 2]

Options:

A. $\frac{q}{48\epsilon_0}$

B. $\frac{q}{4\epsilon_0}$

C. $\frac{q}{8\epsilon_0}$

D. $\frac{q}{24\epsilon_0}$

Answer: D

Solution:

Given, charge q is at one of the corner of the cube.

$\therefore$ Contribution of q in cube will be $q_{\text{enclosed}} = q/8$

As, only 3 faces of cube is allowing the flux lines to pass through it.

$$\therefore \text{Flux } (\phi) = \frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{1}{3} \frac{q/8}{\epsilon_0} = \frac{q}{24\epsilon_0}$$

Question88

The electric field in a region is given $\mathbf{E} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j} \right) \frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area 0.2m^2 (parallel to YZ-plane) to that of the surface of area 0.3m^2 (parallel to XZ - plane) is $a : b$, where $a = \dots\dots\dots$

[Here $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along X, Y and Z -axes, respectively]
[25 Feb 2021 Shift 1]

Answer: 1

Solution:

$$\text{Given, } \mathbf{E} = \frac{3E_0}{5}\hat{i} + \frac{4E_0}{5}\hat{j},$$

$$A_1 = 0.2\text{m}^2\hat{i} \text{ and } A_2 = 0.3\text{m}^2\hat{j}$$

Let ϕ_1 and ϕ_2 be the flux linked with area A_1 and A_2 , respectively. As we know that,

$$\phi = \oint \mathbf{E} \cdot d\mathbf{S} = \mathbf{E} \cdot \mathbf{A}$$

$$\Rightarrow \phi_1 = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j} \right) \cdot 0.2\hat{i} = \frac{3}{5}E_0 \times 0.2$$

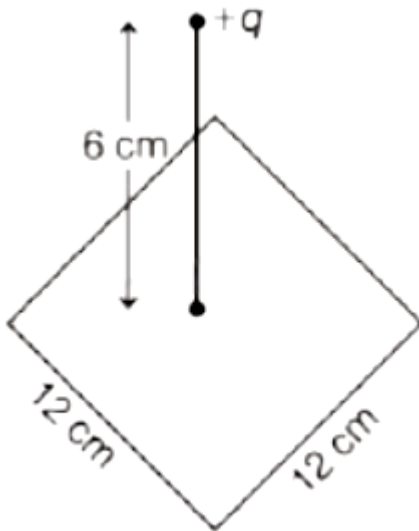
$$\text{and similarly, } \phi_2 = \frac{4}{5}E_0 \times 0.3$$

$$\text{Now, } \frac{\phi_1}{\phi_2} = \frac{\frac{3}{5}E_0 \times 0.2}{\frac{4}{5}E_0 \times 0.3} = \frac{0.6}{1.2} = \frac{1}{2}$$

$$\therefore a = 1$$

Question89

A point charge of $+12\mu\text{C}$ is at a distance 6cm vertically above the centre of a square of side 12cm as shown in figure. The magnitude of the electric flux through the square will be $\dots\dots\dots \times 10^3 \text{N} - \text{m}^2/\text{C}$



[24 Feb 2021 Shift 2]

Answer: 226

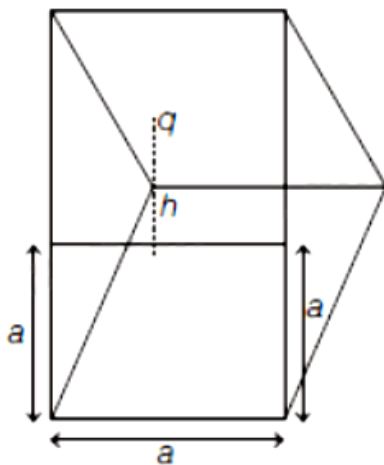
Solution:

Given, charge, $q = 12\mu\text{C} = 12 \times 10^{-6}\text{C}$

Height of charge from surface, $h = 6\text{cm} = 6 \times 10^{-2}\text{m}$

and side of square, $a = 12\text{cm} = 12 \times 10^{-2}\text{m}$

From figure, it is clear that the given square is one of the face of a cube of side 12cm and $+12\mu\text{C}$ charge is placed at its centre. Then, by Gauss's theorem,



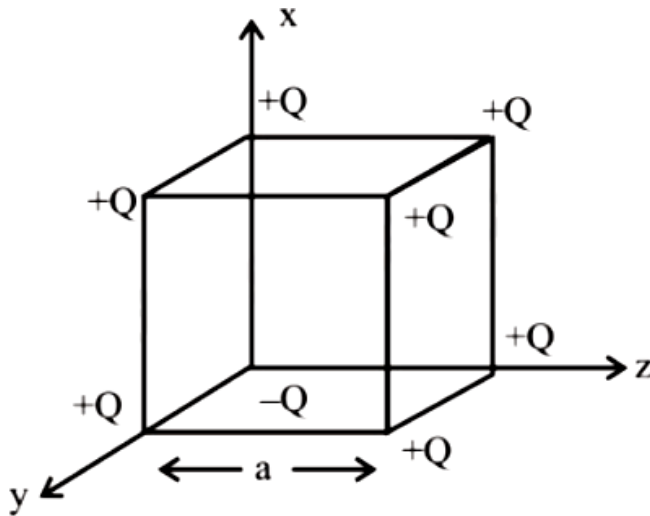
Flux through any face, $\phi = \frac{q}{6\epsilon_0}$

$$\Rightarrow \phi = \frac{12 \times 10^{-6}}{6 \times 8.854 \times 10^{-12}} = 0.226 \times 10^6 \text{N} \cdot \text{m}^2/\text{C}$$

$$= 226 \times 10^3 \text{N} \cdot \text{m}^2/\text{C}$$

Question90

A cube of side ' a ' has point charges +Q located at each of its vertices except at the origin where the charge is $-Q$. The electric field at the centre of cube is :



{24Feb2021 Shift1}

Options:

A. $\frac{3Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$

B. $\frac{-2Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$

C. $\frac{2Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$

D. $\frac{Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$

Answer: B

Solution:

Electric field due to all charges will cancel out except two charges $+Q$ and $-Q$ placed along body diagonal.

$$\vec{E}_{-Q} = \frac{-Q}{4\pi\epsilon_0 \frac{3a^2}{4}} \frac{(\hat{x} + \hat{y} + \hat{z})}{\sqrt{3}}$$

$$\vec{E}_{+Q} = \frac{-Q}{3\pi\epsilon_0 a^2} \frac{(\hat{x} + \hat{y} + \hat{z})}{\sqrt{3}}$$

Net electric field at the centre of cube is

$$\begin{aligned}\vec{E}_{\text{net}} &= \vec{E}_{-Q} + \vec{E}_{+Q} \\ &= \frac{-2Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})\end{aligned}$$

Question91

An infinite number of point charges, each carrying $1\mu\text{C}$ charge, are placed along the Y -axis at $y = 1\text{m}, 2\text{m}, 4\text{m}, 8\text{m}$. The total force on a 1C point charge, placed at the origin, is $x \times 10^3\text{N}$. The value of x to the nearest integer, is

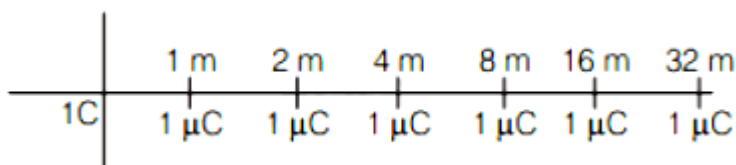
(Take, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\text{N} - \text{m}^2/\text{C}^2$)

[18 Mar 2021 Shift 2]

Answer: 12

Solution:

The 1C charge present in the origin and other $1\mu\text{C}$ charge are placed at $1\text{m}, 2\text{m}, 4\text{m}, 8\text{m}, \dots$



Using the Coulomb's law,

$$F = \frac{kq_1q_2}{r^2}$$

$$F = 9 \times 10^9 \times (1) \times 10^{-6} \left[1 + \frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{8^2} + \dots \right]$$

$$= 9 \times 10^3 \times \left[\frac{1}{1 - \frac{1}{4}} \right] = 12 \times 10^3 \text{N}$$

The value of x to the nearest integer is 12 .

Question92

Find out the surface charge density at the intersection of point X = 3m plane and X -axis, in the region of uniform line charge of 8nC/m lying along the Z -axis in free space [16 Mar 2021 Shift 2]

Options:

A. 0.424nCm^{-2}

B. 47.88nCm^{-2}

C. 0.07nCm^{-2}

D. 4.0nCm^{-2}

Answer: A

Solution:

Given,

Linear charge density, $\lambda = 8 \text{nC/m} = 8 \times 10^{-9} \text{C/m}$

The relation between surface charge density and linear charge density can be given as

$$\frac{2k\lambda}{r} = \frac{\sigma}{\epsilon_0} \dots\dots(i)$$

where,

k = Coulomb's constant,

λ = linear charge density,

σ = surface charge density,
 ϵ_0 = absolute electrical permittivity of free space
 and r = distance.
 Substituting the values in Eq. (i), we get

$$\begin{aligned}\sigma &= \frac{2k\lambda\epsilon_0}{r} \\ &= \frac{2 \times 9 \times 10^9 \times 8 \times 10^{-9} \times 8.85 \times 10^{-12}}{3} \\ &= 0.424 \times 10^{-9} \text{Cm}^{-2} = 0.424 \text{nCm}^{-2}\end{aligned}$$

Question93

An oil drop of radius 2mm with a density 3gcm^{-3} is held stationary under a constant electric field $3.55 \times 10^5 \text{V m}^{-1}$ in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess? (Take, $g = 9.81 \text{m/s}^2$)
[18 Mar 2021 Shift 1]

Options:

- A. 48.8×10^{11}
- B. 1.73×10^{10}
- C. 17.3×10^{10}
- D. 1.73×10^{12}

Answer: B

Solution:

Given, the radius of oil drop, $r = 2\text{mm} = 0.002\text{m}$
 The density of oil drop, $\rho = 3\text{g/cm}^3 = 3 \times 10^3 \text{kg/m}^3$
 The constant electric field, $E = 3.55 \times 10^5 \text{V m}^{-1}$
 The constant electric field, $E = 3.55 \times 10^5 \text{V m}^{-1}$
 Under stationary condition of oil drop,
 $mg = qE$
 $\Rightarrow \rho V g = qE \quad [\because m = \rho V]$
 $\Rightarrow \rho \left(\frac{4}{3} \pi r^3 \right) g = neE$
 $\left[\because V = \frac{4}{3} \pi r^3 \text{ and } q = ne \right]$

$$n = \frac{\rho \left(\frac{4}{3} \pi r^3 \right) g}{eE}$$

$$\Rightarrow n = \frac{3 \times 10^3 \times \left(\frac{4}{3} \pi (0.002)^3 \right) 9.81}{1.6 \times 10^{-19} \times (3.55 \times 10^5)}$$

$$\Rightarrow n = 1.73 \times 10^{10}$$

Question94

The electric field in a region is given by $\mathbf{E} = \frac{2}{5} \mathbf{E}_0 \hat{i} + \frac{3}{5} \mathbf{E}_0 \hat{j}$ with $E_0 = 4.0 \times 10^3 \text{ N/C}$. The flux of this field through a rectangular surface area 0.4 m^2 parallel to the yz -plane is $\text{N} \cdot \text{m}^2 \text{C}^{-1}$.
[17 Mar 2021 Shift 2]

Answer: 640

Solution:

Given,

The electric field in the region, $\mathbf{E} = \frac{2}{5} \mathbf{E}_0 \hat{i} + \frac{3}{5} \mathbf{E}_0 \hat{j}$

Here, $E_0 = 4 \times 10^3 \frac{\text{N}}{\text{C}}$

Area of the rectangular surface, $A = 0.4 \text{ m}^2$

The direction of electric field vector and area vector is same, so the angle between the electric field vector and area vector is 0° .

As we know the expression of electric flux,

$$\phi = \mathbf{E} \cdot \mathbf{A} \cos \theta \dots (i)$$

Here, $\mathbf{E}$ is the electric field vector, and A is the surface area of the surfaces.

Consider the surface parallel to the $Y - Z$ plane, so the area vector, $A = 0.4 \hat{i} \text{ m}^2$

Substituting the values in Eq. (i), we get

$$\phi = \mathbf{E} \cdot \mathbf{A} \cos 0^\circ \Rightarrow \phi = \frac{2}{5} E_0 (0.4)$$

$$\Rightarrow \phi = \frac{2}{5} (4 \times 10^3) (0.4) = 640 \text{ N m}^2 \text{C}^{-1}$$

Hence, the electric flux of the surface parallel to the $Y - Z$ plane is $640 \text{ N m}^2 \text{C}^{-1}$.

Question95

Two identical tennis balls each having mass ' m ' and charge ' q ' are suspended from a fixed point by threads of length ' l '. What is the equilibrium separation when each thread makes a small angle ' θ ' with the vertical ?

[27 Jul 2021 Shift 1]

Options:

A. $x = \left(\frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{\frac{1}{2}}$

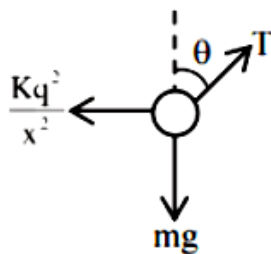
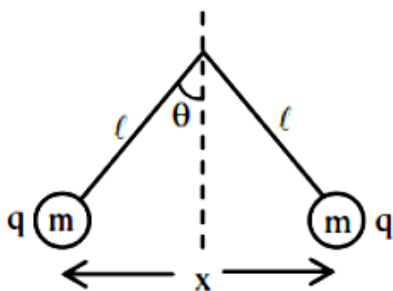
B. $x = \left(\frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{\frac{1}{3}}$

C. $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{\frac{1}{2}}$

D. $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{\frac{1}{2}}$

Answer: B

Solution:



$$T \cos \theta = mg$$

$$T \sin \theta = \frac{kq^2}{x^2}$$

$$\tan \theta = \frac{kq^2}{x^2 mg}$$

$$\text{as } \tan \theta \approx \sin \theta \approx \frac{x}{2L}$$

$$\frac{x}{2L} = \frac{K q^2}{x^2 mg}$$

$$x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$$

Question96

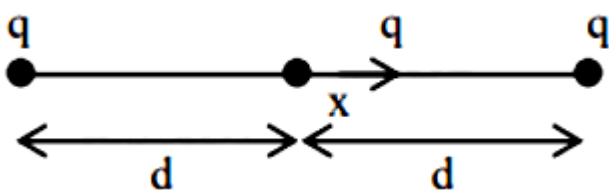
A particle of mass 1mg and charge q is lying at the mid-point of two stationary particles kept at a distance ' 2 m ' when each is carrying same charge ' q '. If the free charged particle is displaced from its equilibrium position through distance ' x ' ($x \ll 1\text{m}$). The particle executes SHM. Its angular frequency of oscillation will be _____

$$\times 10^5 \text{ rad/s if } q^2 = 10\text{C}^2$$

[25 Jul 2021 Shift 1]

Answer: 6

Solution:



Net force on free charged particle

$$F = \frac{kq^2}{(d+x)^2} - \frac{kq^2}{(d-x)^2}$$

$$F = -kq^2 \left[\frac{4dx}{(d^2 - x^2)^2} \right]$$

$$a = -\frac{4kq^2 d}{m} \left(\frac{x}{d^4} \right)$$

$$a = -\left(\frac{4kq^2}{md^3} \right) x$$

So, angular frequency

$$\omega = \sqrt{\frac{4kq^2}{md^3}}$$

$$\omega = \sqrt{\frac{4 \times 9 \times 10^9 \times 10}{1 \times 10^{-6} \times 1^3}}$$

$$\omega = 6 \times 10^8 \text{ rad/sec}$$

Question97

An electric dipole is placed on x-axis in proximity to a line charge of linear charge density $3.0 \times 10^{-6} \text{ C/m}$. Line charge is placed on z-axis and positive and negative charge of dipole is at a distance of 10mm and 12mm from the origin respectively. If total force of 4N is exerted on the dipole, find out the amount of positive or negative charge of the dipole.

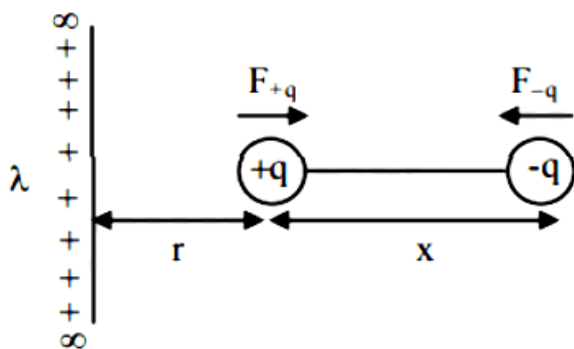
[22 Jul 2021 Shift 2]

Options:

- A. 815.1nC
- B. $8.8 \mu\text{C}$
- C. 0.485mC
- D. $4.44 \mu\text{C}$

Answer: D

Solution:



$$r = 10\text{mm}, x = 2,$$

$$|\vec{F}_q| = \frac{2k\lambda}{r} \cdot q$$

$$\left| \vec{F}_{-q} \right| = \frac{2k\lambda}{r+x} \cdot q$$

$$\Rightarrow \left| \vec{F}_{\text{net}} \right| = \frac{2k\lambda q}{r} - \frac{2k\lambda q}{r+x}$$

$$\left| \vec{F}_{\text{net}} \right| = \frac{2k\lambda q \cdot x}{r(r+x)}$$

$$4 = \frac{2 \times 9 \times 10^9 \times 3 \times 10^{-6} \times q \times 2\text{mm}}{10\text{mm} \cdot 12\text{mm}}$$

$$\Rightarrow q = 4.44 \mu\text{C}$$

Question98

A certain charge Q is divided into two parts q and $(Q - q)$. How should the charges Q and q be divided so that q and $(Q - q)$ placed at a certain distance apart experience maximum electrostatic repulsion?

[20 Jul 2021 Shift 1]

Options:

A. $Q = \frac{q}{2}$

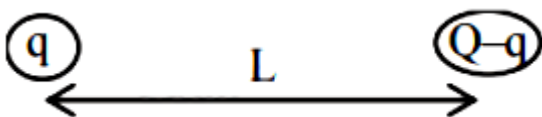
B. $Q = 2q$

C. $Q = 4q$

D. $Q = 3q$

Answer: B

Solution:



$$F_q = \frac{kq(Q-q)}{L^2} = \frac{k}{L^2}(qQ - q^2)$$

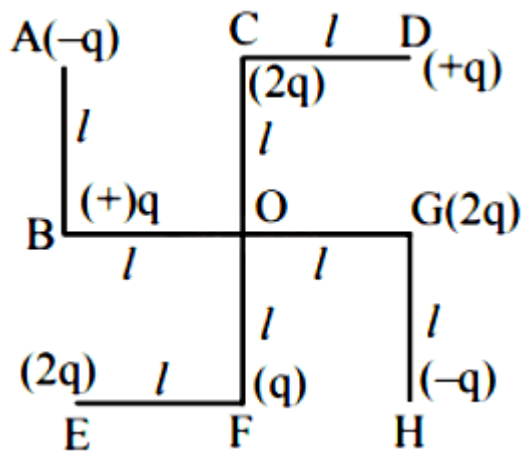
$$\frac{dF}{dq} = 0 \text{ when force is maximum}$$

$$\frac{dF}{dq} = \frac{k}{L^2}[Q - 2q] = 0$$

$$\Rightarrow Q - 2q = 0 \Rightarrow Q = 2q$$

Question99

What will be the magnitude of electric field at point O as shown in figure? Each side of the figure is l and perpendicular to each other?



[27 Jul 2021 Shift 2]

Options:

- A. $\frac{1}{4\pi\epsilon_0} \frac{q}{l^2}$
- B. $\frac{1}{4\pi\epsilon_0} \frac{q}{(2l^2)} (2\sqrt{2} - 1)$
- C. $\frac{q}{4\pi\epsilon_0 (2l)^2}$
- D. $\frac{1}{4\pi\epsilon_0} \frac{2q}{2l^2} (\sqrt{2})$

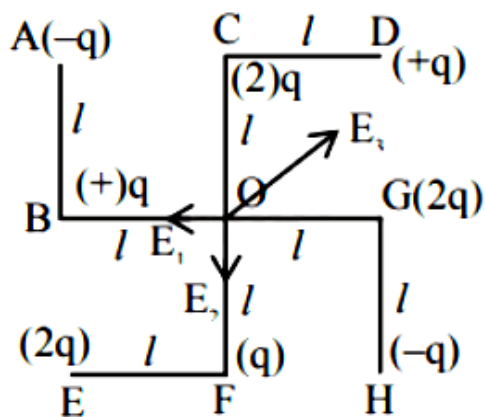
Answer: B

Solution:

$$E_1 = \frac{kq}{l^2} = E_2$$

$$E_3 = \frac{kq}{(\sqrt{2}l)^2} = \frac{kq}{2l^2}$$

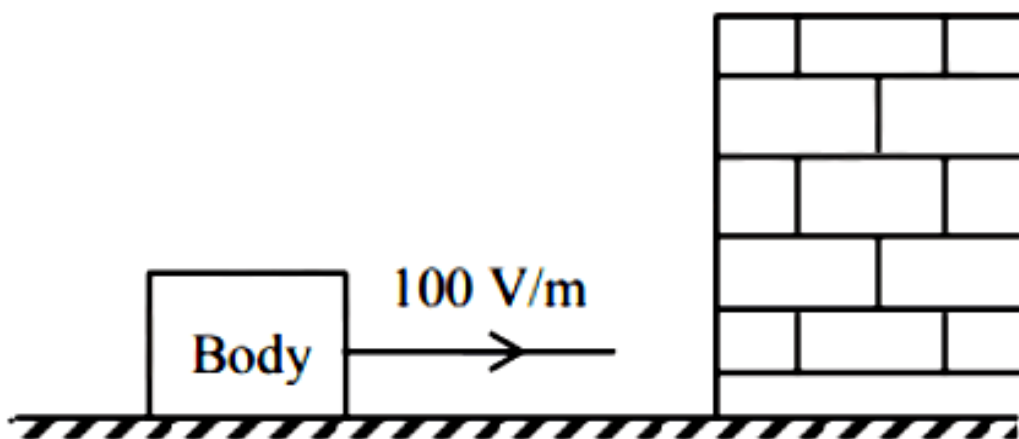
$$E = \frac{\sqrt{2}kq}{l^2} - \frac{kq}{2l^2} = \frac{kq}{2l^2} (2\sqrt{2} - 1)$$



Question100

A body having specific charge $8\mu\text{C/g}$ is resting on a frictionless plane at a distance 10cm from the wall (as shown in the figure). It starts moving towards the wall when a uniform electric field of 100V/m is applied horizontally towards the wall.

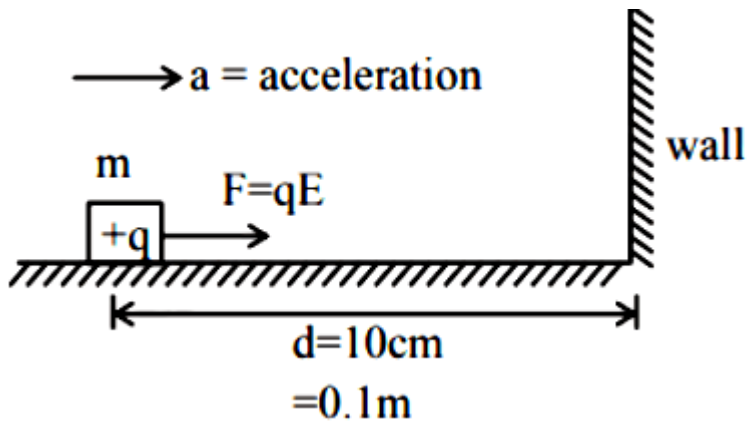
If the collision of the body with the wall is perfectly elastic, then the time period of the motion will be _____ s.



[20 Jul 2021 Shift 1]

Answer: 1

Solution:



$$F = ma$$

$$qE = ma$$

$$a = \frac{qE}{m}$$

$$\text{Now } d = \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2d}{a}}$$

$$t = \sqrt{\frac{2d}{\left(\frac{qE}{m}\right)}}$$

$$t = \sqrt{\frac{2 \times 0.1}{\left(\frac{8 \times 10^{-6}}{10^{-3}}\right) \times 100}} = \frac{1}{2}$$

$$\therefore \text{Time period} = 2t = 1\text{sec}$$

Question101

The total charge enclosed in an incremental volume of $2 \times 10^{-9}\text{m}^3$ located at the origin is $n\text{C}$, if electric flux density of its field is found as $\mathbf{D} = e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k} \text{C/m}^2$.
[22 Jul 2021 Shift 2]

Answer: 4

Solution:

Electric flux density

$$(D) = \frac{\text{charge}}{\text{Area}} \times \hat{r} = \frac{Q}{4\pi r^2} \hat{r} = \epsilon_0 \left(\frac{Q}{4\pi \epsilon_0 r^2} \hat{r} \right)$$

$$\Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k}}{\epsilon_0}$$

Also by Gauss's law

$$\frac{\rho}{\epsilon_0} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \vec{E} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \frac{\vec{D}}{\epsilon_0}$$

$$\Rightarrow \rho = \frac{\partial}{\partial x}(e^{-x} \sin y) + \frac{\partial}{\partial y}(-e^{-x} \cos y) + \frac{\partial}{\partial z}(2z)$$

$$\rho = -e^{-x} \sin y + e^{-x} \sin y + 2$$

$$\text{At origin } \rho = -e^{-0} \sin 0 + e^{-0} \sin 0 + 2$$

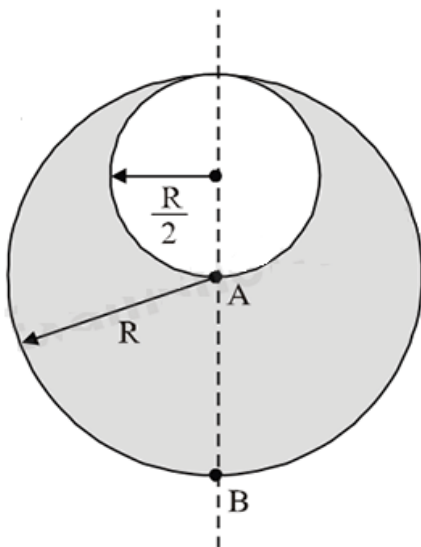
$$\rho = 2 \text{ C/m}^3$$

$$\text{Charge} = \rho \times \text{volume} = 2 \times 2 \times 10^{-9} = 4 \times 10^{-9} = 4 \text{ nC}$$

Question102

Consider a sphere of radius R which carries a uniform charge density ρ . If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the

ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is:



[9 Jan. 2020, I]

Options:

A. $\frac{21}{34}$

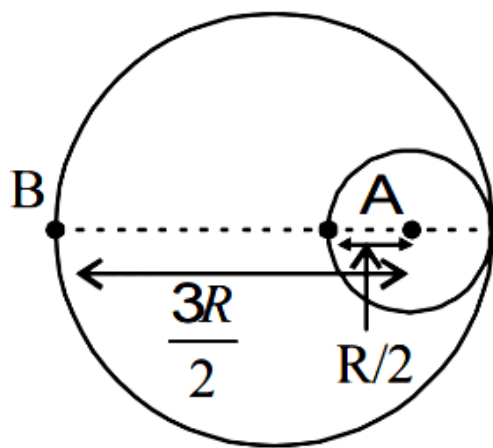
B. $\frac{18}{34}$

C. $\frac{17}{54}$

D. $\frac{18}{54}$

Answer: B

Solution:



Electric field at A $\left(R' = \frac{R}{2}\right)$

$$E_A \cdot d s = \frac{q}{\epsilon_0}$$

$$\Rightarrow \vec{E}_A = \frac{\rho \times \frac{4}{3}\pi \left(\frac{R}{2}\right)^3}{\epsilon_0 \cdot 4\pi \left(\frac{R}{2}\right)^2}$$

$$\Rightarrow \vec{E}_A = \frac{\sigma(R/2)}{3\epsilon_0} = \left(\frac{\sigma R}{6\epsilon_0}\right)$$

Electric fields at 'B'

$$\vec{E}_B = \frac{k \times \rho \times \frac{4}{3}\pi R^3}{R^2} - \frac{k \times \rho \times \frac{4}{3}\pi \left(\frac{R}{2}\right)^3}{\left(\frac{3R}{2}\right)^2}$$

$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\epsilon_0} - \left(\frac{1}{4\pi\epsilon_0}\right) \frac{(\sigma)}{\left(\frac{3R}{2}\right)^2} \frac{4\pi}{3} \left(\frac{R}{2}\right)^3$$

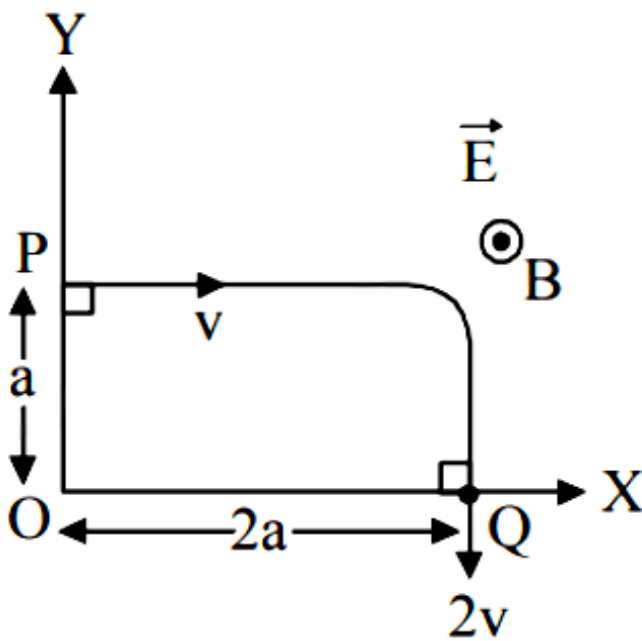
$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\epsilon_0} - \frac{\sigma R}{54\epsilon_0}$$

$$\Rightarrow E_B = \frac{17}{54} \left(\frac{\sigma R}{\epsilon_0} \right)$$

$$\left| \frac{E_A}{E_B} \right| = \frac{1 \times 54}{6 \times 17} = \left(\frac{9}{17} \right) = \frac{9}{17} \times \frac{2}{2} = \frac{18}{34}$$

Question103

An electric dipole of moment $\vec{p} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29} \text{C} \cdot \text{m}$ is at the origin $(0,0,0)$. The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to:



[9 Jan. 2020, I]

Options:

A. $(+\hat{i} - 3\hat{j} - 2\hat{k})$

B. $(-\hat{i} + 3\hat{j} - 2\hat{k})$

C. $(+\hat{i} + 3\hat{j} - 2\hat{k})$

D. $(-\hat{i} - 3\hat{j} + 2\hat{k})$

Answer: C

Solution:

Since $\vec{r} \cdot \vec{p} = 0$

$\vec{E}$ must be antiparallel to $\vec{p}$

$\therefore \hat{E}$ is parallel to $(+\hat{i} + 3\hat{j} - 2\hat{k})$

Question 104

A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\hat{i}$ and $-2v\hat{j}$. Then which of the following statements (A, B, C, D) are the correct? (Trajectory shown is schematic and not to scale)

(A) $E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4} \left(\frac{mv^2}{a} \right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is $2 m a v$.

[9 Jan. 2020, I]

Options:

A. (A), (C), (D)

B. (B), (C), (D)

C. (A), (B), (C)

D. (A), (B), (C), (D)

Answer: C

Solution:

(A) By work energy theorem

$$W_{\text{mg}} + W_{\text{ele}} = \frac{1}{2}m(2v)^2 - \frac{1}{2}m(v)^2$$

$$0 + qE_0 2a = \frac{3}{2}mv^2 \Rightarrow E_0 = \frac{3mv^2}{4qa}$$

(B) Rate of work done at P = power of electric force = $qE_0 V = \frac{3mv^3}{4a}$

(C) At, Q, $\frac{dw}{dt} = 0$ for both the fields

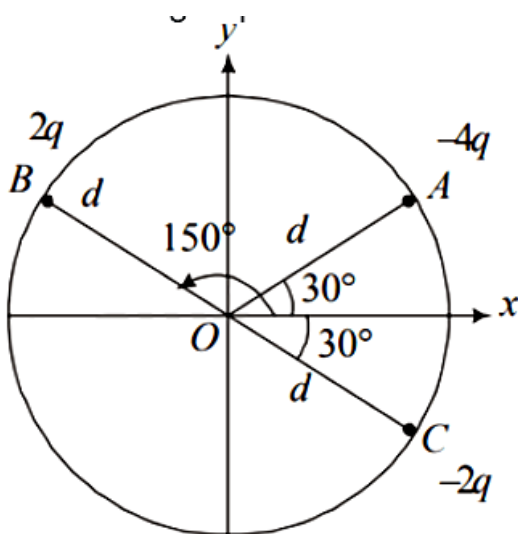
(D) The difference of magnitude of angular momentum of the particle at P and Q,

$$\Delta \vec{L} = (-m2v2a\hat{k}) - (-mva\hat{k})$$

$$|\Delta \vec{L}| = 3mva$$

Question105

Three charged particles



A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d . The charged particles A, C

and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x -direction is:
[8 Jan. 2020, I]

Options:

A. $\frac{\sqrt{3}q}{\pi \epsilon_0 d^2}$

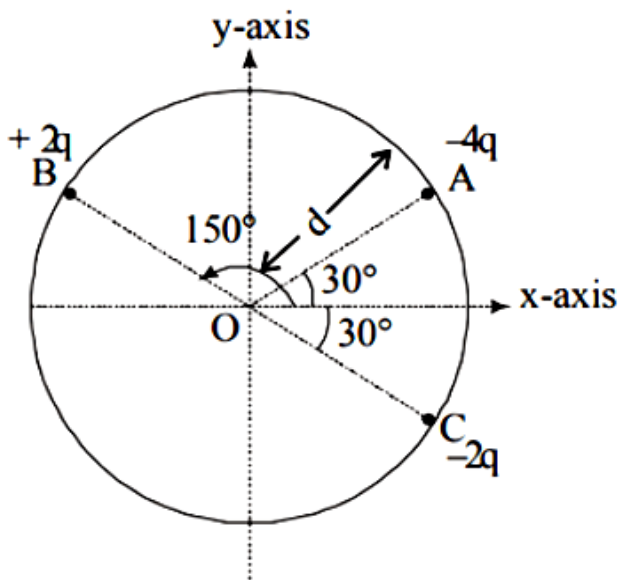
B. $\frac{2\sqrt{3}q}{\pi \epsilon_0 d^2}$

C. $\frac{\sqrt{3}q}{4\pi \epsilon_0 d^2}$

D. $\frac{3\sqrt{3}q}{4\pi \epsilon_0 d^2}$

Answer: A

Solution:



Electric field due to charge +2q at centre O

$$\vec{E}_1 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

Electric field due to charge -2q at centre O

$$\vec{E}_2 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

Electric field due to charge -4q at centre O

$$\vec{E}_3 = \frac{1}{4\pi\epsilon_0} \times \frac{4q}{d^2} \left[\frac{\sqrt{3}\hat{i} + \hat{j}}{2} \right]$$

∴ Net electric field at point O

$$\vec{E}_0 = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \frac{\sqrt{3}q}{\pi\epsilon_0 d^2} \hat{i}$$

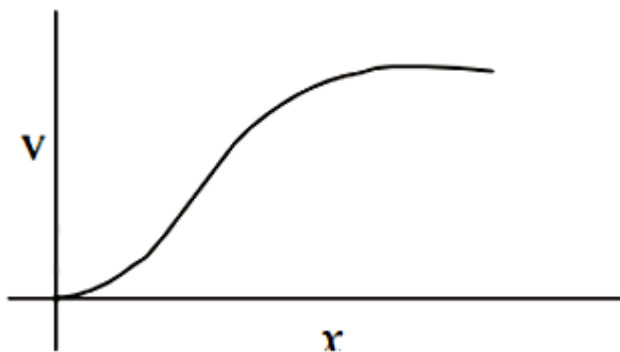
Question106

A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)

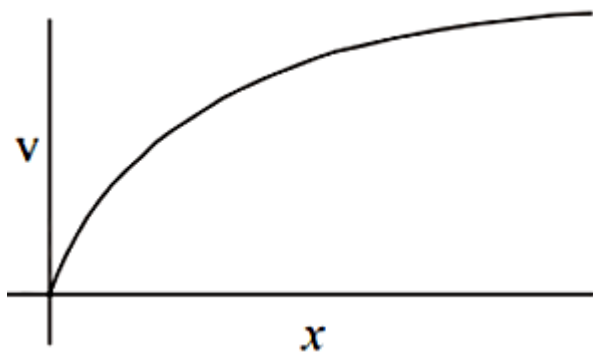
[8 Jan. 2020, II]

Options:

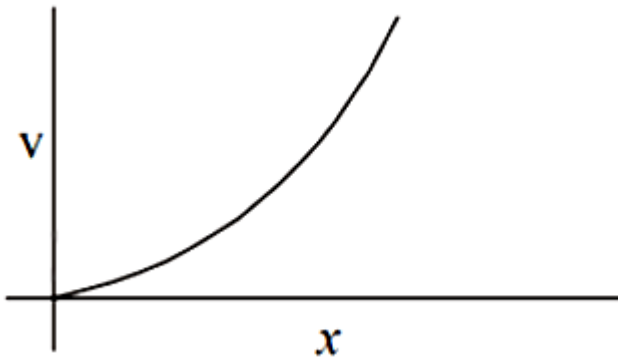
A.



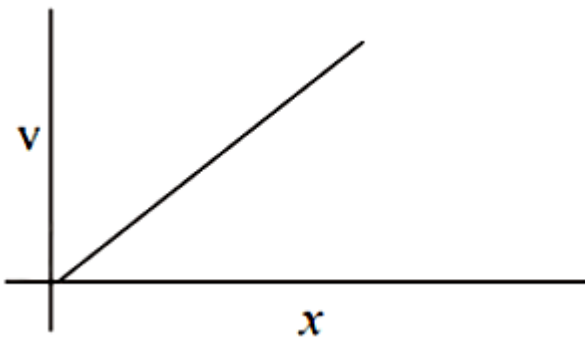
B.



C.

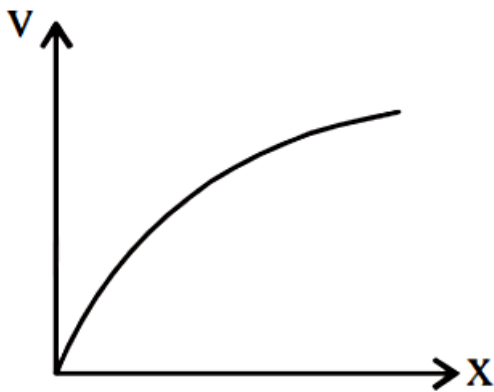


D.



Answer: B

Solution:



Using

$$v^2 - u^2 = 2aS \dots\dots(i)$$

Here, $u = 0$, $s = x$

Also, $F_{\text{electric}} = ma$

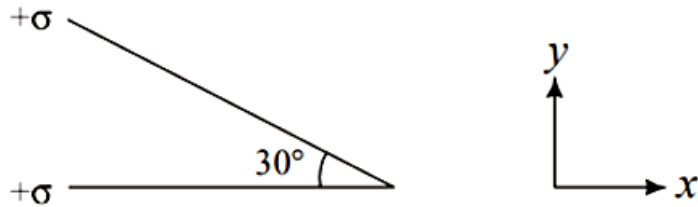
$$\Rightarrow qE = ma \Rightarrow a = \frac{qE}{m} \Rightarrow a = \frac{qE}{m}$$

Substituting the values in (i) we get

$$v^2 = \frac{2qE}{m}x$$

Question107

Two infinite planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 30° . The electric field in the region shown between them is given by:



[7 Jan. 2020, I]

Options:

A. $\frac{\sigma}{2\epsilon_0} \left[(1 + \sqrt{3})\hat{y} - \frac{\hat{x}}{2} \right]$

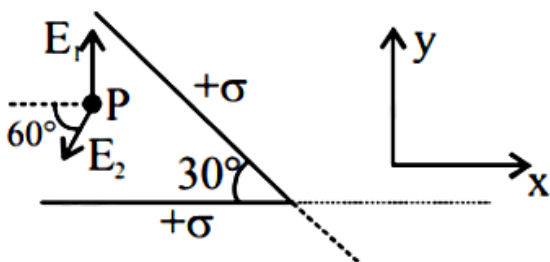
B. $\frac{\sigma}{\epsilon_0} \left[\left(1 + \frac{\sqrt{3}}{2} \right) \hat{y} + \frac{\hat{x}}{2} \right]$

C. $\frac{\sigma}{2\epsilon_0} \left[(1 + \sqrt{3})\hat{y} + \frac{\hat{x}}{2} \right]$

D. $\frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$

Answer: D

Solution:



From figure,

$$\vec{E}_1 = \frac{\sigma}{2\epsilon_0} \hat{y} \text{ and } \vec{E}_2 = \frac{\sigma}{2\epsilon_0} (-\cos 60^\circ \hat{x} - \sin 60^\circ \hat{y})$$

$$= \frac{\sigma}{2\epsilon_0} \left(-\frac{1}{2}\hat{x} - \frac{\sqrt{3}}{2}\hat{y} \right)$$

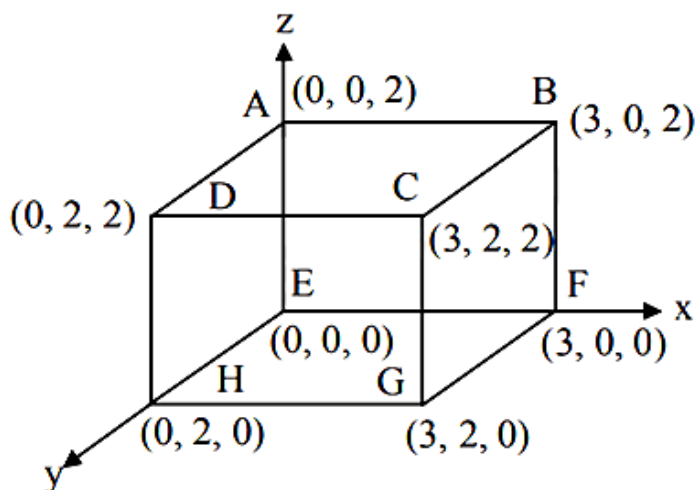
Electric field in the region shown in figure (P)

$$\vec{E}_P = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{2\epsilon_0} \left[-\frac{1}{2}\hat{x} + \left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} \right]$$

$$\text{or, } \vec{E}_P = \frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$$

Question108

An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j} \text{ N/C}$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_1 and ϕ_{11} respectively. The difference between $(\phi_1 - \phi_{11})$ is (in $\text{N m}^2/\text{C}$) _____.



[9 Jan 2020, II]

Answer: -48

Solution:

Flux of electric field $\vec{E}$ through any area $\vec{A}$ is defined as

$$\phi = \int \vec{E} \cdot \vec{A} \cos \theta$$

Here, θ = angle between electric field and area vector of a surface

For surface ABCD Angle, $\theta = 90^\circ$

$$\therefore \phi_1 = \int \vec{E} \cdot \vec{A} \cos 90^\circ = 0$$

$$\text{For surface BCGF } \phi_n = \int \vec{E} \cdot d\vec{A}$$

$$\therefore \phi_{11} = [4 \times \hat{i} - (y^2 + 1)\hat{j}] \cdot 4\hat{i} = 16x$$

$$\phi_{11} = 48 \frac{\text{Nm}^2}{\text{C}}$$

$$\phi_1 - \phi_{11} = -48$$

Question 109

In finding the electric field using Gauss law the formula $|\vec{E}| = \frac{q_{\text{enc}}}{\epsilon_0 A}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation?

[8 Jan 2020, I]

Options:

A. Only when the Gaussian surface is an equipotential surface and $|\vec{E}|$ is constant on the surface.

B. equipotential surface and $|\vec{E}|$ is constant on the surface.

C. Only when $|\vec{E}| = \text{constant}$ on the surface.

D. For any choice of Gaussian surface.

Answer: A

Solution:

According to Gauss' Law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

Where $\vec{E}$ is the electric field and $d\vec{A}$ is the elemental areal vector for any small elemental area on the gaussian surface.

$$\left| \vec{E}_1 \right| \left| d\vec{A} \right| \cos \theta = \frac{q_{enc}}{\epsilon_0}$$

Given that the formula $\left| \vec{E} \right| = \frac{q_{enc}}{\epsilon_0 \left| A \right|}$ is applicable.

$$\therefore \left| \vec{E} \right| \left| A \right| = \frac{q_{enc}}{\epsilon_0}$$

$$\therefore \left| \vec{E} \right| \left| A \right| = \oint \left| \vec{E}_1 \right| \left| d\vec{A} \right| \cos \theta$$

This relation will be valid when $\vec{E}_1$ is a constant and $\left| \vec{E}_1 \right| = \left| \vec{E} \right|$ and also when $\theta = 0^\circ$

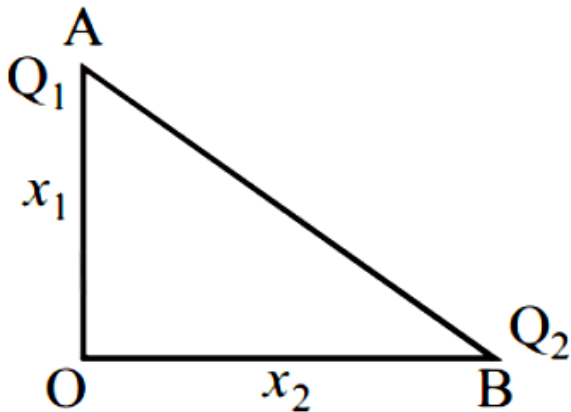
$$\oint \left| \vec{E}_1 \right| \left| d\vec{A} \right| \cos \theta \text{ becomes } \left| \vec{E} \right| \oint \left| d\vec{A} \right| = \left| \vec{E} \right| \left| A \right|$$

The condition $\theta = 0$ means that the components of electric field parallel to the elemental surface are $E_1 \sin \theta = 0$.

This means that the potential gradient over the surface is zero. Hence the surface is equipotential.

Question 110

Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to :



[Sep. 06, 2020 (I)]

Options:

A. $\frac{x_1^3}{x_2^3}$

B. $\frac{x_2}{x_1}$

C. $\frac{x_1}{x_2}$

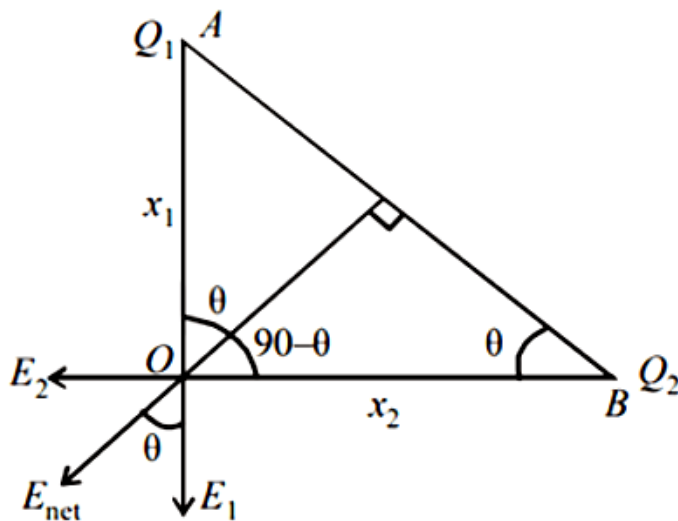
D. $\frac{x_2^2}{x_1^2}$

Answer: C

Solution:

Electric field due charge Q_2 , $E_2 = \frac{kQ_2}{x_2^2}$

Electric field due charge Q_1 , $E_1 = \frac{kQ_1}{x_1^2}$



From figure,

$$\tan \theta = \frac{E_2}{E_1} = \frac{x_1}{x_2} \Rightarrow \frac{kQ_2}{x_2^2 \times \frac{kQ_1}{x_1^2}} = \frac{x_1}{x_2}$$

$$\Rightarrow \frac{Q_2 x_1^2}{Q_1 x_2^2} = \frac{x_1}{x_2} \Rightarrow \frac{Q_2}{Q_1} = \frac{x_2}{x_1} \text{ or, } \frac{Q_1}{Q_2} = \frac{x_1}{x_2}$$

Question111

Consider the force F on a charge ‘ q ’ due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ‘ q ’ is placed at distance r from the centre of the shell?

[Sep. 06, 2020 (II)]

Options:

A. $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$

B. $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2} > F > 0$ for $r < R$

C. $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r > R$

D. $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for all r

Answer: C

Solution:

For spherical shell

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \text{ (if } r \geq R \text{)}$$

$$= 0 \text{ (if } r < R \text{)}$$

Force on charge in electric field, $F = qE$

$$\therefore F = 0 \text{ (For } r < R \text{)}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \text{ (For } r > R \text{)}$$

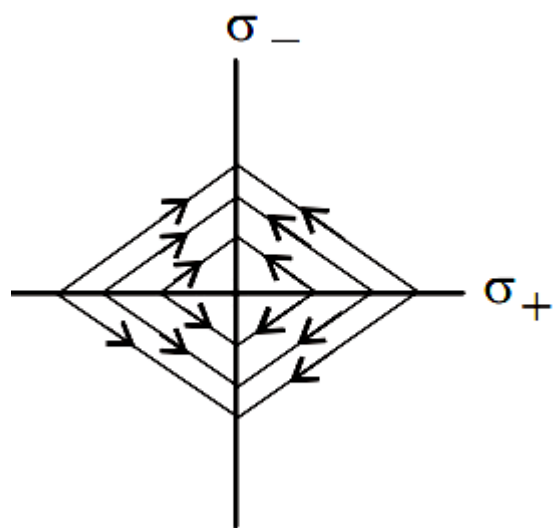
Question112

Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_- , where $|\sigma_+| > |\sigma_-|$, intersect at right angle. Which of the following best represents the electric field lines for this system?

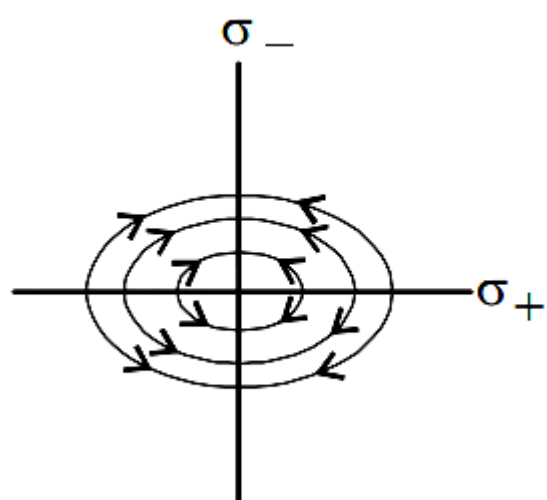
[Sep. 04, 2020 (I)]

Options:

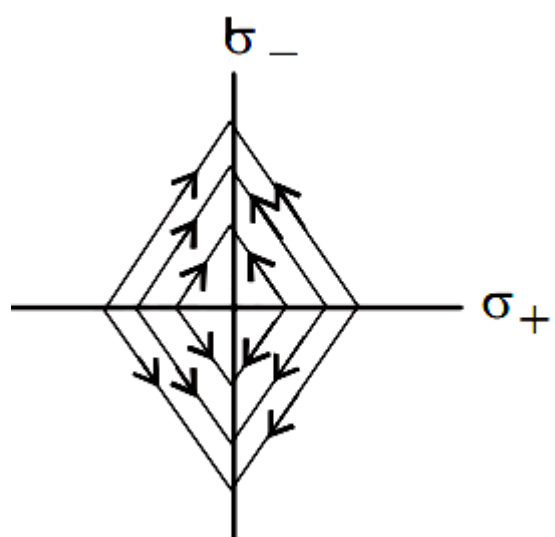
A.



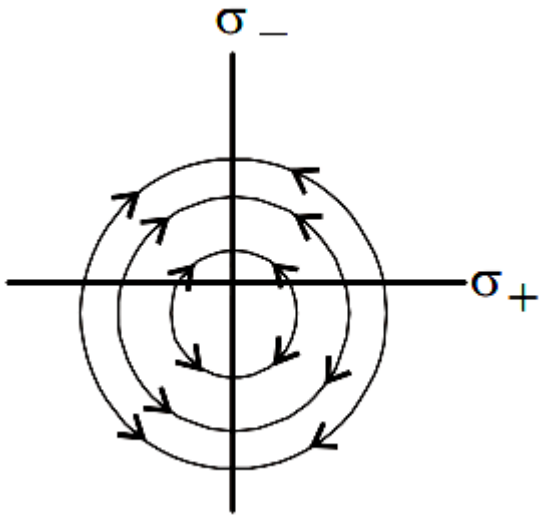
B.



C.



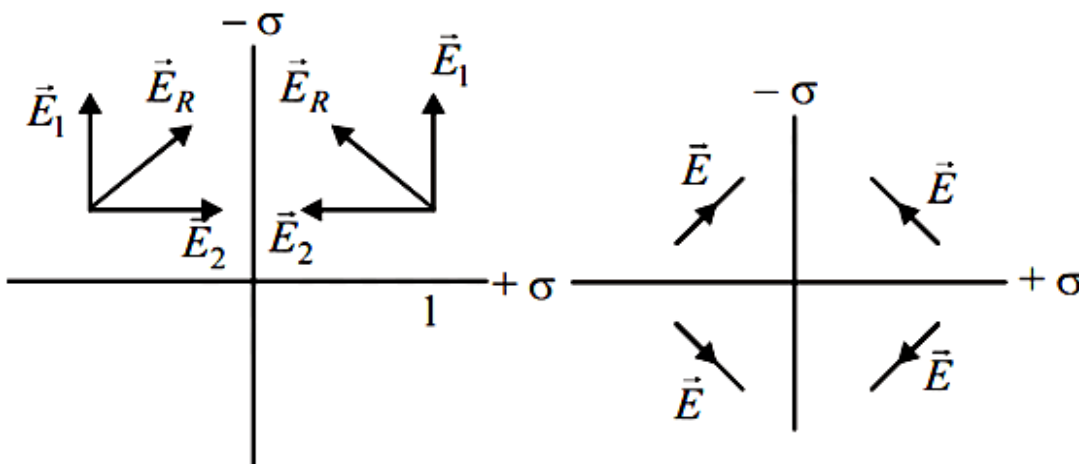
D.



Answer: C

Solution:

The electric field produced due to uniformly charged infinite plane is uniform. So option (b) and (d) are wrong. And +ve charge density σ_+ is bigger in magnitude so its field along Y direction will be bigger than field of -ve charge density σ_- in X direction. Hence option (c) is correct.



Question113

A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x -direction, where a and E_0 are constants. Initially the particle was at rest at $x = 0$. Other than the initial position the kinetic energy of the particle becomes zero when the

distance of the particle from the origin is :
[Sep. 04,2020 (II)]

Options:

A. a

B. $\sqrt{\frac{2}{a}}$

C. $\sqrt{\frac{3}{a}}$

D. $\sqrt{\frac{1}{a}}$

Answer: C

Solution:

$$f = qE = qE_0(1 - ax^2)$$

$$\Rightarrow mv \frac{dv}{dx} = qE_0(1 - ax^2)$$

$$v dv = \frac{qE_0(1 - ax^2)}{m} dx$$

∫ both sides;

$$\int_0^v v dv = \frac{qE_0}{m} \int_0^x (1 - ax^2) dx$$

$$\frac{v^2}{2} = \frac{qE_0}{m} \left(x - \frac{ax^3}{3} \right)$$

K.E of particle becomes zero

$$\frac{1}{2}mv^2 = qe_0 \cdot \left(x - \frac{ax^3}{3} \right) = 0$$

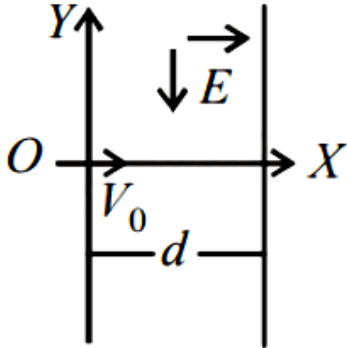
$$x = \frac{ax^3}{3}$$

$$\frac{3}{a} = x^2$$

$$x = \sqrt{\frac{3}{a}}$$

Question114

A charged particle (mass m and charge q) moves along X axis with velocity V_0 . When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E \hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is :



[Sep. 02, 2020 (I)]

Options:

A. $y = \frac{qEd}{mV_0^2}(x - d)$

B. $y = \frac{qEd}{mV_0^2} \left(\frac{d}{2} - x \right)$

C. $y = \frac{qEd}{mV_0^2}x$

D. $y = \frac{qEd^2}{mV_0^2}x$

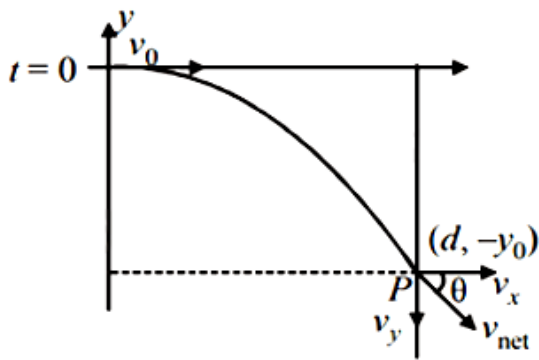
Answer: B

Solution:

$$F_x = 0, a_x = 0, (v)_x = \text{constant}$$

$$\text{Time taken to reach at 'P'} = \frac{d}{v_0} = t_0 \text{ (let)(i)}$$

$$(\text{Along } -y), y_0 = 0 + \frac{1}{2} \cdot \frac{qE}{m} \cdot t_0^2 \text{(ii)}$$



$$\tan \theta = \frac{v_y}{v_x} = \frac{qE t_0}{m \cdot v_0}, \left(t = \frac{d}{v_0} \right)$$

$$\tan \theta = \frac{qE d}{m \cdot v_0^2}, \text{ Slope} = \frac{-qE d}{m v_0^2}$$

No electric field $\Rightarrow F_{\text{net}} = 0, \vec{v} = \text{const.}$

$$y = mx + c, \left\{ \begin{array}{l} m = \frac{qE d}{m v_0^2} \\ (d, -y_0) \end{array} \right\}$$

$$-y_0 = \frac{-qE d}{m v_0^2}, d + c \Rightarrow c = -y_0 + \frac{qE d^2}{m v_0^2}$$

$$y = \frac{-qE d}{m v_0^2} x - y_0 + \frac{qE d^2}{m v_0^2}$$

$$y_0 = \frac{1}{2} \cdot \frac{qE}{m} \left(\frac{d}{v_0} \right)^2 = \frac{1}{2} \frac{qE d^2}{m v_0^2}$$

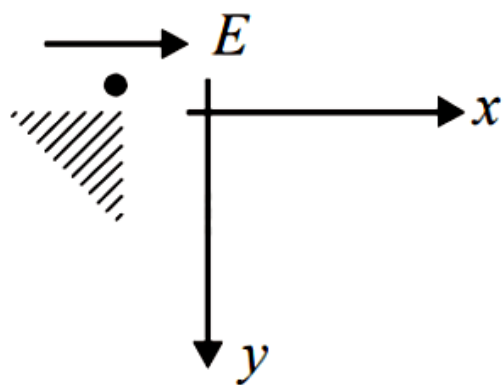
$$y = \frac{-qE d x}{m v_0^2} - \frac{1}{2} \frac{qE d^2}{m v_0^2} + \frac{qE d^2}{m v_0^2}$$

$$y = \frac{-qE d}{m v_0^2} + \frac{1}{2} \frac{qE d^2}{m v_0^2} \Rightarrow y = \frac{qE d}{m v_0^2} \left(\frac{d}{2} - x \right)$$

Question115

A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass ? (Curves are

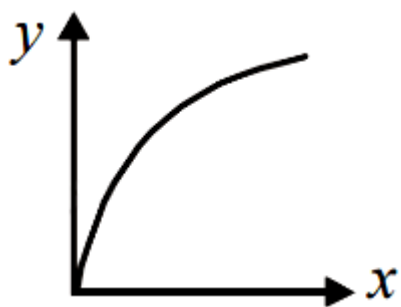
drawn schematically and are not to scale).



[Sep. 02, 2020 (I)]

Options:

A.



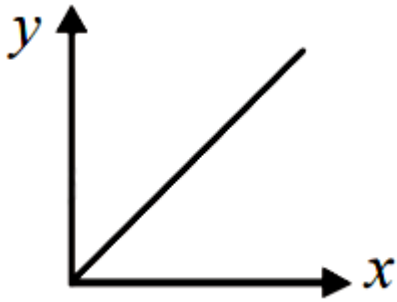
B.



C.



D.



Answer: D

Solution:

Net force acting on the particle,

$$\vec{F} = qE \hat{i} + mg \hat{j}$$

Net acceleration of particle is constant, initial velocity is zero therefore path is straight line.

$$a_x = \frac{2E}{m}$$

$$a_y = g$$

$$a = \sqrt{\left(\frac{2E}{m}\right)^2 + g^2}$$

Question116

A particle of mass m and charge q has an initial velocity $\vec{v} = v_0 \hat{j}$. If an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ act on the particle, its speed will double after a time:
[2020, I]

Options:

A. $\frac{2mv_0}{qE_0}$

B. $\frac{3mv_0}{qE_0}$

C. $\frac{\sqrt{3}mv_0}{qE_0}$

–

D. $\frac{\sqrt{2}mv_0}{qE_0}$

Answer: C

Solution:

Electric field will increase the speed of particle in x direction.

$$F_x = qE$$

$$\therefore a = \frac{qE}{m}$$

$$\text{Also } V_x = at = \frac{qE}{m}t$$

$$v_x^2 + v_y^2 = v^2$$

$$\Rightarrow v_x^2 + v_0^2 = (2v_0)^2$$

$$\Rightarrow V_x = \sqrt{3}v_0$$

$$\therefore \frac{qE}{m}t = \sqrt{3}v_0$$

$$\Rightarrow t = \frac{\sqrt{3}mv_0}{qE_0}$$

Question117

Two identical electric point dipoles have dipole moments $\vec{P}_1 = P\hat{i}$ and $\vec{P}_2 = -P\hat{i}$ and are held on the x axis at distance 'a' from each other.

When released, they move along x axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is :

[Sep. 06, 2020 (II)]

Options:

A. $\frac{P}{a} \sqrt{\frac{1}{\pi\epsilon_0 ma}}$

B. $\frac{P}{a} \sqrt{\frac{1}{2\pi\epsilon_0 ma}}$

C. $\frac{P}{a} \sqrt{\frac{2}{\pi\epsilon_0 ma}}$

$$D. \frac{p}{a} \sqrt{\frac{2}{2\pi\epsilon_0 ma}}$$

Answer: B

Solution:

Let v be the speed of dipole.

Using energy conservation

$$K_i + U_i = K_f + U_f$$

$$\Rightarrow 0 - \frac{2k \cdot p_1}{r^3} p_2 \cos(180^\circ) = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + 0$$

$$\left(\because \text{Potential energy of interaction between dipole} = \frac{-2p_1 p_2 \cos \theta}{4\pi \epsilon_0 r^3} \right)$$

$$\Rightarrow mv^2 = \frac{2kp_1 p_2}{r^3} \Rightarrow v = \sqrt{\frac{2kp_1 p_2}{mr^3}}$$

When $p_1 = p_2 = p$ and $r = a$

$$v = \frac{p}{a} \sqrt{\frac{1}{2\pi \epsilon_0 ma}}$$

Question118

Three charges $+Q$, q , $+Q$ are placed respectively, at distance, $\frac{d}{2}$ and d from the origin, on the x -axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is:

[9 Jan. 2019 I]

Options:

A. $-Q/4$

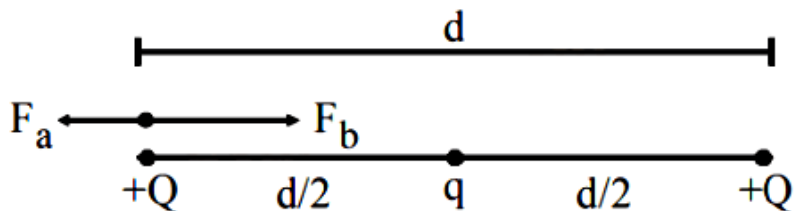
B. $+Q/2$

C. $+Q/4$

D. $-Q/2$

Answer: A

Solution:



Force due to charge $+Q$

$$F_a = \frac{K Q Q}{d^2}$$

Force due to charge q

$$F_b = \frac{K Q q}{\left(\frac{d}{2}\right)^2}$$

For equilibrium,

$$\vec{F}_a + \vec{F}_b = 0$$

$$\Rightarrow \frac{kQQ}{d^2} + \frac{kQq}{(d/2)^2} = 0 \therefore q = -\frac{Q}{4}$$

Question119

Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$ where A and a are constants. If Q is the total charge of this charge distribution, the radius R is:
[9 Jan. 2019, II]

Options:

A. $a \log \left(1 - \frac{Q}{2\pi a A} \right)$

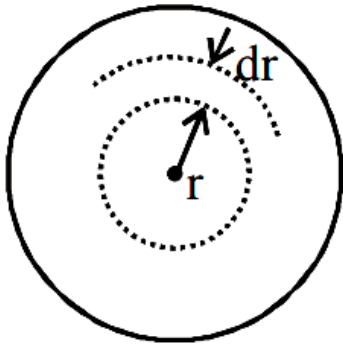
B. $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$

C. $a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$

D. $\frac{a}{2} \log \left(1 - \frac{Q}{2\pi a A} \right)$

Answer: B

Solution:



$$\begin{aligned} Q &= \int \rho \, dv = \int_0^R \frac{A}{r^2} e^{-2r/a} (4\pi r^2 \, dr) \\ &= 4\pi A \int_0^R e^{-2r/a} \, dr = 4\pi A \left(\frac{e^{-2r/a}}{-\frac{2}{a}} \right) \Big|_0^R \\ &= 4\pi A \left(-\frac{a}{2} \right) (e^{-2R/a} - 1) \\ Q &= 2\pi a A (1 - e^{-2R/a}) \\ R &= \frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right) \end{aligned}$$

Question120

For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is:

[9 Jan. 2019 I]

Options:

A. $\frac{R}{\sqrt{5}}$

B. $\frac{R}{\sqrt{2}}$

C. R

D. $R\sqrt{2}$

Answer: B

Solution:

Electric field on the axis of a ring of radius R at a distance h from the centre,

$$E = \frac{kQh}{(h^2 + R^2)^{3/2}}$$

Condition: for maximum electric field $\frac{dE}{dh} = 0$

$$\Rightarrow \frac{d}{dh} \left[\frac{kQh}{(R^2 + h^2)^{3/2}} \right] = 0$$

By using the concept of maxima and minima we get,

$$h = \frac{R}{\sqrt{2}}$$

Question121

Two point charges $q_1(\sqrt{10}\mu\text{C})$ and $q_2(-25\mu\text{C})$ are placed on the x - axis at $x = 1\text{m}$ and $x = 4\text{m}$ respectively. The electric field (in V / m) at a point $y = 3\text{m}$ on y -axis is

$$\left[\text{take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{N m}^2\text{C}^{-2} \right]$$

[9 Jan 2019, II]

Options:

A. $(63\hat{i} - 27\hat{j}) \times 10^2$

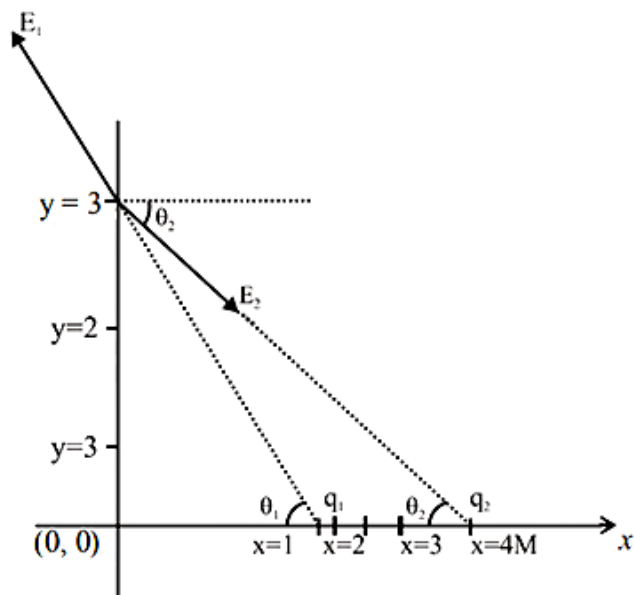
B. $(-63\hat{i} + 27\hat{j}) \times 10^2$

C. $(81\hat{i} - 81\hat{j}) \times 10^2$

D. $(-81\hat{i} + 81\hat{j}) \times 10^2$

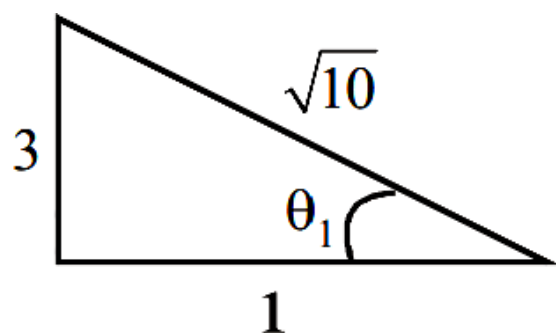
Answer: A

Solution:



Let $\vec{E}_1$ and $\vec{E}_2$ are the values of electric field due to charge, q_1 and q_2 respectively magnitude of

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1^2}$$



$$= \frac{1}{4\pi\epsilon_0} \frac{\sqrt{10} \times 10^{-6}}{(1^2 + 3^2)}$$

$$= (9 \times 10^9) \times \sqrt{10} \times 10^{-7}$$

$$= 9\sqrt{10} \times 10^2$$

$$\therefore \vec{E}_1 = 9\sqrt{10} \times 10^2 \left[\cos \theta_1 (-\hat{i}) + \sin \theta_1 \hat{j} \right]$$

$$\Rightarrow E_1 = 9 \times \sqrt{10} \times 10^2 \left[\frac{1}{\sqrt{10}} (-\hat{i}) + \frac{3}{\sqrt{10}} \hat{j} \right]$$

$$\Rightarrow E_1 = 9 \times 10^2 [-\hat{i} + 3\hat{j}] = [-9\hat{i} + 27\hat{j}] 10^2$$

$$\text{Similarly, } E_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2}$$

$$E_2 = \frac{9 \times 10^9 \times (25) \times 10^{-6}}{(4^2 + 3^2)} \quad E_2 = 9 \times 10^3 \text{ V/m}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 (\cos \theta_2 \hat{i} - \sin \theta_2 \hat{j}) \quad \because \tan \theta_2 = \frac{3}{4}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 \left(\frac{4}{5} \hat{i} - \frac{3}{5} \hat{j} \right) = (72\hat{i} - 54\hat{j}) \times 10^2$$

$$\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 = (63\hat{i} - 27\hat{j}) \times 10^2 \text{ V/m}$$

Question122

An electric field of 1000V / m is applied to an electric dipole at angle of 45° . The value of electric dipole moment is 10^{-29}C . m . What is the potential energy of the electric dipole?

[11 Jan 2019, II]

Options:

A. $-20 \times 10^{-18}\text{J}$

B. $-7 \times 10^{-27}\text{J}$

C. $-10 \times 10^{-29}\text{J}$

D. $-9 \times 10^{-20}\text{J}$

Answer: B

Solution:

Potential energy of a dipole is given by

$$U = -\vec{P} \cdot \vec{E}$$
$$= -PE \cos \theta$$

[Where θ = angle between dipole and perpendicular to the field]

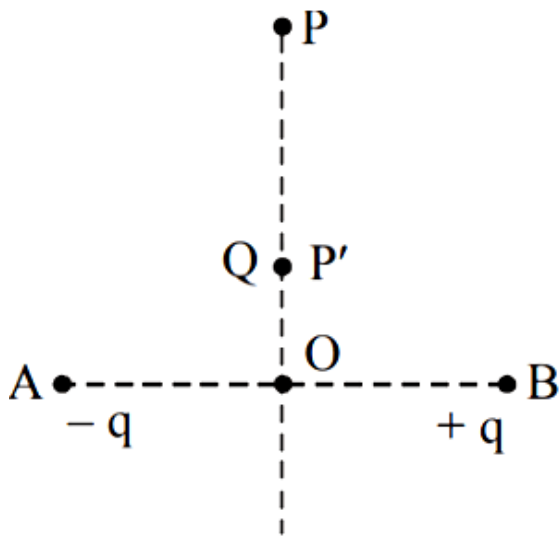
$$= -(10^{-29})(10^3) \cos 45^\circ$$

$$= -0.707 \times 10^{-26}\text{J} = -7 \times 10^{-27}\text{J}$$

Question123

Charges $-q$ and $+q$ located at A and B, respectively constitute an electric dipole. Distance $AB = 2a$, O is the mid point of the dipole and OP is perpendicular to AB. A charge Q is placed at P where $OP = y$ and $y \gg 2a$. The charge Q experiences an electrostatic force F. If Q is now moved along the equatorial line to P' such that OP'

$= \left(\frac{y}{3} \right)$, the force on Q will be close to: $\left(\frac{y}{3} \gg 2a \right)$



[10 Jan 2019, II]

Options:

A. $3F$

B. $\frac{F}{3}$

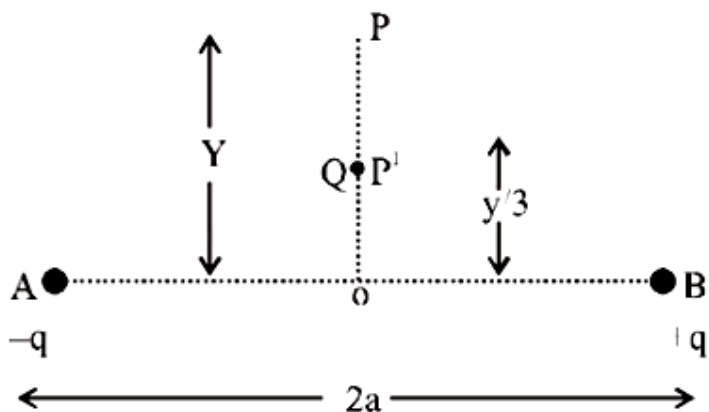
C. $9F$

D. $27F$

Answer: D

Solution:

Electric field of equatorial plane of dipole $= -\frac{K \vec{P}}{r^3}$

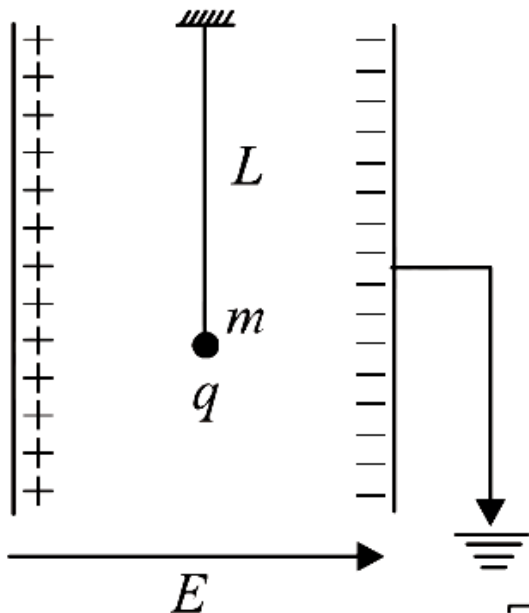


$$\therefore \text{At point P, } = +\frac{K P Q}{y^3} Q$$

$$\text{At Point P}^1, F^1 = +\frac{K P Q}{(y/3)^3} = 27F .$$

Question124

A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E as shown in figure. Its bob has mass m and charge q . The time period of the pendulum is given by:



[10 April 2019, II]

Options:

A. $2\pi \sqrt{\frac{L}{\left(g + \frac{qE}{m}\right)}}$

B. $2\pi \sqrt{\frac{L}{\sqrt{\left(g^2 - \frac{q^2 E^2}{m^2}\right)}}}$

$$C. 2\pi \sqrt{\frac{L}{\left(g - \frac{qE}{m}\right)}}$$

$$D. 2\pi \sqrt{\frac{L}{\sqrt{\left(g^2 + \frac{q^2 E^2}{m^2}\right)}}}$$

Answer: D

Solution:

Time period of the pendulum (T) is given by

$$T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}}$$

$$g_{\text{eff}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$\Rightarrow g_{\text{eff}} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \Rightarrow T = 2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

Question125

Four point charges $-q$, $+q$, $+q$ and $-q$ are placed on y -axis at $y = -2d$, $y = -d$, $y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will behave as:

[9 April 2019, II]

Options:

$$A. E \propto \frac{1}{D^3}$$

$$B. E \propto \frac{1}{D}$$

$$C. E \propto \frac{1}{D^4}$$

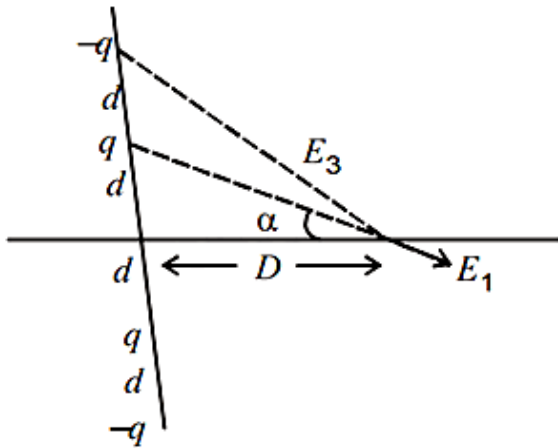
$$D. E \propto \frac{1}{D^2}$$

Answer: D

Solution:

$$\vec{E} = (\vec{E}_1 + \vec{E}_2) + (\vec{E}_3 + \vec{E}_4)$$

or $E = 2E \cos \alpha - 2E \cos \beta$



$$= \frac{2kq}{(D^2 + d^2)} \times \frac{D}{\sqrt{D^2 + d^2}} - \frac{2kq}{(D^2 + (2d)^2)} \times \frac{D}{\sqrt{D^2 + (2d)^2}}$$

$$= \frac{2kqD}{(D^2 + d^2)^{3/2}} - \frac{2kqD}{[D^2 + (2d)^2]^{3/2}}$$

For $d \ll D$

$$E \propto \frac{D}{D^3} \propto \frac{1}{D^2}$$

Question126

The bob of a simple pendulum has mass 2g and a charge of $5.0 \frac{1}{4} \text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000V / m . At equilibrium, the angle that the pendulum makes with the vertical is :

(take $g = 10 \text{m/s}^2$)
[8 April 2019 I]

Options:

A. $\tan^{-1}(2.0)$

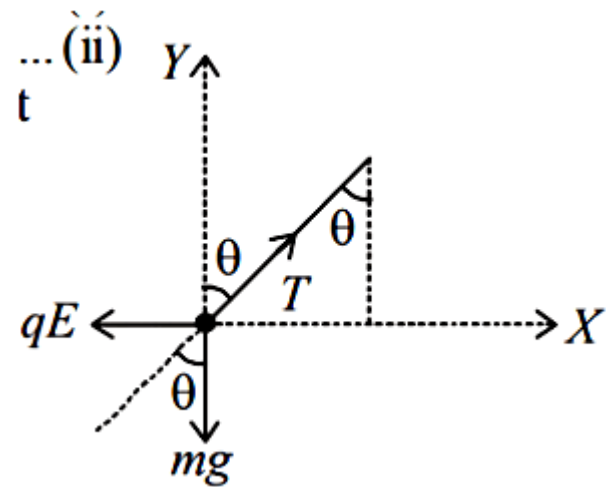
B. $\tan^{-1}(0.2)$

C. $\tan^{-1}(5.0)$

D. $\tan^{-1}(0.5)$

Answer: D

Solution:



At equilibrium resultant force on bob must be zero, so $T \cos \theta = mg$ (i)

$T \sin \theta = qE$ (ii)

Solving (i) and (ii) we get

$$\tan \theta = \frac{qE}{mg}$$

$$\tan \theta = \frac{5 \times 10^{-6} \times 2000}{2 \times 10^{-3} \times 10} = \frac{1}{2}$$

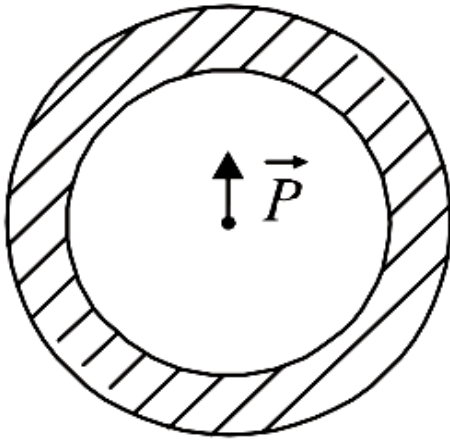
[Here, $q = 5 \times 10^{-6} \text{C}$, $E = 2000 \text{V/m}$, $m = 2 \times 10^{-3} \text{kg}$]

$$\Rightarrow \tan^{-1} \left(\frac{1}{2} \right)$$

Question127

Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b , and carries charge Q . At its centre is a

dipole $\vec{p}$ as shown. In this case :



[12 April 2019, I]

Options:

- A. surface charge density on the inner surface is uniform and equal to $\frac{Q/2}{4\pi a^2}$
- B. electric field outside the shell is the same as that of a point charge at the centre of the shell.
- C. surface charge density on the outer surface depends on $|\vec{p}|$
- D. surface charge density on the inner surface of the shell is zero everywhere.-

Answer: B

Solution:

Surface charge density depends only due to Q. Also

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_1 \lambda}{\epsilon_0}$$

$$\text{or } E \times 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}, r \geq R$$

Question128

Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B, of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre.

**If A and B do not experience any force, then.
[12 April 2019, II]**

Options:

A. $a = 8^{-1/4}R$

B. $a = \frac{3R}{2^{1/4}}$

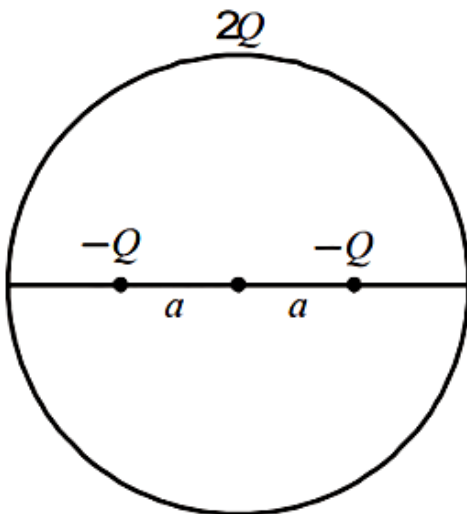
C. $a = 2^{-1/4}R$

D. $a = R/\sqrt{3}$

Answer: A

Solution:

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$



$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int S(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int_0^r (kr)(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{4\pi k}{\epsilon_0} \left(\frac{r^4}{4} \right)$$

$$\therefore E = \frac{k}{4\epsilon_0} r^2 \dots\dots(i)$$

$$\text{Also } 2Q = \int_0^R (kr)(4\pi r^2) dr = 4\pi k \left| \frac{r^4}{4} \right|_0^R$$

$$Q = \frac{\pi k R^4}{2} \dots\dots(ii)$$

From above equations,

$$E = \frac{Qr^2}{2\pi\epsilon_0 R^4} \dots\dots\dots(iii)$$

According to given condition

$$= E Q \frac{Q^4}{4\pi\epsilon_0 (20)^2} \dots\dots\dots(iv)$$

From equations (iii) and (iv), we have

$$a = 8^{-1/4} R.$$

Question129

An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is :

[8 April 2019, II]

Options:

A. $\sqrt{\frac{qE}{md}}$

B. $\sqrt{\frac{2qE}{md}}$

C. $2\sqrt{\frac{qE}{md}}$

D. $\sqrt{\frac{qE}{2md}}$

Answer: B

Solution:

$$\tau = -PE \sin \theta$$

$$\text{or } I \alpha = -PE (\theta)$$

$$\alpha = \frac{PE}{I}(-\theta)$$

On comparing with

$$\alpha = -\omega^2 \theta$$

$$\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{qdE}{2m\left(\frac{d}{2}\right)^2}} = \sqrt{\frac{2qE}{md}}$$

Question130

Two identical conducting spheres A and B, carry equal charge. They are separated by a distance much larger than their diameter, and the force between them is F . A third identical conducting sphere, C, is uncharged. Sphere C is first touched to A, then to B, and then removed. As a result, the force between A and B would be equal to [Online April 16,2018]

Options:

A. $\frac{3F}{4}$

B. $\frac{F}{2}$

C. F

D. $\frac{3F}{8}$

Answer: D

Solution:

Spheres A and B carry equal charge say 'q'

$$\therefore \text{Force between them, } F = \frac{kqq}{r^2}$$

When A and C are touched, charge on both $q_A = q_C = \frac{q}{2}$

Then when B and C are touched, charge on B

$$q_B = \frac{\frac{q}{2} + q}{2} = \frac{3q}{4}$$

Now, the force between charge q_A and q_B

$$F' = \frac{kq_A q_B}{r^2} = \frac{k \times \frac{q}{2} \times \frac{3q}{4}}{r^2} = \frac{3}{8} \frac{kq^2}{r^2} = \frac{3}{8} F$$

Question131

A body of mass M and charge q is connected to a spring of spring constant k . It is oscillating along x - direction about its equilibrium position, taken to be at $x = 0$, with an amplitude A . An electric field E is applied along the X -direction. Which of the following statements is correct?

[Online April 15, 2018]

Options:

- A. The total energy of the system is $\frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$
- B. The new equilibrium position is at a distance: $\frac{2qE}{k}$ from $x = 0$
- C. The new equilibrium position is at a distance: $\frac{qE}{2k}$ from $x = 0$
- D. The total energy of the system is $\frac{1}{2}m\omega^2 A^2 - \frac{1}{2} \frac{q^2 E^2}{k}$

Answer: A

Solution:

Equilibrium position will shift to point where resultant force = 0

$$kx_{eq} = qE \Rightarrow x_{eq} = \frac{qE}{k}$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2}kx_{eq}^2$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$$

Question132

A solid ball of radius R has a charge density ρ given by $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$

for $0 \leq r \leq R$. The electric field outside the ball is:

[Online April 15, 2018]

Options:

A. $\frac{\rho_0 R^3}{\epsilon_0 r^2}$

B. $\frac{4\rho_0 R^3}{3\epsilon_0 r^2}$

C. $\frac{3\rho_0 R^3}{4\epsilon_0 r^2}$

D. $\frac{\rho_0 R^3}{12\epsilon_0 r^2}$

Answer: D

Solution:

Charge density, $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$

$$dq = \rho dv$$

$$q_{\text{in}} = \int dq = \rho dv$$

$$= \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr \quad (\because dv = 4\pi r^2 dr)$$

$$= 4\pi\rho_0 \int_0^R \left(1 - \frac{r}{R}\right) r^2 dr$$

$$= 4\pi\rho_0 \int_0^R r^2 dr - \frac{r^2}{R} dr$$

$$= 4\pi\rho_0 \left[\left[\frac{r^3}{3} \right]_0^R - \left[\frac{r^4}{4R} \right]_0^R \right] = 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^4}{4R} \right]$$

$$= 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^3}{4} \right] = 4\pi\rho_0 \left[\frac{R^3}{12} \right]$$

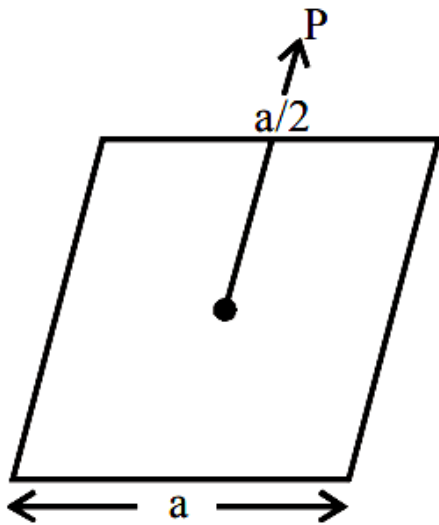
$$q = \frac{\pi\rho_0 R^3}{3}$$

$$E \cdot 4\pi r^2 = \left(\frac{\pi\rho_0 R^3}{3\epsilon_0} \right)$$

$$\therefore \text{Electric field outside the ball, } E = \frac{\rho_0 R^3}{12 \epsilon_0 r^2}$$

Question133

A charge Q is placed at a distance $a/2$ above the centre of the square surface of edge a as shown in the figure. The electric flux through the square surface is:



[Online April 15, 2018]

Options:

A. $\frac{Q}{3\epsilon_0}$

B. $\frac{Q}{6\epsilon_0}$

C. $\frac{Q}{2\epsilon_0}$

D. $\frac{Q}{\epsilon_0}$

Answer: B

Solution:

When cube is of side a and point charge Q is at the center of the cube then the total electric flux due to this charge will pass evenly through the six faces of the cube.
So, the electric flux through one face will be equal to $1/6$ of the total electric flux due to this charge.

$$\text{Flux through 6 faces} = \frac{Q}{\epsilon_0}$$

$$\therefore \text{Flux through 1 face} = \frac{Q}{6\epsilon_0}$$

Question134

An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x-axis. When subjected to an electric field $\vec{E}_1 = E \hat{i}$, it experiences a torque $\vec{T}_1 = \tau \hat{i}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E \hat{j}$ it experiences torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is:
[2017]

Options:

- A. 60°
- B. 90°
- C. 30°
- D. 45°

Answer: A

Solution:

$T = pE \sin \theta$ Torque experienced by the dipole in an electric field, $\vec{T} = \vec{p} \times \vec{E}$

$$\vec{p} = p \cos \theta \hat{i} + p \sin \theta \hat{j}$$

$$\vec{E}_1 = E \hat{i}$$

$$\vec{T}_1 = \vec{p} \times \vec{E}_1 = (p \cos \theta \hat{i} + p \sin \theta \hat{j}) \times E (\hat{i})$$

$$\tau \hat{k} = pE \sin \theta (-\hat{k}) \dots\dots(i)$$

$$\vec{E}_2 = \sqrt{3}E \hat{j}$$

$$\vec{T}_2 = p \cos \theta \hat{i} + p \sin \theta \hat{j} \times \sqrt{3}E \hat{j}$$

$$\tau \hat{k} = \sqrt{3}pE \cos \theta \hat{k} \dots\dots(ii)$$

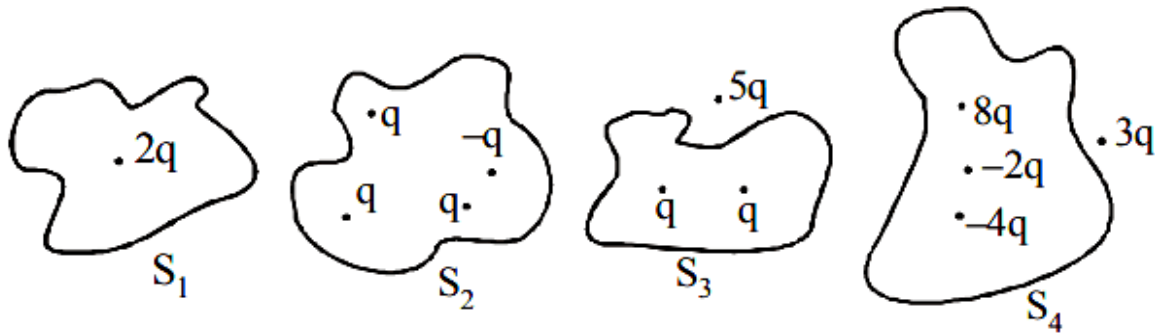
From eqns. (i) and (ii)

$$pE \sin \theta = \sqrt{3}pE \cos \theta$$

$$\tan \theta = \sqrt{3} \quad \therefore \theta = 60^\circ$$

Question135

Four closed surfaces and corresponding charge distributions are shown below.



Let the respective electric fluxes through the surfaces be Φ_1 , Φ_2 , Φ_3 , and Φ_4 . Then :

[Online April 9, 2017]

Options:

- A. $\Phi_1 < \Phi_2 = \Phi_3 > \Phi_4$
- B. $\Phi_1 > \Phi_2 > \Phi_3 > \Phi_4$
- C. $\Phi_1 = \Phi_2 = \Phi_3 = \Phi_4$
- D. $\Phi_1 > \Phi_3$; $\Phi_2 < \Phi_4$

Answer: C

Solution:

The net flux linked with closed surfaces S_1 , S_2 , S_3 & S_4 are

For surface S_1 , $\phi_1 = \frac{1}{\epsilon_0}(2q)$

For surface S_2 , $\phi_2 = \frac{1}{\epsilon_0}(q + q + q - q) = \frac{1}{\epsilon_0}2q$

For surface S_3 , $\phi_3 = \frac{1}{\epsilon_0}(q + q) = \frac{1}{\epsilon_0}(2q)$

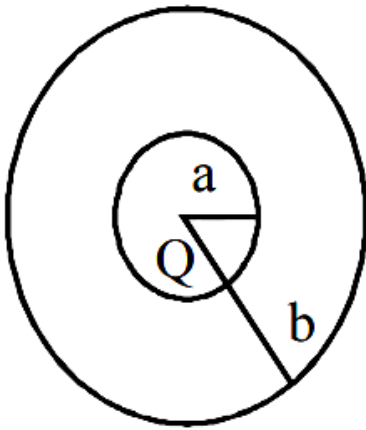
For surface S_4 , $\phi_4 = \frac{1}{\epsilon_0}(8q - 2q - 4q) = \frac{1}{\epsilon_0}(2q)$

Hence, $\phi_1 = \phi_2 = \phi_3 = \phi_4$ i.e. net electric flux is same for all surfaces.

Keep in mind, the electric field due to a charge outside (S_3 and S_4), the Gaussian surface contributes zero net flux through the surface, because as many lines due to that charge enter the surface as leave it.

Question136

The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), have volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant, is:



[2016]

Options:

A. $\frac{2Q}{\pi(a^2 - b^2)}$

B. $\frac{2Q}{\pi a^2}$

C. $\frac{Q}{2\pi a^2}$

D. $\frac{Q}{2\pi(b^2 - a^2)}$

Answer: C

Solution:

Applying Gauss's law

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\therefore E \times 4\pi r^2 = \frac{Q + 2\pi A r^2 - 2\pi A a^2}{\epsilon_0}$$

$$\rho = \frac{dr}{dV}$$

$$Q = \rho 4\pi r^2$$

$$Q = \int_a^r \frac{A}{r} 4\pi r^2 dr = 2\pi A[r^2 - a^2]$$

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{Q - 2\pi Aa^2}{r^2} + 2\pi A \right]$$

For E to be independent of 'r'

$$Q - 2\pi Aa^2 = 0$$

$$\therefore A = \frac{Q}{2\pi a^2}$$

Question137

Shown in the figure are two point charges $+Q$ and $-Q$ inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on the outer surface and Q_2 net charge on it then :

[Online April 10, 2015]

Options:

A. $\sigma_1 \neq 0, Q_1 = 0$

$\sigma_2 = 0, Q_2 = 0$

B. $\sigma_1 \neq 0, Q_1 = 0$

$\sigma_2 \neq 0, Q_2 = 0$

C. $\sigma_1 = 0, Q_1 = 0$

$\sigma_2 = 0, Q_2 = 0$

D. $\sigma_1 \neq 0, Q_1 \neq 0$

$\sigma_2 \neq 0, Q_2 \neq 0$

Answer: C

Solution:

Inside the cavity net charge is zero.

$$\therefore Q_1 = 0 \text{ and } \sigma_1 = 0$$

There is no effect of point charges $+Q$, $-Q$ and induced charge on inner surface on the outer surface.

$$\therefore Q_2 = 0 \text{ and } \sigma_2 = 0$$

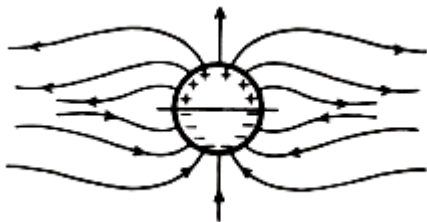
Question138

A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in : (figures are schematic and not drawn to scale)

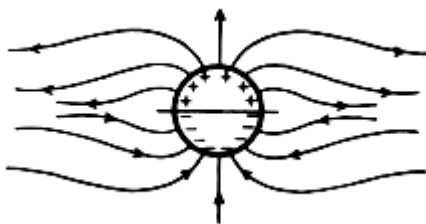
[2015]

Options:

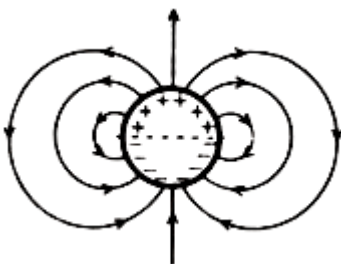
A.



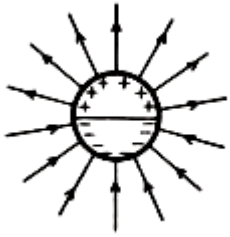
B.



C.



D.



Answer: C

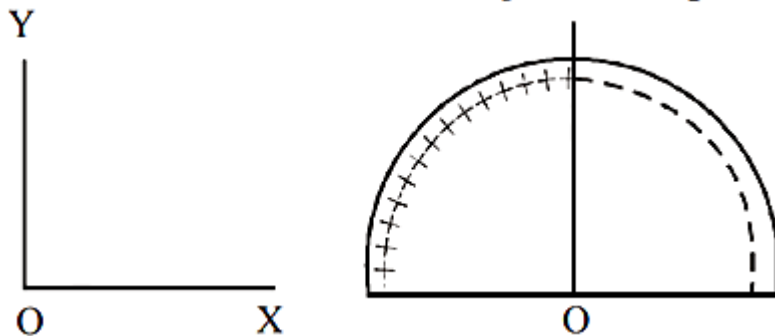
Solution:

Field lines originate perpendicular from positive charge and terminate perpendicular at negative charge.

Further this system can be treated as an electric dipole.

Question139

A wire of length $L (= 20\text{cm})$, is bent into a semicircular arc. If the two equal halves of the arc were each to be uniformly charged with charges $\pm Q$, [$|Q| = 10^3 \epsilon_0$. Coulomb where ϵ_0 is the permittivity (in SI units) of free space] the net electric field at the centre O of the semicircular arc would be :



[Online April 11, 2015]

Options:

- A. $(50 \times 10^3 \text{N} / \text{C}) \hat{j}$
- B. $(50 \times 10^3 \text{N} / \text{C}) \hat{i}$
- C. $(25 \times 10^3 \text{N} / \text{C}) \hat{j}$
- D. $(25 \times 10^3 \text{N} / \text{C}) \hat{i}$

Answer: D

Solution:

Given: Length of wire $L = 20\text{cm}$

charge $Q = 10^3 \epsilon_0$

We know, electric field at the centre of the semicircular arc

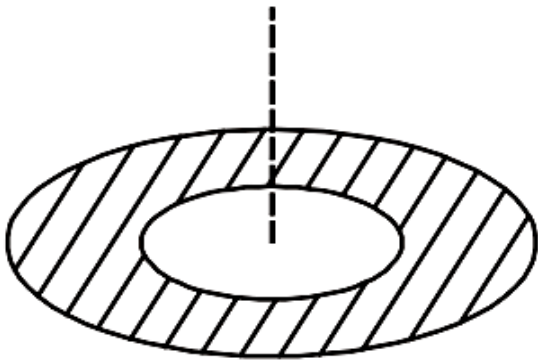
$$E = \frac{2K\lambda}{r}$$

$$\text{or, } E = 2K \left(\frac{2Q}{\pi r} \right) r \left[\text{As } \lambda = \frac{2Q}{\pi r} \right]$$

$$= \frac{4KQ}{\pi r^2} = \frac{4KQ\pi^2}{\pi L^2} = \frac{4\pi KQ}{L^2} = 25 \times 10^3 \text{N/C}$$

Question140

A thin disc of radius $b = 2a$ has a concentric hole of radius ' a ' in it (see figure). It carries uniform surface charge ' σ ' on it. If the electric field on its axis at height ' h ' ($h \ll a$) from its centre is given as ' Ch ' then value of ' C ' is :



[Online April 10, 2015]

Options:

A. $\frac{\sigma}{4a\epsilon_0}$

B. $\frac{\sigma}{8a\epsilon_0}$

C. $\frac{\sigma}{a\epsilon_0}$

D. $\frac{\sigma}{2a\epsilon_0}$

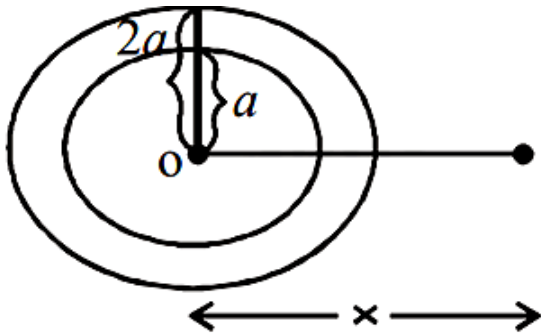
Answer: A

Solution:

Electric field due to complete disc ($R = 2a$) at a distance x and on its axis

$$E_1 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] \quad E_1 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{\sqrt{4a^2 + h^2}} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{2a} \right] \text{ [here } x = h \text{ and, } R = 2a]$$



Similarly, electric field due to disc ($R = a$)

$$E_2 = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{h}{a} \right)$$

Electric field due to given disc

$$E = E_1 - E_2$$

$$\frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{2a} \right] - \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{a} \right] = \frac{\sigma h}{4\epsilon_0 a}$$

$$\text{Hence, } c = \frac{\sigma}{4a\epsilon_0}$$

Question141

The electric field in a region of space is given by, $\vec{E} = E_0 \hat{i} + 2E_0 \hat{j}$ where $E_0 = 100 \text{ N/C}$. The flux of the field through a circular surface of radius 0.02 m parallel to the YZ plane is nearly:
[Online April 19, 2014]

Options:

A. $0.125 \text{ N m}^2/\text{C}$

B. $0.02 \text{ N m}^2 / \text{C}$

C. $0.005 \text{ N m}^2 / \text{C}$

D. $3.14 \text{ N m}^2 / \text{C}$

Answer: A

Solution:

$$\vec{E} = E_0 \hat{i} + 2E_0 \hat{j}$$

Given, $E_0 = 100 \text{ N /c}$

So, $\vec{E} = 100\hat{i} + 200\hat{j}$

Radius of circular surface = 0.02 m

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 0.02 \times 0.02$$

$$= 1.25 \times 10^{-3} \hat{i} \text{ m}^2 \text{ [Loop is parallel to Y-Z plane]}$$

Now, flux (ϕ) = $E A \cos \theta$

$$= (100\hat{i} + 200\hat{j}) \cdot 1.25 \times 10^{-3} \hat{i} \cos \theta^\circ [\theta = 0^\circ]$$

$$= 125 \times 10^{-3} \text{ N m}^2 / \text{c}$$

$$= 0.125 \text{ N m}^2 / \text{c}$$

Question142

A spherically symmetric charge distribution is characterised by a charge density having the following variations:

$$\rho(r) = \rho_0 \left(1 - \frac{r}{R} \right) \text{ for } r < R$$

$$\rho(r) = 0 \text{ for } r \geq R$$

Where r is the distance from the centre of the charge distribution ρ_0

is a constant. The electric field at an internal point ($r < R$) is:

[Online April 12,2014]

Options:

A. $\frac{\rho_0}{4\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$

B. $\frac{\rho_0}{\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$

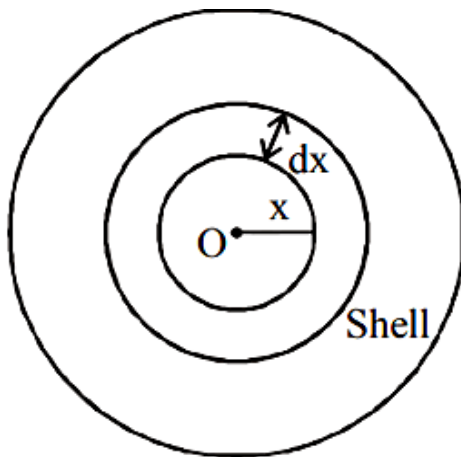
C. $\frac{\rho_0}{3\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$

D. $\frac{\rho_0}{12\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$

Answer: B

Solution:

Let us consider a spherical shell of radius x and thickness dx .



Charge on this shell

$$dq = \rho \cdot 4\pi x^2 dx = \rho_0 \left(1 - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$

$$\therefore \text{Total charge in the spherical region from centre to } r (q = \int dq = 4\pi\rho_0 \int_0^r \left(1 - \frac{x}{R} \right) x^2 dx$$

$$= 4\pi\rho_0 \left[\frac{x^3}{3} - \frac{x^4}{4R} \right]_0^r = 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4R} \right] = 4\pi\rho_0 r^3 \left[\frac{1}{3} - \frac{r}{4R} \right]$$

$$\therefore \text{Electric field at } r, E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{4\pi\rho_0 r^3}{r^2} \left[\frac{1}{3} - \frac{r}{4R} \right] = \frac{\rho_0}{\epsilon_0} \left[\frac{r}{3} - \frac{r^2}{4R} \right]$$

Question143

The magnitude of the average electric field normally present in the atmosphere just above the surface of the Earth is about 150N/C , directed inward towards the center of the Earth. This gives the total

net surface charge carried by the Earth to be:

[Given $\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2/\text{N} - \text{m}^2$, $R_E = 6.37 \times 10^6 \text{m}$]

[Online April 9, 2014]

Options:

A. +670kC

B. -670kC

C. -680kC

D. +680kC

Answer: C

Solution:

Given,

Electric field $E = 150 \text{ N/C}$

Total surface charge carried by earth $q = ?$

or, $q = \epsilon_0 E A$

$= \epsilon_0 E \pi r^2$

$= 8.85 \times 10^{-12} \times 150 \times (6.37 \times 10^6)^2$.

$\simeq 680 \text{K c}$

As electric field directed inward hence

$q = -680 \text{K c}$

Question144

Two charges, each equal to q , are kept at $x = -a$ and $x = a$ on the x - axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to
[2013]

Options:

A. y

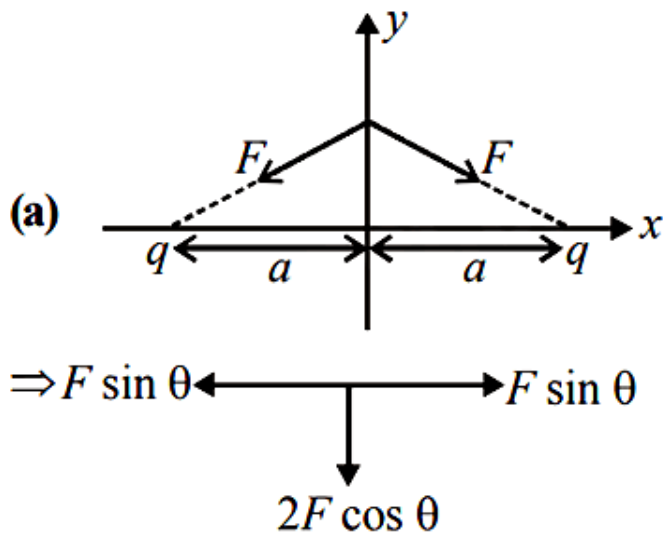
B. $-y$

C. $\frac{1}{y}$

D. $-\frac{1}{y}$

Answer: A

Solution:



$$\Rightarrow F_{\text{net}} = 2F \cos \theta$$

$$F_{\text{net}} = \frac{2kq\left(\frac{q}{2}\right)}{\left(\sqrt{y^2 + a^2}\right)^2} \cdot \frac{y}{\sqrt{y^2 + a^2}}$$

$$F_{\text{net}} = \frac{2kq\left(\frac{q}{2}\right)y}{(y^2 + a^2)^{3/2}} (\because y < a)$$

$$\Rightarrow \frac{kq^2 y}{a^3} \text{ So, } F \propto y$$

Question145

Two balls of same mass and carrying equal charge are hung from a fixed support of length l . At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, x between the balls is proportional to:

[Online April 9, 2013]

Options:

A. 1

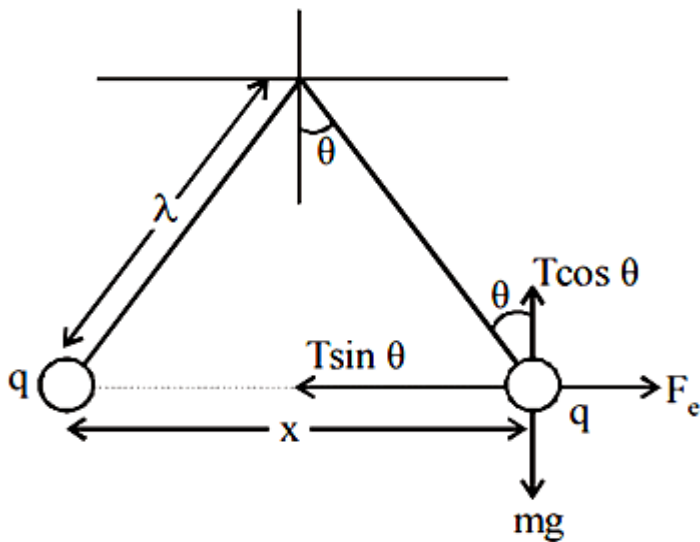
B. 1^2

C. $1^{2/3}$

D. $1^{1/3}$

Answer: D

Solution:



In equilibrium, $F_e = T \sin \theta$

$mg = T \cos \theta$

$$\tan \theta = \frac{F_e}{mg} = \frac{q^2}{4\pi \epsilon_0 x^2 \times mg}$$

$$\text{also } \tan \theta \approx \sin \theta = \frac{x/2}{l}$$

$$\text{Hence, } \frac{x}{2l} = \frac{q^2}{4\pi \epsilon_0 x^2 \times mg}$$

$$\Rightarrow x^3 = \frac{2q^2 l}{4\pi \epsilon_0 mg}$$

$$\therefore x = \left(\frac{q^2 l}{2\pi \epsilon_0 mg} \right)^{1/3}$$

Therefore $x \propto l^{1/3}$

Question 146

The surface charge density of a thin charged disc of radius R is σ . The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc :

[Online April 25, 2013]

Options:

A. reduces by 70.7%

B. reduces by 29.3%

C. reduces by 9.7%

D. reduces by 14.6%

Answer: A

Solution:

Electric field intensity at the centre of the disc.

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{given})$$

Electric field along the axis at any distance x from the centre of the disc

$$E' = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

From question, $x = R$ (radius of disc)

$$\therefore E' = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{R}{\sqrt{R^2 + R^2}} \right)$$

$$= \frac{\sigma}{2\epsilon_0} \left(\frac{\sqrt{2}R - R}{\sqrt{2}R} \right)$$

$$= \frac{4}{14}E$$

$\therefore$ % reduction in the value of electric field

$$= \frac{\left(E - \frac{4}{14}E \right) \times 100}{E} = \frac{1000}{14}\% \simeq 70.7\%$$

Question147

A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5kV m^{-1} . The density of liquid is

$1.26 \times 10^3 \text{ kg m}^{-3}$. The radius of the drop is (neglect buoyancy).
[Online April 23, 2013]

Options:

A. $4.3 \times 10^{-7} \text{ m}$

B. $7.8 \times 10^{-7} \text{ m}$

C. $0.078 \times 10^{-7} \text{ m}$

D. $3.4 \times 10^{-7} \text{ m}$

Answer: B

Solution:

$$F = qE = mg (q = 6e = 6 \times 1.6 \times 10^{-19})$$

$$\text{Density (d)} = \frac{\text{mass}}{\text{volume}} = \frac{m}{\frac{4}{3}\pi r^3}$$

$$\text{or } r^3 = \frac{m}{\frac{4}{3}\pi d}$$

$$\text{Putting the value of d and } m \left(= \frac{qE}{g} \right) \text{ and solving we get } r = 7.8 \times 10^{-7} \text{ m}$$

Question148

Two point dipoles of dipole moment $\vec{p}_1$ and $\vec{p}_2$ are at a distance x from each other and $\vec{p}_1 \parallel \vec{p}_2$. The force between the dipoles is:
[Online April 9, 2013]

Options:

A. $\frac{1}{4\pi\epsilon_0} \frac{4p_1p_2}{x^4}$

B. $\frac{1}{4\pi\epsilon_0} \frac{3p_1p_2}{x^3}$

C. $\frac{1}{4\pi\epsilon_0} \frac{6p_1p_2}{x^4}$

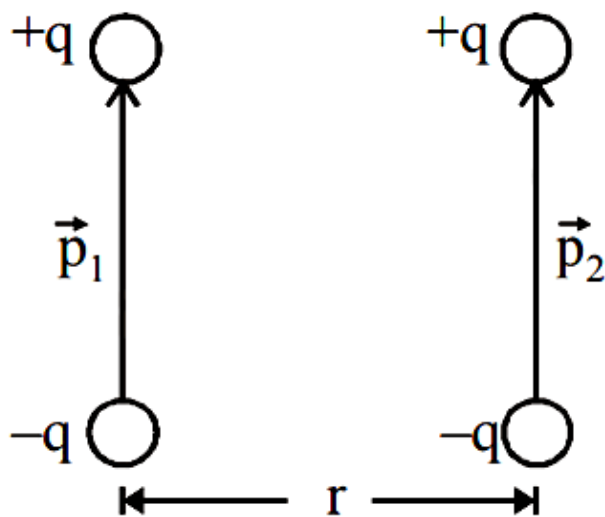
D. $\frac{1}{4\pi\epsilon_0} \frac{8p_1p_2}{x^4}$

Answer: C

Solution:

Force of interaction

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{6p_1p_2}{r^4}$$

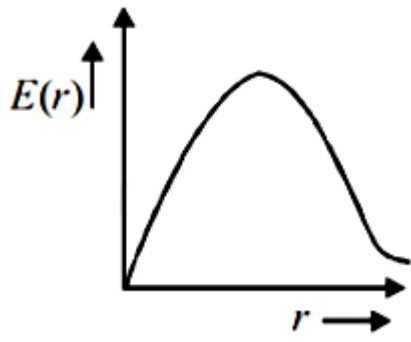


Question149

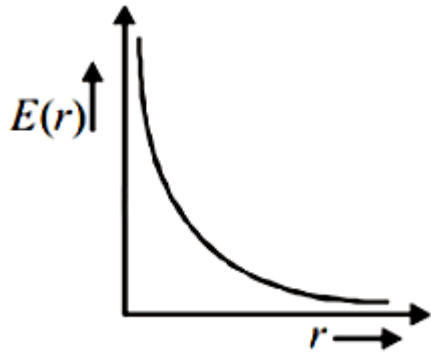
In a uniformly charged sphere of total charge Q and radius R , the electric field E is plotted as function of distance from the centre, The graph which would correspond to the above will be:
[2012]

Options:

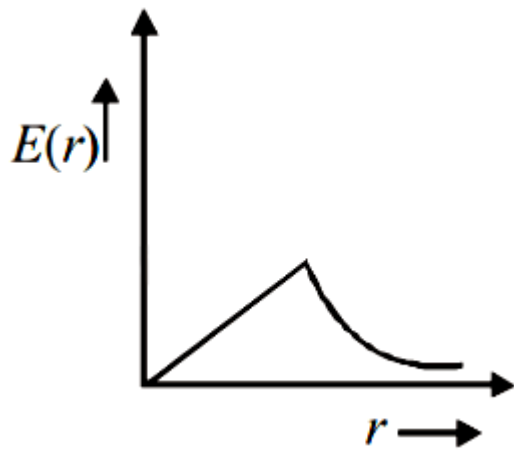
A.



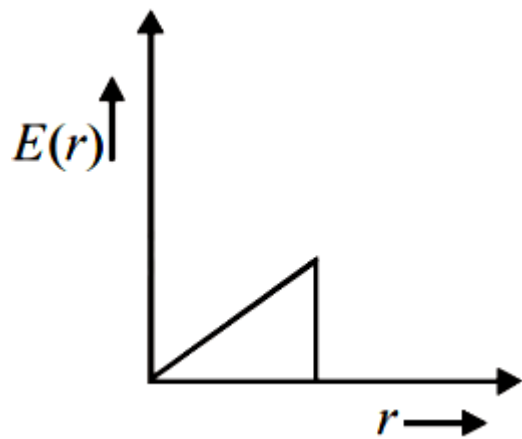
B.



C.



D.



Answer: C

Solution:

$$E_{\text{in}} \propto r$$

$$E_{\text{out}} \propto \frac{1}{r^2}$$

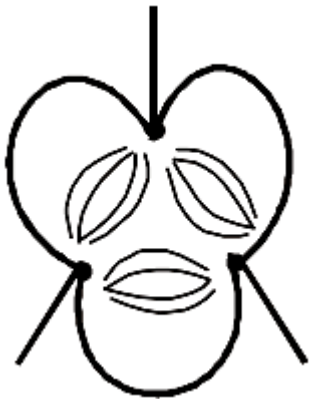
Question150

Three positive charges of equal value q are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in

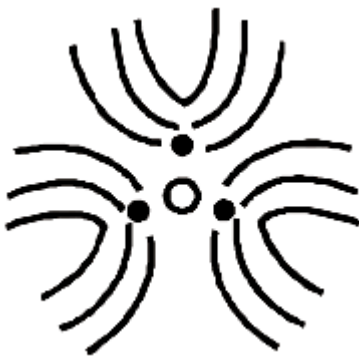
[Online May 26, 2012]

Options:

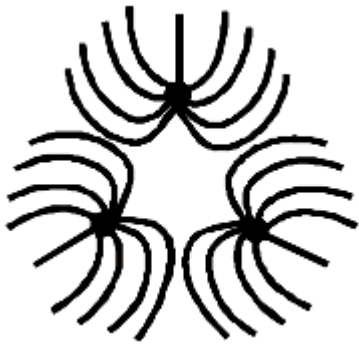
A.



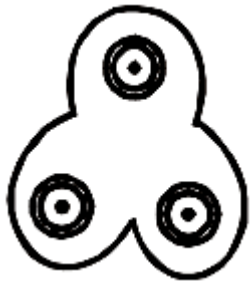
B.



C.



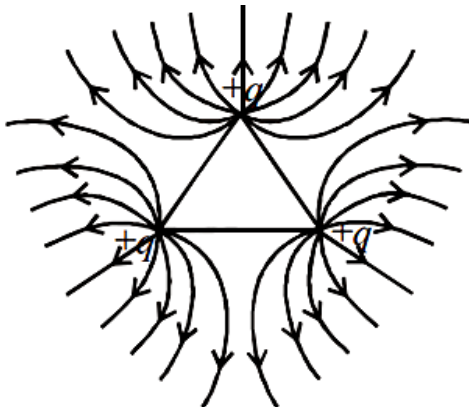
D.



Answer: C

Solution:

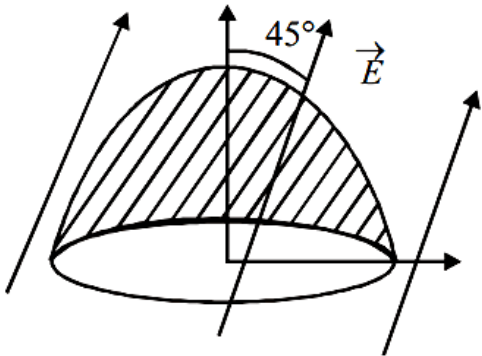
Electric lines of force due to a positive charge is spherically symmetric. All the charges are positive and equal in magnitude. So repulsion takes place. Due to which no lines of force are present inside the equilateral triangle and the resulting lines of force obtained as shown:



Question151

The flat base of a hemisphere of radius a with no charge inside it lies in a horizontal plane. A uniform electric field $\vec{E}$ is applied at an angle

$\frac{\pi}{4}$ with the vertical direction. The electric flux through the curved surface of the hemisphere is



[Online May 19, 2012]

Options:

- A. $\pi a^2 E$
- B. $\frac{\pi a^2 E}{\sqrt{2}}$
- C. $\frac{\pi a^2 E}{2\sqrt{2}}$
- D. $\frac{(\pi + 2)\pi a^2 E}{(2\sqrt{2})^2}$

Answer: B

Solution:

We know that,

$$\phi = \oint \vec{E} \cdot d\vec{S} = E \oint dS \cos 45^\circ$$

In case of hemisphere

$$\phi_{\text{curved}} = \phi_{\text{circular}}$$

$$\text{Therefore, } \phi_{\text{curved}} = E \pi a^2 \cdot \frac{1}{\sqrt{2}} = \frac{E \pi a^2}{\sqrt{2}}$$

Question152

Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance d ($d \ll l$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result charges approach each other with a velocity v . Then as a function of distance x between them,
[2011]

Options:

A. $v \propto x^{-1}$

B. $v \propto x^{1/2}$

C. $v \propto x$

D. $v \propto x^{-1/2}$

Answer: D

Solution:

From figure

$$T \cos \theta = mg \dots\dots(i)$$

$$T \sin \theta = F_e \dots\dots(ii)$$

Dividing equation (ii) by (i), we get

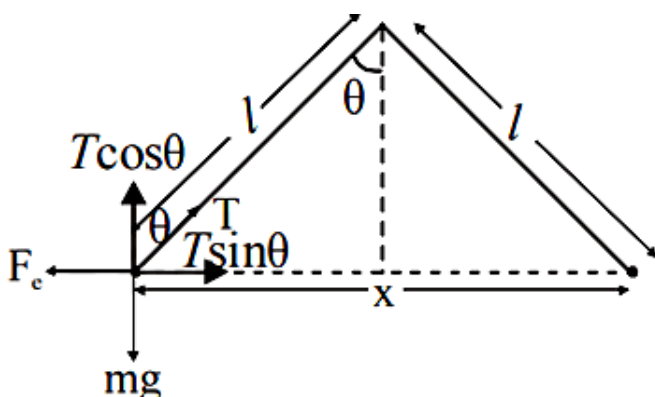
$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{F_e}{mg} \Rightarrow F_e = mg \tan \theta$$

$$\Rightarrow \frac{kq^2}{x^2} = mg \tan \theta \Rightarrow q^2 = \frac{x^2 mg \tan \theta}{k}$$

Since θ is small

$$\therefore \tan \theta \approx \sin \theta = \frac{x}{2l}$$

$$\therefore q^2 = \frac{x^3 mg}{2kl} \Rightarrow q^2 \propto x^{3/2}$$



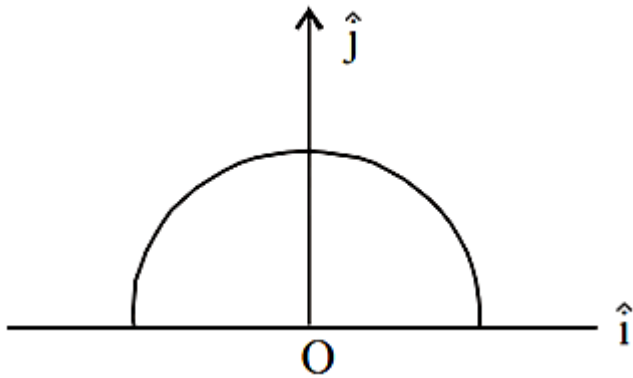
$$\Rightarrow \frac{dq}{dt} \propto \frac{3}{2} \sqrt{x} \frac{dx}{dt} = \frac{3}{2} \sqrt{x} V$$

$$\text{Since } \frac{dq}{dt} = \text{const.}$$

$$\Rightarrow v \propto x^{-1/2} \quad [\because q^2 \propto x^3]$$

Question 153

A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is



[2010]

Options:

A. $\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$

B. $-\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$

C. $-\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$

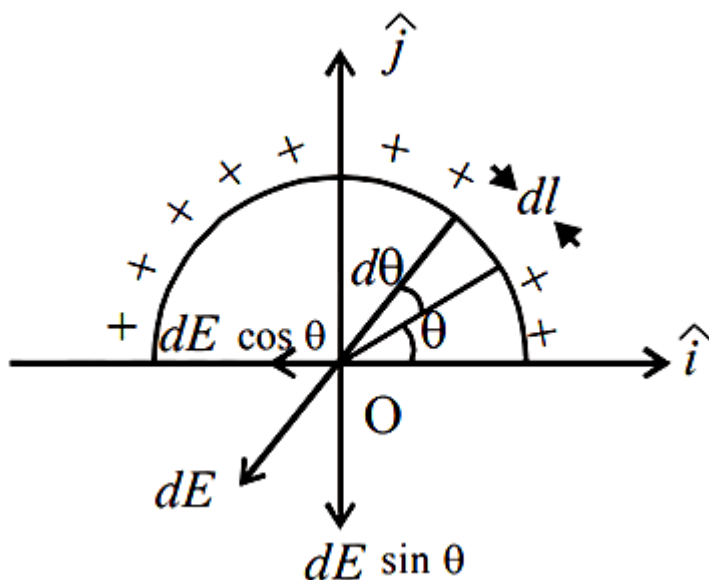
D. $\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$

Answer: C

Solution:

Let us consider a differential element dl subtending an angle $d\theta$ at the centre O as shown in the figure. Linear charge density

$$\lambda = \frac{q}{Qr}$$



Charge on the element, $dq = \left(\frac{q}{\pi r}\right) dl$

$$= \frac{q}{\pi r} (rd\theta) (\because dl = rd\theta)$$

$$= \left(\frac{q}{\pi}\right) d\theta$$

Electric field at the center O due to dq is

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\pi r^2} d\theta$$

Resolving dE into two rectangular component, we find the component $dE \cos \theta$ will be counter balanced by another element on left portion. Hence resultant field at O is the resultant of the component $dE \sin \theta$ only.

$$\therefore E = \int dE \sin \theta = \int_0^\pi \frac{q}{4\pi^2 r^2 \epsilon_0} \sin \theta d\theta$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} [-\cos \theta]_0^\pi$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} (+1 + 1) = \frac{q}{2\pi^2 r^2 \epsilon_0}$$

The direction of E is towards negative y -axis.

$$\therefore \vec{E} = -\frac{q}{2\pi^2 r^2 \epsilon_0} \hat{j}$$

Question 154

Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R}\right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance r ($r < R$) from the origin is given by
[2010]

Options:

A. $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$

B. $\frac{4\rho_0 r}{3\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$

C. $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

D. $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

Answer: A

Solution:

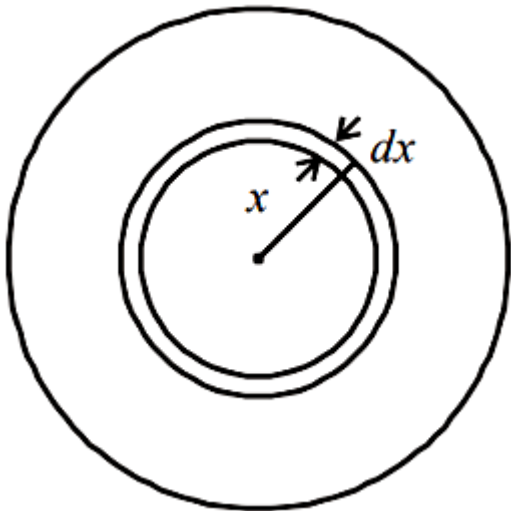
Let us consider a spherical shell of radius x and thickness dx .

Due to spherically symmetric charge distribution, the charge on the spherical surface of radius x is

$$dq = dV \rho = 4\pi x^2 dx \rho = \rho_0 \left(\frac{5}{4} - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$

$\therefore$ Total charge in the spherical region from centre to r ($r < R$) is

$$q = \int dq = 4\pi \rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R} \right) x^2 dx$$



$$= 4\pi \rho_0 \left[\frac{5}{4} \cdot \frac{r^3}{3} - \frac{1}{R} \cdot \frac{r^4}{4} \right] = \pi \rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right)$$

$\therefore$ Electric field intensity at a point on this spherical surface

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\pi \rho_0 r^3}{r^2} \left(\frac{5}{3} - \frac{r}{R} \right) = \frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$$

Question155

A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corners. If the net electrical force on Q is zero, then Q/q equals:

[2009]

Options:

A. -1

B. 1

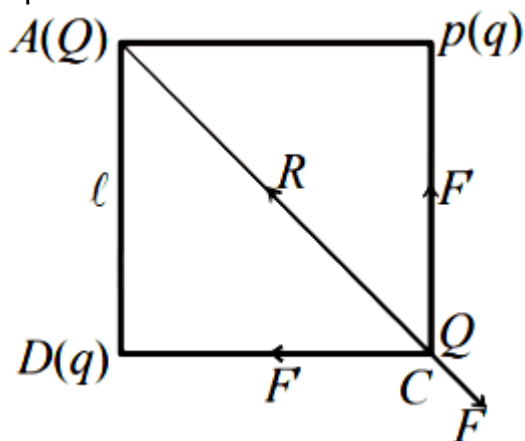
C. $-\frac{1}{\sqrt{2}}$

D. $-2\sqrt{2}$

Answer: D

Solution:

Let F be the force between Q and Q . The force between q and Q should be attractive for net force on Q to be zero. Let F' be the force between Q and q . The resultant of F' and F is R . For equilibrium



Net force on Q at C is zero

$$\therefore \vec{R} + \vec{F} = 0 \Rightarrow \sqrt{2}F' = -F$$

$$\Rightarrow \sqrt{2} \times k \frac{Qq}{l^2} = -k \frac{Q^2}{(\sqrt{2}l)^2}$$

$$\Rightarrow \frac{Q}{q} = -2\sqrt{2}$$

Question156

This question contains Statement- 1 and Statement- 2 . Of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1: For a charged particle moving from point P to point Q, the net work done by an electrostatic field on the particle is independent of the path connecting point P to point Q.

Statement- 2 : The net work done by a conservative force on an object moving along a closed loop is zero.

[2009]

Options:

- A. Statement- 1 is true, Statement- 2 is true; Statement-2 is the correct explanation of Statement- 1 .
- B. Statement- 1 is true, Statement- 2 is true; Statement-2 is not the correct explanation of Statement- 1 .
- C. Statement- 1 is false, Statement- 2 is true.
- D. Statement- 1 is true, Statement- 2 is false.

Answer: A

Question157

Let $\rho(r) = \frac{Q}{\pi R^4}r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'P ' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is :

[2009]

Options:

A. $\frac{Q}{4\pi \epsilon_0 r_1^2}$

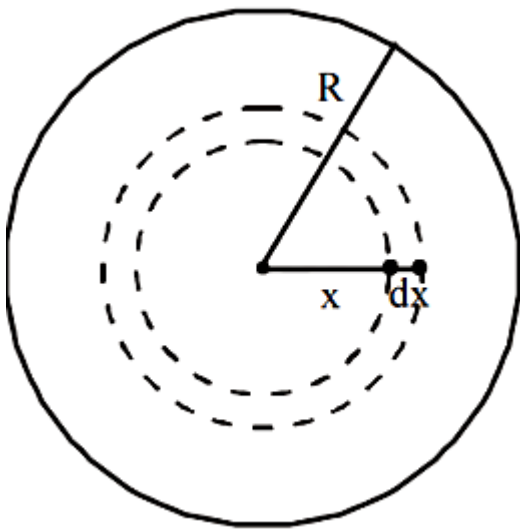
B. $\frac{Qr_1^2}{4\pi\epsilon_0 R^4}$

C. $\frac{Qr_1^2}{3\pi\epsilon_0 R^4}$

D. 0

Answer: B

Solution:



Let us consider a spherical shell of thickness dx and radius x . The area of this spherical shell $= 4\pi x^2$.

The volume of this spherical shell $= 4\pi x^2 dx$. The charge enclosed within shell

$$dq = \left[\frac{Q \cdot x}{\pi R^4} \right] [4\pi x^2 dx] = \frac{4Q}{R^4} x^3 dx$$

The charge enclosed in a sphere of radius r_1 can be calculated by

$$Q = \int dq = \frac{4Q}{R^4} \int_0^{r_1} x^3 dx = \frac{4Q}{R^4} \left[\frac{x^4}{4} \right]_0^{r_1} = \frac{Q}{R^4} r_1^4$$

$\therefore$ The electric field at point P inside the sphere at a distance r_1 from the centre of the sphere is

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_1^2}$$

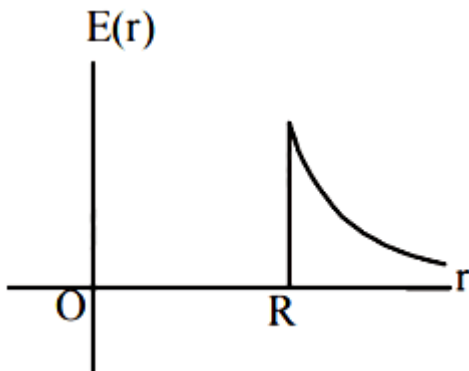
$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{\left[\frac{Q}{R^4} r_1^4 \right]}{r_1^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^4} r_1^2$$

Question158

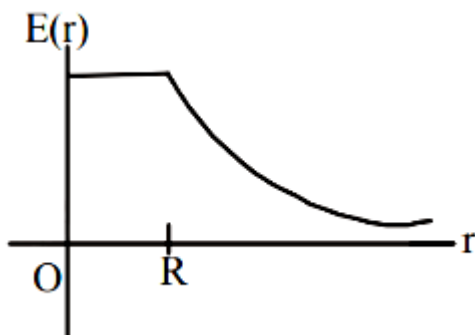
A thin spherical shell of radius R has charge Q spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$ produced by the shell in the range $0 \leq r < \infty$, where r is the distance from the centre of the shell?
[2008]

Options:

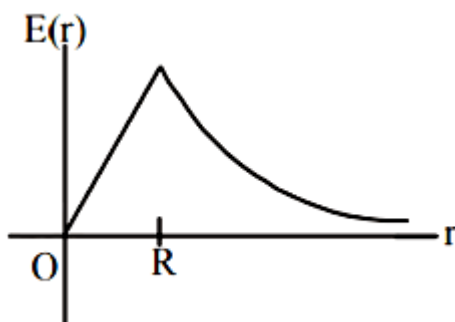
A.



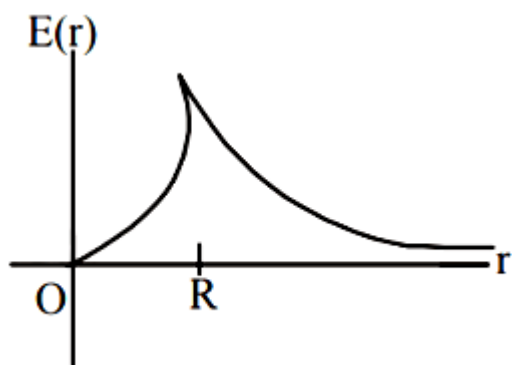
B.



C.



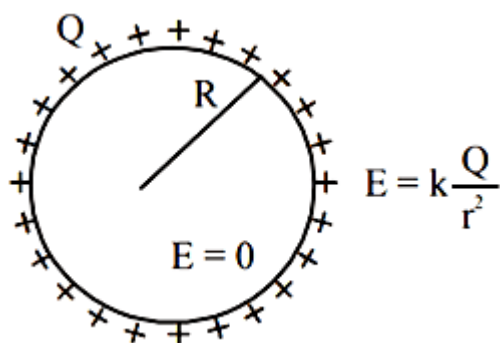
D.



Answer: A

Solution:

The electric field inside a thin spherical shell of radius R has charge Q spread uniformly over its surface is zero.



Outside the shell the electric field is $E = k \frac{Q}{r^2}$. These characteristics are represented by graph (a).

Question159

If g_E and g_M are the accelerations due to gravity on the surfaces of the earth and the moon respectively and if Millikan's oil drop experiment could be performed on the two surfaces, one will find the **ratio**

$\frac{\text{electronic charge on the moon}}{\text{electronic charge on the earth}}$ **to be**

[2007]

Options:

A. g_M / g_E

B. 1

C. 0

D. g_E / g_M

Answer: B

Solution:

It is obvious that by charge conservaiton law, electronic charge does not depend on acceleration due to gravity as it is a universal constant.

So, electronic charge on earth = electronic charge on moon

$\therefore$ Required ratio = 1.

Question160

Two spherical conductors A and B of radii 1mm and 2mm are separated by a distance of 5cm and are uniformly charged. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of spheres A and B is

[2006]

Options:

A. 4: 1

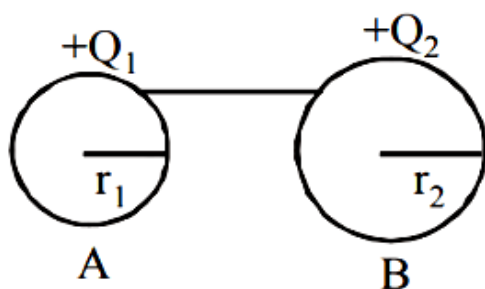
B. 1: 2

C. 2: 1

D. 1: 4

Answer: C

Solution:



When the two conducting spheres are connected by a conducting wire, charge will flow from one to other till both acquire same potential.

∴ After connection, $V_1 = V_2$

$$\Rightarrow k \frac{Q_1}{r_1} = k \frac{Q_2}{r_2} \Rightarrow \frac{Q_1}{r_1} = \frac{Q_2}{r_2}$$

The ratio of electric fields $\frac{E_1}{E_2} = \frac{k \frac{Q_1}{r_1^2}}{k \frac{Q_2}{r_2^2}} \Rightarrow \frac{E_1}{E_2} = \frac{Q_1}{r_1^2} \times \frac{r_2^2}{Q_2}$

$$\Rightarrow \frac{E_1}{E_2} = \frac{r_1 \times r_2^2}{r_1^2 \times r_2} \Rightarrow \frac{E_1}{E_2} = \frac{r_2}{r_1} = \frac{2}{1}$$

Question 161

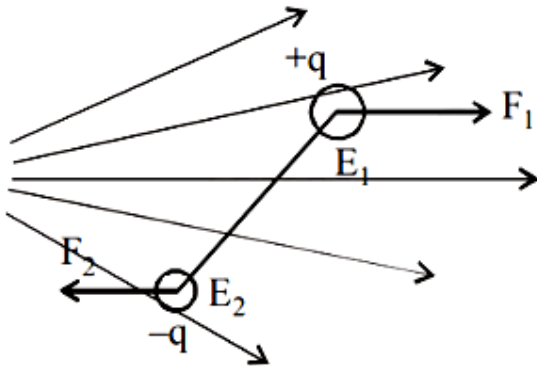
An electric dipole is placed at an angle of 30° to a nonuniform electric field. The dipole will experience [2006]

Options:

- A. a translational force only in the direction of the field
- B. a translational force only in a direction normal to the direction of the field
- C. a torque as well as a translational force
- D. a torque only

Answer: C

Solution:



As the dipole is placed in non-uniform field, so the force acting on the dipole will not cancel each other. This will result in a force as well as torque.

Question162

Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x axis at which the net electric field due to these two point charges is zero is
[2005]

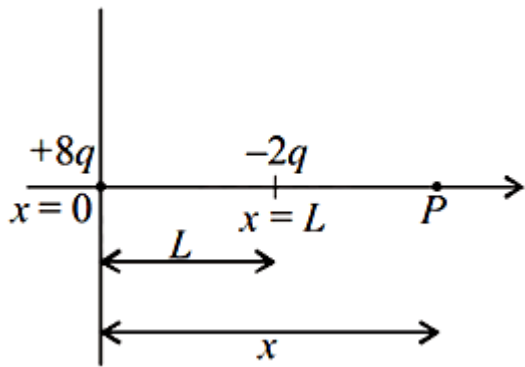
Options:

- A. $\frac{L}{4}$
- B. $2L$
- C. $4L$
- D. $8L$

Answer: B

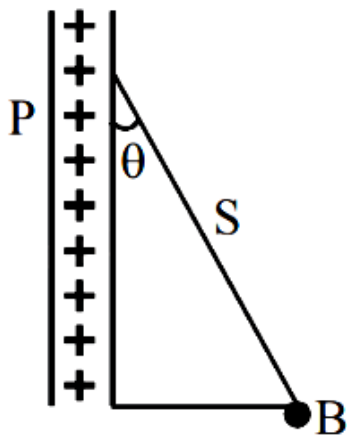
Solution:

$$\begin{aligned} \text{At P } \frac{-K 2q}{(x-L)^2} + \frac{K 8q}{x^2} &= 0 \\ \Rightarrow \frac{1}{(x-L)^2} &= \frac{4}{x^2} \\ \text{or } \frac{1}{x-L} &= \frac{2}{x} \\ \Rightarrow x &= 2x - 2L \text{ or } x = 2L \end{aligned}$$



Question163

A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to



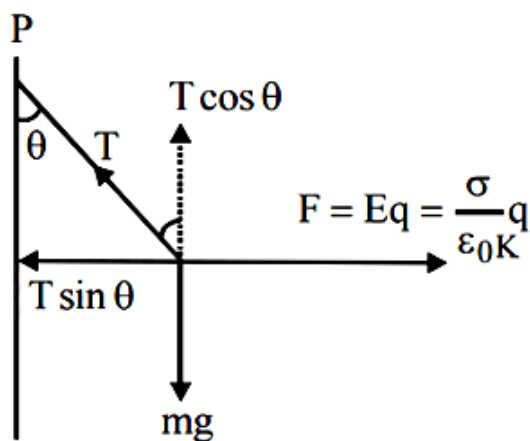
[2005]

Options:

- A. $\cot \theta$
- B. $\cos \theta$
- C. $\tan \theta$
- D. $\sin \theta$

Answer: C

Solution:



$$T \sin \theta = qE \dots\dots(i)$$

$$T \cos \theta = mg \dots\dots(ii)$$

Dividing (i) by (ii),

$$\tan \theta = \frac{qE}{mg} = \frac{q}{mg} \left(\frac{\sigma}{\epsilon_0 K} \right) \frac{\sigma q}{\epsilon_0 K \cdot mg}$$

$$\therefore \sigma \propto \tan \theta$$

Question164

Two spherical conductors B and C having equal radii and carrying equal charges on them repel each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that B but uncharged is brought in contact with B, then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is
[2004]

Options:

A. $F/8$

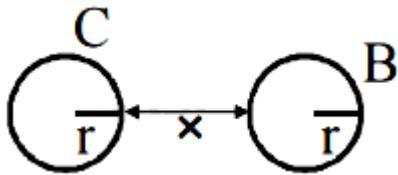
B. $3F/4$

C. $F/4$

D. $3F/8$

Answer: D

Solution:



Initial force, $F = K \frac{Q_B Q_C}{x^2}$ x is distance between the spheres. When third spherical conductor comes in contact with B charge on B is halved i.e., $\frac{Q}{2}$ and charge on third sphere becomes $\frac{Q}{2}$. Now it is touched to C, charge then equally distributes themselves to make potential same, hence charge on C becomes $\left(Q + \frac{Q}{2}\right) \frac{1}{2} = \frac{3Q}{4}$.

$$\therefore F_{\text{new}} = k \frac{Q_C' Q_B'}{x^2} = k \frac{\left(\frac{3Q}{4}\right) \left(\frac{Q}{2}\right)}{x^2} = k \frac{3Q^2}{8x^2}$$

$$\text{or } F_{\text{new}} = \frac{3}{8}F$$

Question 165

Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium the value of q is
[2004]

Options:

A. $-\frac{Q}{2}(1 + 2\sqrt{2})$

B. $\frac{Q}{4}(1 + 2\sqrt{2})$

C. $-\frac{Q}{4}(1 + 2\sqrt{2})$

D. $\frac{Q}{2}(1 + 2\sqrt{2})$

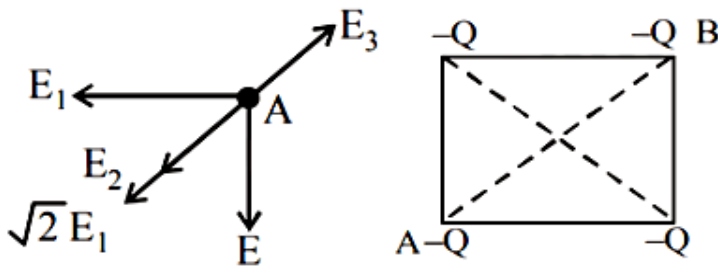
Answer: B

Solution:

For the system to be equilibrium, net field at A should be zero

$$\sqrt{2}E_1 + E_2 = E_3$$

$$\therefore \frac{kQ \times \sqrt{2}}{a^2} + \frac{kQ}{(\sqrt{2}a)^2} = \frac{kq}{\left(\frac{a}{\sqrt{2}}\right)^2}$$



$$\Rightarrow \frac{Q\sqrt{2}}{1} + \frac{Q}{2} = 2q \Rightarrow q = \frac{Q}{4}(2\sqrt{2} + 1)$$

Question 166

A charged oil drop is suspended in a uniform field of 3×10^4 v/m so that it neither falls nor rises. The charge on the drop will be (Take the mass of the charge = 9.9×10^{-15} kg and $g = 10 \text{ m/s}^2$) [2004]

Options:

- A. $1.6 \times 10^{-18} \text{ C}$
- B. $3.2 \times 10^{-18} \text{ C}$
- C. $3.3 \times 10^{-18} \text{ C}$
- D. $4.8 \times 10^{-18} \text{ C}$

Answer: C

Solution:

Given, Electric field, $E = 3 \times 10^4$

Mass of the drop, $m = 9.9 \times 10^{-15} \text{ kg}$

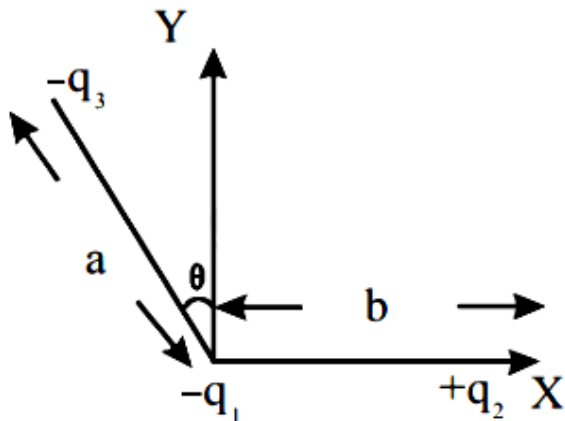
At equilibrium, coulomb force on drop balances weight of drop.

$$qE = mg$$

$$\Rightarrow q = \frac{mg}{E} \Rightarrow q = \frac{9.9 \times 10^{-15} \times 10}{3 \times 10^4} = 3.3 \times 10^{-18} \text{ C}$$

Question167

Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in the figure. The x - component of the force on $-q_1$ is proportional to



[2003]

Options:

A. $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$

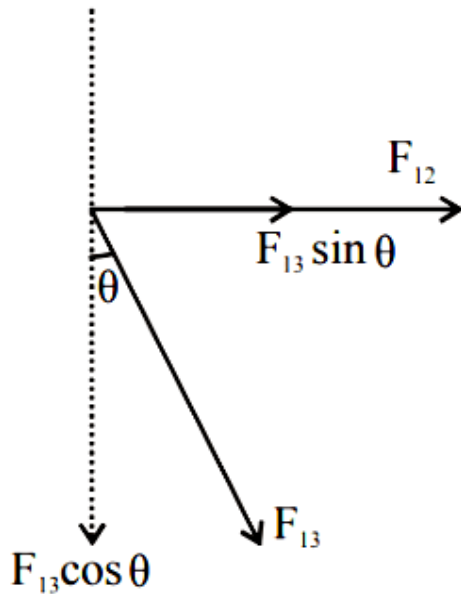
B. $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$

C. $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$

D. $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$

Answer: B

Solution:



Force applied by charge q_2 on q_1

$$F_{12} = k \frac{q_1 q_2}{b^2}$$

Force applied by charge q_3 on q_1 $F_{13} = k \frac{q_1 q_3}{a^2}$

The X -component of net force (F_x) on

q_1 is $F_{12} + F_{13} \sin \theta$

$$\therefore F_x = k \frac{q_1 q_2}{b^2} + k \frac{q_1 q_3}{a^2} \sin \theta$$

$$\therefore F_x \propto \frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$$

Question168

If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , the electric charge inside the surface will be

[2003]

Options:

A. $(\phi_2 - \phi_1)\epsilon_0$

B. $(\phi_1 - \phi_2)/\epsilon_0$

C. $(\phi_2 - \phi_1)/\epsilon_0$

D. $(\phi_1 - \phi_2)\epsilon_0$

Answer: A

Solution:

The electric flux ϕ_1 entering an enclosed surface is taken as negative and the electric flux ϕ_2 leaving the surface is taken as positive, by convention. Therefore the net flux leaving the enclosed surface,
 $\phi = \phi_2 - \phi_1$

According to Gauss theorem

$$\phi = \frac{q}{\epsilon_0} \Rightarrow q = \epsilon_0 \phi = \epsilon_0 (\phi_2 - \phi_1)$$

Question169

If a charge q is placed at the centre of the line joining two equal charges Q such that the system is in equilibrium then the value of q is

[2002]

Options:

A. $\frac{Q}{2}$

B. $-\frac{Q}{2}$

C. $\frac{Q}{4}$

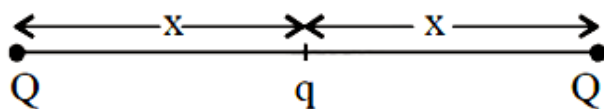
D. $-\frac{Q}{4}$

Answer: D

Solution:

At equilibrium net force is zero,

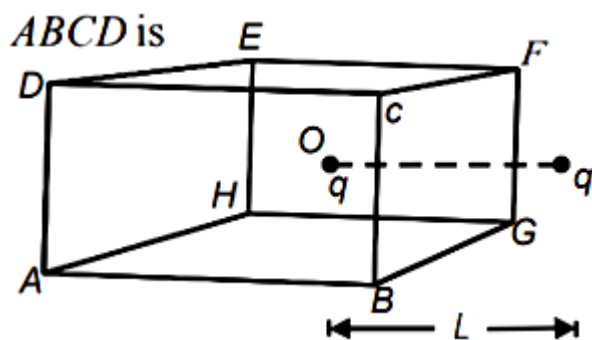
$$\therefore k \frac{Q \times Q}{(2x)^2} + k \frac{Qq}{x^2} = 0$$



$$\Rightarrow q = -\frac{Q}{4}$$

Question170

A charged particle q is placed at the centre O of cube of length L (ABCD E F GH). Another same charge q is placed at a distance L from O . Then the electric flux through ABCD is



[2002]

Options:

A. $q/4\pi \epsilon_0 L$

B. zero

C. $q/2\pi \epsilon_0 L$

D. $q/3\pi \epsilon_0 L$

E. (None)

Answer: E

Solution:

Electric flux due to charge placed outside is zero. But for the charge inside the cube, flux due to each face is $\frac{1}{6} \left[\frac{q}{\epsilon_0} \right]$ which is not in option.

